

Chapter 8: Plasma operation and control

To cite this article: ITER Physics Expert Group on Disruptions, Plasma Control, and MHD *et al* 1999
Nucl. Fusion **39** 2577

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M N Rosenbluth

Chapter 8: Plasma operation and control

ITER Physics Expert Group on Disruptions, Plasma Control, and MHD*

ITER Physics Expert Group on Energetic Particles,
Heating and Current Drive[†]

ITER Physics Expert Group on Diagnostics[‡]

ITER Physics Basis Editors[§]

ITER EDA, Naka Joint Work Site, Mukouyama,
Naka-machi, Naka-gun, Ibaraki-ken, Japan

Abstract. Wall conditioning of fusion devices involves removal of desorbable hydrogen isotopes and impurities from interior device surfaces to permit reliable plasma operation. Techniques used in present devices include baking, metal film gettering, deposition of thin films of low- Z material, pulse discharge cleaning, glow discharge cleaning, radio frequency discharge cleaning, and *in situ* limiter and divertor pumping. Although wall conditioning techniques have become increasingly sophisticated, a reactor scale facility will involve significant new challenges, including the development of techniques applicable in the presence of a magnetic field and of methods for efficient removal of tritium incorporated into co-deposited layers on plasma facing components and their support structures. The current status of various approaches is reviewed, and the implications for reactor scale devices are summarized. Creation and magnetic control of shaped and vertically unstable elongated plasmas have been mastered in many present tokamaks. The physics of equilibrium control for reactor scale plasmas will rely on the same principles, but will face additional challenges, exemplified by the ITER/FDR design. The absolute positioning of outermost flux surface and divertor strike points will have to be precise and reliable in view of the high heat fluxes at the separatrix. Long pulses will require minimal control actions, to reduce accumulation of AC losses in superconducting PF and TF coils. To this end, more complex feedback controllers are envisaged, and the experimental validation of the plasma equilibrium response models on which such controllers are designed is encouraging. Present simulation codes provide an adequate platform on which equilibrium response techniques can be validated. Burning plasmas require kinetic control in addition to traditional magnetic shape and position control. Kinetic control refers to measures controlling density, rotation and temperature in the plasma core as well as in plasma periphery and divertor. The planned diagnostics (Chapter 7) serve as sensors for kinetic control, while gas and pellet fuelling, auxiliary power and angular momentum input, impurity injection, and non-inductive current drive constitute the control actuators. For example, in an ignited plasma, core density controls fusion power output. Kinetic control algorithms vary according to the plasma state, e.g. H- or L-mode. Generally, present facilities have demonstrated the kinetic control methods required for a reactor scale device. Plasma initiation – breakdown, burnthrough and initial current ramp – in reactor scale tokamaks will not involve physics differing from that found in present day devices. For ITER, the induced electric field in the chamber will be $\sim 0.3 \text{ V} \cdot \text{m}^{-1}$ – comparable to that required by breakdown theory but somewhat smaller than in present devices. Thus, a start-up 3 MW electron cyclotron heating system will be employed to assure burnthrough. Simulations show that plasma current ramp up and termination in a reactor scale device can follow procedures developed to avoid disruption in present devices. In particular, simulations remain in the stable area of the l_i - q plane. For design purposes, the resistive V-s consumed during initiation is found, by experiments, to follow the Ejima expression, $0.45\mu_0 RI_p$. Advanced tokamak control has two distinct goals. First, control of density, auxiliary power, and inductive current ramping to attain reverse shear q profiles and internal transport barriers, which persist until dissipated by magnetic flux diffusion. Such internal transport barriers can lead to transient ignition. Second, combined use poloidal field shape control with non-inductive current drive and NBI angular momentum injection to create and control steady state, high bootstrap fraction, reverse shear discharges. Active $n = 1$ magnetic feedback and/or driven rotation will be required to suppress resistive wall modes for steady state plasmas that must operate in the wall stabilized regime for reactor levels of $\beta \geq 0.03$.

* ITER Physics Expert Group on Disruptions, Plasma Control, and MHD: S. Mirnov (Troitsk, chair), J. Wesley

(ITER JCT, co-chair), N. Fujisawa (ITER JCT), Yu. Gribov (ITER JCT), O. Gruber (IPP-Garching), T. Hender

Contents

1. Surface and wall conditioning	2579
1.1. ITER wall/surface conditioning requirements and constraints	2579
1.2. Conditioning methods used in present tokamaks	2580
1.3. Tritium retention and removal	2584
1.4. Implications for ITER: summary	2587
2. Plasma control	2588
2.1. Magnetic control and plasma disturbances	2590
2.1.1. Physics basis for plasma control	2590
2.1.2. Dynamic equilibrium control and plasma disturbances	2591
2.1.3. Magnetic control in present tokamaks and ITER	2592
2.1.4. Intelligent or adaptive magnetics control	2595
2.1.5. ITER controller design simulations and predicted response	2595
2.1.6. Summary and implications for ITER	2596
2.2. Kinetics control and divertor control	2597
2.2.1. Introduction and background	2598
2.2.2. Kinetics control requirements for ITER	2598
2.2.3. Present experience and accomplishments	2601
2.2.4. ITER kinetics control simulations	2604
2.2.5. Summary and implications for ITER	2605
2.3. Plasma operation scenario and related considerations	2605
2.3.1. Plasma initiation	2605
2.3.2. Plasma current ramp up	2610
2.3.3. Plasma current ramp down	2613
2.3.4. Fast plasma shutdown: requirements and options	2614
2.3.5. ITER scenario concept and simulations	2615
2.4. Advanced tokamaks: control of the current and pressure profiles	2617
2.4.1. Candidate parameters for ‘advanced performance’ operation in ITER	2618
2.4.2. Advanced performance control in present experiments and projections to ITER	2621
2.4.3. Simulations of ITER current drive and plasma operating point control	2622
2.4.4. Summary and future R&D recommendations	2622

(UKAEA-Culham), N. Ivanov (RRC-Kurchatov), S. Jardin (PPPL), J. Lister (EPFL), F. Perkins (ITER JCT), M. Rosenbluth (ITER JCT), N. Sauthoff (PPPL), T. Taylor (General Atomics), S. Tokuda (JAERI), K. Yamazaki (NIFS), R. Yoshino (JAERI). *Additional contributing author:* V. Phillips (TEXTOR-Jülich).

[†] ITER Physics Expert Group on Energetic Particles, Heating and Current Drive: J. Jacquinot (JET, chair), S. Putvinski (ITER JCT, co-chair), G. Bosia (ITER JCT), A. Fukuyama (Okayama Univ.), R. Hemsworth (CEA-Cadarache), S. Kononov (RRC-Kurchatov), W. M. Nevins (LLNL), F. Perkins (ITER JCT), K. Rasumova (RRC-Kurchatov), F. Romanelli (ENEA-Frascati), K. Tobita (JAERI), K. Ushigusa (JAERI), J. W. Van Dam (Univ. Texas), V. Vdovin (RRC-Kurchatov), S. Zweben (PPPL).

[‡] ITER Physics Expert Group on Diagnostics: K.M. Young (PPPL, chair), A.E. Costley (ITER JCT, co-chair), R. Bartiromo (ENEA-Padova), L. de Kock (ITER JCT), E.S. Marmar (MIT), V.S. Mukhovatov (ITER JCT), K. Muraoka (Kyushu Univ.), A. Nagashima (JAERI), M.P. Petrov (Ioffe Inst.), P.E. Stott (JET), V. Strelkov (RRC-Kurchatov), S. Yamamoto (ITER JCT).

[§] ITER Physics Basis Editors: F.W. Perkins (ITER JCT), D.E. Post (ITER JCT), N.A. Uckan (ORNL), M. Azumi (JAERI), D.J. Campbell (NET), N. Ivanov (RRC-Kurchatov), N. Sauthoff (PPPL), M. Wakatani (Kyoto Univ.). *Additional contributing editors:* W.M. Nevins (LLNL), M. Shimada (JAERI), J. Van Dam (Univ. Texas).

Preamble

Plasma operation and control constitute the ultimate manifestation of the science of tokamak operation, and in the broadest sense encompass the practical application of the physics basis knowledge and 30 years of plasma operation experience that is embodied in Chapters 2–7 of this Article. This Chapter addresses the implementation of plasma operation and control means, with emphasis on the operational techniques that are used to create and control plasmas in tokamaks so as to obtain maximum plasma performance or conduct scientific and technology validation studies. Assessing how this experience will extrapolate to ITER and reactor tokamaks is the rationale for this Chapter.

Control and monitoring of plasma operation must act to effect protection of tokamak systems against the normal and abnormal effects of plasma operation. This aspect of plasma control – already of some import in present tokamaks – will assume a higher

level of importance for reactor tokamaks and ITER, since the plasma energies and surface energy deposition levels inherent in reactor regime operation have a higher potential to effect plasma facing component surface damage, and the need for comprehensive protection of reactor tokamak systems against plasma operation produced damage is arguably higher – for both economic and safety reasons – in a reactor scale tokamak.

Hardware system considerations also enter into plasma control and operation and, in many cases, it is these design specific hardware considerations – PF coil configuration, power system controllability and response time, available auxiliary power, deposited power profiles, as well as the availability of diagnostic data – that ultimately determine how plasma control in reactor tokamaks can be effected and the degree to which the underlying characteristics of the plasma can be controlled – or not controlled. In the discussion that follows, both physics and hardware considerations are addressed.

The content of this Chapter is organized into two Sections. Section 1 addresses wall conditioning, an aspect of plasma operation necessary to obtaining the very low impurity release conditions that are essential for tokamak plasma operation in general and high fusion performance operation in ITER in particular. The subject of tritium retention and removal of retained in-vessel tritium, an operational and safety matter of great importance for reactor tokamaks and ITER, is also addressed in Section 1. Section 2 introduces what can be thought of as the kernel of tokamak plasma operation. More specialized details of the control of the plasma magnetic configuration and control of the basic plasma kinetic properties (density, fusion power, power and particle exhaust and divertor conditions) are, respectively, addressed in Sections 2.1 and 2.2. Specialized aspects of the scenario, including the details of plasma start-up, current ramp up and ramp down and current termination phases are presented in Section 2.3. The emerging topics of control and optimization means for ‘advanced performance’ and/or steady state plasma operation – where modification and active control of the naturally occurring plasma current profile is required (see Section 2.7 of Chapter 3) – are addressed in Section 2.4. The presentation here also briefly addresses the further considerations of implementation and control of the various possible internal or edge transport barriers that such ‘advanced performance’ modes may incorporate.

1. Surface and wall conditioning

The plasma facing surfaces in ITER will need to be conditioned before plasma operation commences. For ITER, pre-operation conditioning is required not only to establish a stable discharge, but also to avoid contamination of a burning DT plasma with appreciable levels of both higher- Z impurities and also hydrogen (H). In ITER, excessive levels of either of these impurities will lead to degradation of DT burn and fusion power capability and ultimately termination of sustainable fusion burn and plasma disruption. This Section summarizes ITER requirements for surface conditioning, reviews present tokamak condition methods, and presents how these methods will extrapolate to the ITER reactor regime. In the latter regard, extrapolation of presently used tokamak wall and surface conditioning methods to ITER and to reactor tokamaks in general is not fully straightforward, since operational limitations inherent in a practical ITER/reactor design will preclude or limit utilization of several of the presently most used surface conditioning techniques. In addition, unlike in most present tokamaks, plasma operation – with sustained current and fusion power – in ITER will comprise the majority of the tokamak operation cycle time. This plasma operation duty factor distinction is expected to result in a substantially different wall conditioning situation than in present tokamaks, where the plasma operation duty factor is very low.

1.1. ITER wall/surface conditioning requirements and constraints

As in present tokamaks, the plasma facing surfaces in ITER will need to be conditioned before plasma operation, after torus vacuum vessel openings, vents and major air or water leaks, and possibly after the occurrence of major disruptions. ITER plasma facing surfaces may also have to be conditioned – in a quasi-continuous manner – during the actual course of plasma operation. Conditioning – which, broadly defined, means achieving surface conditions in which the amount of desorbable hydrogenic species (H, D and/or T in ITER) is limited and in which the amount of desorbable non-hydrogenic impurities is reduced to very low levels – is necessary in ITER for a variety of reasons. These reasons divide into five categories:

- (1) maintenance of a low torus vacuum base pressure;

- (2) limitation of the release of non-hydrogenic impurities at start-up and during the ensuing plasma discharge so as to maintain acceptable plasma purity;
- (3) limitation of the release of surface absorbed H, D and/or T so as to maintain plasma density, isotope and fusion power control and acceptable edge plasma characteristics;
- (4) control of surface generated dust production and inventory; and
- (5) control/limitation of the tritium (T) inventory in the surface layers of the plasma facing components.

Consideration (3) also reflects the important nuance for DT burning tokamaks that volume entrained H – a ubiquitous component of many fabricated metals – constitutes an impurity that can significantly reduce plasma DT reactivity. Considerations (4) and (5) reflect the fact that surface generated dust (particulates, loosely bound layers, etc.) and surface absorbed or bulk absorbed T can constitute appreciable in-vessel inventories of radioactivated or inherently radioactive material. The mobilization and release of these inventories in certain operational failures could raise plant personnel and public safety concerns. In addition, long term in-vessel T accumulation is critical to the efficient utilization of the limited amounts of T available for ITER.

The detailed operational requirements for ITER plasma facing surface conditioning are determined both by the basic plasma operation requirements – pre-discharge base pressure, allowable plasma impurity and H levels, allowable uncontrolled DT release levels, etc. – and by the plasma facing surface materials and the conditions under which they are used. These latter aspects are design specific. The present ITER design incorporates a mixture of in-vessel materials and surfaces: carbon fibre composite (CFC) divertor plates (total in-vessel surface area $\sim 100 \text{ m}^2$), tungsten liners in the divertor and baffle ($\sim 400 \text{ m}^2$), and a Be clad first wall ($\sim 1200 \text{ m}^2$) and divertor dome. The torus vacuum vessel and major internal structural components are austenitic stainless steel, alloy SS316L ($\sim 7200 \text{ m}^2$). All of the plasma facing components are mounted on water cooled Cu alloy heat sinks. The maximum bakeout temperature of the in-vessel components will be limited to 240°C . This limit is set by the 4 MPa maximum pressure that can be allowed in the water cooling

pipes. There may be a possibility of baking in-vessel components to higher temperatures with hot gas, but the time required to completely remove water, introduce gas and then reintroduce water will limit such enhanced temperature baking to major commissioning and recommissioning periods.

The use of superconducting magnets in ITER will also significantly restrict surface conditioning methods. Owing to the time required to ramp the toroidal field up or down ($\sim 3 \text{ h}$) and the desire to avoid unnecessary TF magnet stress cycles, the ITER toroidal field will be maintained at or near its nominal 5.7 T value for weeks at a time, and present guidelines specify that the TF magnets can be cycled to zero 1000 times during the ~ 30 year life of the machine. If half of these magnet cycles are allocated to glow conditioning, this implies about 15 conditioning periods per calendar year. Given that scheduled ‘ready for plasma’ operation time in ITER will typically be about 25%, weekly (but not daily) glow conditioning appears possible. In addition, since rapid and frequent changes in the poloidal field of the type needed for rapid pulse discharge cleaning or high repetition rate low current tokamak pulses will cause unacceptable heating in the PF and TF coil cases and other cryogenic temperature structures, the use of pulse discharge cleaning and repetitive low current conditioning discharges will not be possible.

The means used for conditioning in present tokamaks and the underlying surface conditioning that they effect are briefly summarized in Section 1.2 below. The application of these methods to ITER is addressed in Section 1.4.

1.2. Conditioning methods used in present tokamaks

Conditioning of the tokamak plasma facing surfaces has been important in reducing the influx of both impurities and hydrogenic species in tokamak and has been proven to be an essential aspect of maximizing the plasma parameter operation space [1] (see also Sections 2 and 3 of Chapter 3) and for minimizing the frequency of disruptions (see Section 4 of Chapter 3).

The effects of surface and wall conditioning can be categorized into two major areas: reduction of impurity influxes; and control of hydrogenic fuelling from plasma facing surfaces. Historically, impurity control was the first area addressed in wall conditioning [2,3] followed, more recently, by techniques to control hydrogenic wall fuelling. With respect to impurity control, the control of oxygen (O) as a plasma impu-

rity plays a particularly important role [1,2]. Oxygen is present in tokamak wall materials as oxides and as water or other hydroxides. The sources of oxygen contamination in tokamaks are many and include air and water leaks, water vapour absorbed on surfaces and in materials during prolonged torus vacuum vessel openings, and long term diffusion from the bulk to the surface of the plasma facing component and vacuum vessel wall materials. Any wall material has to be critically assessed with respect to a low oxygen content, low affinity to store water, and a high resistance of their oxides against chemical reaction with hydrogen. Oxygen readily forms volatile gases (e.g. CO, CO₂ and H₂O), which can subsequently be released during plasma formation and operation. Oxygen is particularly troublesome during plasma start-up, since oxygen is a strong radiator for plasma temperatures in the 100 eV range. It is therefore necessary to limit oxygen contamination of start-up plasmas in ITER to $\leq 10\%$ (see Section 2.3).

The means available for control of oxygen include vacuum baking (to liberate surface absorbed water, CO and CO₂), various types of plasma discharge cleaning, which can remove more tightly bound oxygen, and gettering, which controls oxygen by passivation to form stable compounds that are not easily dissociated. A figure of merit for candidate getters is a high free energy of formation per oxygen atom for the formed oxides. A wide variety of materials make effective getters: Be and B, with respective free energies of 581 and 397 kJ·mol⁻¹, Al and Si, 527 and 428 kJ·mol⁻¹, and Ti and Ta, 444 and 352 kJ·mol⁻¹.

Control of hydrogenic influxes from the walls is the other important task for wall conditioning, particularly in present tokamaks without active during discharge pumping (see Section 3 of Chapter 3 and Section 2.2.2 below). Wall released hydrogen (H or D) can be particularly troublesome in present day tokamaks with large areas of plasma facing areas made from graphite. In many cases, the highest plasma confinement and DD neutron production have been obtained in low density discharges with low hydrogenic recycling, such as negative central shear (NCS), supershot, hot ion H-mode and VH-mode discharges [4–7].

A variety of tokamak wall conditioning techniques have been developed over the last two decades. The various presently operating tokamaks employ special combinations of several of these techniques to provide the wall conditions that permit high plasma performance operation or recommissioning after vacuum leak recovery or major disruptions.

Baking is effective in removing water, volatile hydrocarbons and hydrogen from the first wall materials. Baking is part of the standard regimen for the ultra-high vacuum practice that forms the starting for tokamak surface conditioning. Thus, most of the fusion devices have the capability of baking vacuum vessels and in-vessel components to temperatures of 250 °C or greater, including DIII-D, TEXTOR, JT-60U, Tore-Supra and JET. Higher temperatures are more effective, particularly when large quantities of graphite or carbon composite in-vessel material are present. As an example, Fig. 1 shows the volatile gas desorption for graphite during baking [8]. Figure 1 also demonstrates that the baking temperature for ITER, which is limited to 240 °C, should allow for adequate removal of water and volatile hydrocarbons from the graphite tiles. This temperature, however, is low when tritium removal is envisaged.

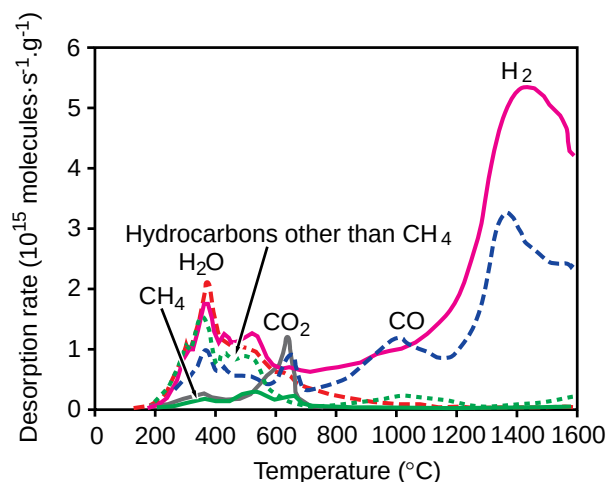


Figure 1. Thermal desorption of volatile gases from graphite (from Ref. [8]).

It has been found from present tokamak operation, however, that baking alone, without simultaneous or following plasma conditioning, does not usually produce satisfactory results.

Metal film gettering, primarily by Ti, Cr, was one of the earliest techniques applied. It passivates oxygen and other volatile impurities and reduces the influx of these impurities into the plasma discharge. However, the gettering material also has to fulfil other constraints, like plasma compatibility, hydrogen uptake and mechanical stability (flaking), which has resulted in abandonment of high-*Z* metal gettering. The JET tokamak successfully uses Be as a getter, which has the additional advantage of having a low *Z*, and a moderate hydrogen uptake, so

potential problems with influxes of the getter material into the plasma and hydrogen uptake are minimized [9]. Be evaporation has reduced the oxygen content of JET plasmas and their Z_{eff} values, and significantly improved the hydrogenic particle control. Besides this, Be gettering also resulted in slower plasma current quenches and more benign disruption effects. The successful operation of JET with beryllium evaporation on the inner wall has led to the decision to clad the inner vessel wall of ITER with beryllium.

Pulsed discharge cleaning. Taylor discharge cleaning (TDC) [10], repetitively pulsed low temperature discharges, is another early wall conditioning technique. Pulsing results in improved impurity removal efficiency: the dwell time between pulses allows for the volatile impurity gases to be pumped out of the vessel. TDC is especially effective in reducing surface oxygen to less than one atomic monolayer in metal wall tokamaks, particularly when it is used at elevated wall temperatures. In many applications, the ohmic heating of resistive torus vessels or in-vessel structures that TDC produces also raises vessel and in-vessel temperatures to high levels; in other cases, hot gas or external heaters are added to obtain the desired temperatures (see below).

In tokamaks in which the *in situ* baking temperature of the torus vessel is limited, thermal desorption of volatile impurities and hydrogenic species has been facilitated by heating only the plasma facing surfaces using disruptive discharge cleaning (DDC). The method has been used extensively in TFTR [11]. A series of low density helium discharges that terminate in disruption are used to provide high transient surface heat and particle fluxes that ‘flash desorb’ impurities and also raise the bulk temperatures of thermally isolated in-vessel components (e.g. graphite tiles) appreciably above the equilibrium temperature of the vacuum vessel wall. Although this DDC process requires operation of all tokamak systems and is more complicated and time consuming than simple baking, it is effective in conditioning the first wall.

Glow discharge cleaning has proven to be useful for both impurity and hydrogen recycling control. This technique is routinely used on many fusion devices, and also between discharges (ASDEX-Upgrade, DIII-D and TEXTOR). In addition, it has been used for disruption recovery on JET, TFTR and JT-60U. It uses a DC anode inserted into the torus vessel to produce a steady state low pres-

sure ‘cathode glow’ discharge (discharge pressures are typically about 0.1 Pa (10^{-3} Torr) that more or less uniformly covers the torus/wall surface. The anode voltage is typically ~ 500 V, and a current limiting series resistor or inductor is used to stabilize the discharge current and to limit current during arcing that can occur during initial periods of glow conditioning. Typical ion flux densities achieved with present systems in fusion devices are in the range $0.1\text{--}0.5\text{ A}\cdot\text{m}^{-2}$. For initial removal of oxygen and other volatile impurities, mostly hydrogen but sometimes also argon is used. Helium glow cleaning is used mainly to decrease the hydrogen from graphite walls by particle induced desorption processes. This is especially important for graphite wall machines, since graphite can be a large reservoir of hydrogenic atoms, providing excessive and unwanted fuelling. Graphite stores only a very small amount of helium and generally has minimal effect on the following plasma discharges. For this purpose, helium glow discharges have been used prior to operations or before every discharge to provide reproducible wall conditions [12]. Desorption of hydrogenic particles only occurs within the range of the He atoms in the near surface region. And for the most effective He conditioning, the energy of the incident ions needs to be greater than a few hundred eV, as shown in Fig. 2.

In DIII-D, between-pulse glow cleaning has been found to significantly improve plasma operation reliability in low field/high- β experiments where $q_{95} < 3$ is required. Between-pulse conditioning is also found to reduce the frequency of disruptions during such experiments [12].

Deposition of thin films of various materials on the entire plasma facing wall is a more direct way of modifying the composition and surface characteristics of the plasma facing surfaces. Techniques to apply these films consist of chemical vapour deposition (CVD), solid target erosion and deposition, and pellet injection [3]. However, the most used technique is the plasma assisted deposition of thin films, initially in the form of amorphous carbon films (carbonization) to reduce sputtering of metal impurities [13] and later by the use of boron films (boronization), which are more effective in simultaneously reducing oxygen, carbon and metal influxes. Boronization is presently used on many fusion devices throughout the world. In the DIII-D tokamak, boronization has also improved the energy confinement (VH-mode) [7], an effect that has not been seen on other toka-

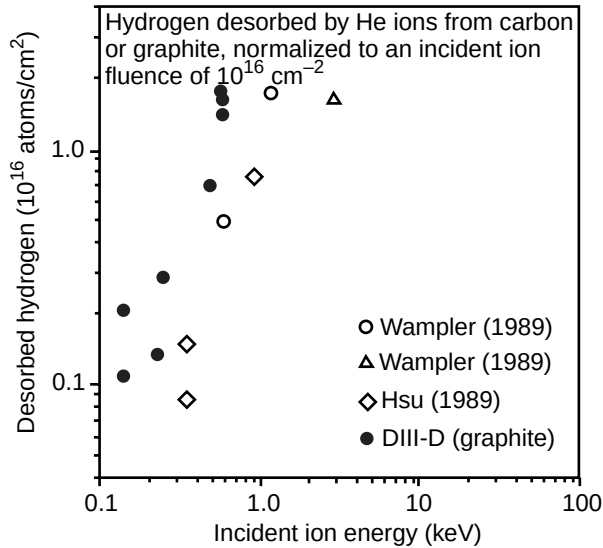


Figure 2. Helium induced desorption of hydrogenic species from hydrogen saturated carbon films (laboratory data, indicated by open symbols, see Ref. [12]) and DIII-D graphite tiles (*in situ* glow conditioning data, filled symbol) as a function of incident He ion energy. The desorption is normalized to a He ion fluence of $1 \times 10^{16} \text{ ions} \cdot \text{cm}^{-2}$.

maks after boronization. Deposition of silicon (silicization) has been done in TEXTOR and, recently, in ASDEX-Upgrade and is very effective in reducing the oxygen influx and recycling. In TEXTOR, it has also increased the edge radiation by physical sputtering of silicon and established radiated improved confinement conditions (RI-mode) without additional impurity seeding [14]. Lithium has also been used as a wall coating material. The most successful application of lithium has been on TFTR, where lithium pellet injection increased the fusion triple product and energy confinement time in super-shot discharges [15].

Radio frequency discharge cleaning has been used for impurity removal in a limited number of applications, generally as an alternative to more conventional glow or pulsed discharge cleaning methods. Electron cyclotron resonant (ECR) heating was successfully applied in the Gamma 10 mirror device, which has a very complex inner geometry, to improve wall conditions [16]. Recently, ECR discharges have been used on Alcator C-Mod for boronization, and discharges produced by pure ion cyclotron resonant heating have been used in the TEXTOR tokamak for in-vessel conditioning [17]. While rf heating discharges can be used basically in a magnetic field, the energy of helium or hydro-

gen atoms striking the wall is usually considerably lower than glow discharges, and hence, as shown in Fig. 2, the desorption efficiency is expected to be lower. In addition, the uniformity of the incident particles produced by these discharges needs to be determined.

The tritium inventory that can accumulate in the target surfaces and in the co-deposits in reactor tokamaks and in ITER is projected to be appreciable ($\sim 1 \text{ kg}$). For this reason, tritium removal becomes an important wall and torus conditioning issue for such tokamaks. It may also be needed for isotopic control [18]. For isotopic ‘switchover’ in JET tritium experiments, glow preconditioning with the desired species followed by ohmic and low power auxiliary heated discharges is found to be sufficient to obtain adequate ‘isotopic purity’ in subsequent operation with 100% T or 50%–50% DT. The physics basis for tritium retention and removal in present day and reactor tokamaks is discussed further in Section 1.3 below.

In situ limiter and divertor pumping. All of the techniques discussed above apply to present day tokamaks, which operate transiently, with plasma on/off duty cycle ratios $\ll 1$. *In situ* pumping can remove impurities on a steady state basis and also provide recycling control, important for long pulse and steady state fusion reactors. Several tokamaks have implemented *in situ* pumps in conjunction with pump limiters (e.g. TEXTOR and Tore-Supra) or in the divertor region (JET, DIII-D and ASDEX-Upgrade). These systems have demonstrated the ability to lower the wall inventory of hydrogenic particles, at least in graphite wall machines [19, 20]. For example, the *in situ* cryopump in DIII-D has been used in lieu of intershot helium glow discharge conditioning to reduce the hydrogenic wall inventory [19]. Wall inventories as low as those maintained by the helium glow were obtained. Pumping has thus demonstrated the ability to achieve low recycling wall conditions equivalent to other conditioning techniques. While reproducible discharges were obtained with pumping in DIII-D, the ability of the cryopump to control impurities, especially after disruptions, is not addressed in detail and is an area requiring further investigations before *in situ* pumping can be considered as a replacement for other techniques, such as He glow conditioning. Nevertheless, pumping is a promising conditioning technique for devices with a continuous magnetic field and long pulses such as ITER.

1.3. Tritium retention and removal

As has been noted above, the amount of tritium retained in the surface and bulk materials facing the plasma in ITER and the means required to periodically remove retained tritium become important considerations for the design of ITER systems and plasma operation procedures. Projections of the amount of retained tritium are required for several reasons: assessment of the radiological hazard associated with routine operation and with potential accident scenarios; determination of the plasma fuelling requirements and DT supply; and establishment of detritiation requirements for coolant water. Present understanding of the mechanisms by which hydrogenic isotopes are retained in PFCs as well as quantitative estimates of the fraction of injected plasma fuel that is retained come from both experiments in existing tokamaks and from laboratory experiments designed to probe fundamental processes [21, 22]. Results from these tokamak and laboratory experiments are being used to validate models that are utilized in predicting tritium retention in ITER (see, for example, [23]).

Some appreciation of the magnitude of the tritium retention effect in ITER can be had from present predictions of these models, which indicate that more than 1 kg of tritium can accumulate in PFCs in less than 1 year of the 'full duty cycle' nuclear testing operation that is planned to take place during the final 5 years of the ITER basic performance phase (BPP). The projected tritium burnup rate will be about 5 kg per year for such operation (1500 pulses per year $\times$ 3.5 g per pulse). For reference, the projected total tritium burnup for the BPP will be about 27 kg, and the annual worldwide tritium production rate projected to be available for ITER will be about 2 kg per year. So the projected 'per annum' rate of tritium retention in ITER will be appreciable compared to the annual utilization rate (to be derived primarily from already accumulated commercial tritium reserves) and will be comparable to the projected production rate in the period in which ITER operates. These supply related considerations alone mandate that retained tritium in ITER be limited to levels of ~ 1 kg and that any long term secular increase in retained tritium be prevented.

1.3.0.1. Tritium retention mechanisms

Hydrogenic isotopes will be retained in plasma facing components of ITER by three principal mechanisms (see [21, 22] and references therein):

- (i) direct implantation of ions escaping from the plasma, which leads to hydrogen retention primarily in a shallow surface region and possibly also diffusion into the bulk (depending on the material used and the temperature);
- (ii) co-deposition of hydrogen isotopes with eroded carbon or beryllium (the latter only if abundant O is present) on plasma facing surfaces, which produces co-deposited surface layers with significant hydrogenic content; and
- (iii) production of tritium by transmutation nuclear reactions in beryllium, which results in tritium inventory within the bulk material, principally in microscopic defect sites and bubbles containing helium.

There is still a significant uncertainty in quantifying the in-vessel tritium inventory of ITER. Hydrogenic retention is influenced by such factors as the materials used (e.g. carbon, beryllium, tungsten), as well as their crystalline structure, the temperature, and neutron irradiation history, by the presence of other impurities such as oxygen, by plasma conditions close to the PFC, and by the spatial distribution of erosion and redeposition. The use of more than one PFC materials (Be, C, W) in ITER will lead to the formation of complex redeposition layers involving several materials with retention properties that are not well known.

1.3.0.2. Retention in present tokamak experiments

Quantitative estimates of hydrogen retention are routinely obtained in tokamak experiments using hydrogen or deuterium. The basic mechanism by which the hydrogen species are retained have been identified in several tokamaks. Under carbon wall conditions the dominant mechanism for hydrogenic retention is co-deposition of carbon with deuterium. In-vessel surveys in TFTR by beta back scattering and ion beam analysis of tiles removed from the vessel show that the dominant regions of redeposition are located away from the plasma-limiter interaction regions, with co-deposited films of up to $10\text{ }\mu\text{m}$ thickness developing in low flux regions of the scrape off layer (SOL). Significant co-deposition was also observed in gaps between the limiter tiles. In TFTR deuterium fuelled discharges over a five year period, the fraction of injected deuterium retained in the first wall was found to be 44% (with uncertainty $\pm 17\%$).

More recently, TFTR and JET have provided precise measurements based on the accounting of tritium in DT experiments. During the DT experiments in TFTR, which lasted 3 years, the fraction of tritium retained in the vessel was found to vary with discharge type, clean-up history and the period studied [24–26]. Overall, the long term retention was in the range 30–55%, depending on discharge history. This fraction is in excellent agreement with the retention observed in deuterium experiments. Of the 2.7 g of tritium introduced by NBI and 1.7 g introduced by gas puffing, 1.6 g remained in the vessel at the end of DT operations.

Several removal techniques including DC discharges in oxygen and deuterium, pulsed discharges, and ventilation with room air were applied to remove tritium during this period, with varying degrees of success. GDC in D2 was found to be less effective than He/O GDC, but He/O-GDC was substantially less efficient than anticipated from laboratory experiments [26]. In addition, later removal campaigns were less effective than the initial campaign, presumably because the majority of the retained tritium was more deeply buried. Following termination of TFTR experiments, the residual tritium retention in the torus after various removal techniques had been exhausted was approximately 1.3 g, with a residual outgassing rate of <1 mg per day.

To evaluate the consequences for ITER of the large hydrogenic retention observed in TFTR, one has to keep in mind that TFTR operated at low wall temperatures of about 300 K and had no strong external pumping comparable with divertor pumping. Low wall temperatures enhance the amount of deuterium stored in the co-deposits and decrease the efficiency of clean-up removal techniques, and a low pumping reduces the gas throughput and increases the fraction of retention. In addition, plasma scenarios with high edge temperatures, as used routinely in TFTR, result in large carbon impurity release rates, which results in thicker carbon deposits.

Although DT experiments in JET have, to date, been of much briefer duration than in TFTR, the use of both beryllium and carbon in PFCs is of particular relevance to ITER. JET operates at wall temperatures of about 600 K with diverted plasmas with much colder plasma temperatures in front of the target. Prior to 1989, PFCs in JET were fabricated from carbon based materials or Inconel (from which the vacuum vessel was constructed). Carbonization [27] was used to cover metallic areas.

With a carbon dominated first wall, 13 mg of deuterium was typically required to fuel a JET discharge, of which a total of 60% was recovered after a full day of plasma operations [28]. Overnight and weekend outgassing increased this recovery to up to 80%. Analysis of in-vessel components [29] revealed similar retention. After beryllium was introduced in 1989, both by evaporation as a gettering medium and in the form of two toroidal belt limiters, increased wall pumping led to a reduced fuelling efficiency but also to an increased recovery rate of deuterium after the discharge, so that the amount of deuterium per plasma pulse retained in the vessel remained approximately the same as with an ‘all carbon’ first wall.

Analysis of carbon divertor tiles removed from the JET torus after the experimental campaign in the MkI pumped divertor indicated a deuterium retention of ~3.3 g for the entire vessel, very similar to that observed in previous campaigns [30]. Moreover, following experiments using a beryllium divertor target [31], similar levels of deuterium were found on beryllium tiles from the inner divertor strike region as on equivalent carbon tiles [30]. Significant carbon deposition was also observed in this region, presumably due to redeposition of carbon eroded from the PFCs of the main plasma chamber. Deuterium deposition on the beryllium tiles of the outer divertor strike region was significantly below that observed in equivalent carbon tiles. Initial analysis of carbon tiles removed from the MkII pumped divertor, which approximates a continuous toroidal target much more closely, showed a very similar pattern of deuterium deposition on the target tiles to that observed in the MkI target, with the dominant deposition in the inner leg of the divertor [32]. However, the average D concentrations on all the plasma exposed areas of the divertor floor had fallen by a factor of 2, possibly due to the higher operating temperature (200–350 °C) of the MkII tiles as compared with that of the MkI tiles (~50 °C, with limited excursions to higher temperatures). Of particular significance was the observation, for the first time, of films and flakes of deuterium saturated material in cooled regions behind the divertor pumping slot. If these deuterium saturated flakes are assumed to be formed with toroidal uniformity, they contain an inventory corresponding to 3% of the throughput.

Analysis of tritium retention and removal was a key aim of the JET preliminary tritium experiment (PTE) in 1991, during which 5.5 mg of tritium was injected into plasmas via the NBI system. Although 67% of the injected tritium remained in the torus two

days after the experiment, specialized clean-up procedures succeeded in reducing this fraction to $\sim 10\%$ [33]. During the DTE1 experiment (1997), a total of 35 g of tritium was introduced into the torus: 0.6 g by NBI, and the remainder by gas puffing. The maximum tritium inventory in the torus during DTE1 just after the last T fuelled pulse was ~ 11.5 g. This tritium inventory could be reduced by a factor of about 2 by pulsing in deuterium plasmas at which point no noticeable further tritium exchange could be observed. This release by isotope exchange is much less than in the PTE experiment. Thus, by the end of the MkIIa divertor campaign, about 6 g of tritium remained in the torus. The vessel was vented to change the divertor tiles remotely, and, at the end of the shutdown, the total amount of tritium released was 2.0 g, i.e. approximately one-third of the tritium inventory at the end of plasma operation [34].

1.3.0.3. Other retention data from present experiments

Studies of hydrogenic retention in other large and medium tokamaks have also been made via analysis of retained tritium produced by DD reactions. However, these tritons are produced with 1 MeV and can impinge directly on the first wall with high energy and, thus, a high retention probability.

In JT-60U, analysis of exhaust gases showed that 80–90% of tritium produced in DD reactions was initially retained in the torus [35]. Hydrogen plasma pulsing and helium DC with the vessel at 150–300 °C reduced this fraction to 70–80%. Analysis of sample tiles from the first wall and divertor accounted for 50% of the tritium produced [36], with the remainder thought to be accounted for by dust, toroidal asymmetries and in sections other than the immediate first wall.

Tritium produced by deuterium plasmas in DIII-D accumulated in a narrow surface layer on the carbon PFCs on the first wall. The fraction of tritium thus retained corresponded to 20% of the tritium production in the 1991–1992 period, or $\sim 10\%$ of the integrated tritium production in DIII-D to that time [37]. It was found that the tritium could be removed from the tiles as DT gas by baking in an oven to 1000 °C.

ASDEX-Upgrade has investigated both graphite and tungsten coated graphite as divertor target materials and the inventory of hydrogenic species following plasma operation has been analysed by quantitative thermal desorption spectroscopy, nuclear reaction analysis and calibrated secondary ion mass spectroscopy [38,39]. The measured inventory, equiv-

alent to $\sim 0.33 \text{ g} \cdot \text{m}^{-2}$ of deuterium, was found to be predominantly in co-deposited layers, which form on the inner divertor strike region, and in near surface regions (10–25 μm) beyond the implantation zone. Inventories at the divertor strike points were a factor of 10–100 larger than at the inner wall limiter. Measurements of the tritium inventory of sample tiles from the inner wall limiter and the outer divertor strike point were consistent with the tritium production estimated from the DD neutron yield. As in the cases of hydrogen and deuterium, the tritium was found to be present at depths (up to 25 μm) well beyond the ion implantation zone, indicating that diffusion into the bulk occurs.

1.3.0.4. Extrapolation of tokamak data to ITER

Great caution is necessary when directly extrapolating the hydrogenic deuterium and tritium retention data from present day machines to ITER. The short pulse lengths of present day devices – typically 1–10 s – and the significantly smaller ratios of plasma volume to wall surface area favour wall pumping compared with external pumping, and hence lead to higher retention. In present day devices, the wall retention fraction decreases with increasing gas throughput (JET). Thus, it seems that extrapolations of present retention behaviour to ITER should be based more upon the absolute amount of fuel retained rather than on the retained fraction. Moreover, owing to the short pulse length, measurements in present devices are influenced disproportionately by transient effects such as impurity generation and transport, and their affect on erosion/redeposition, during the start-up and shutdown periods. Finally, the ITER divertor densities are expected to be substantially higher than those in JET or ASDEX, and with somewhat lower divertor target or edge temperatures (see Chapter 4). Furthermore, the wall surface temperature in the divertor region will be higher.

1.3.0.5. Retention data from laboratory experiments

The interaction of hydrogenic species with plasma facing components has also been extensively studied in laboratory experiments, where individual processes can be investigated and understood in isolation and where conditions are better controlled and diagnosed than in tokamaks. New implantation and co-deposition data are available, particularly for beryllium and for tungsten alloys, for conditions representative of ITER (see, for example, [21]).

1.3.0.6. ITER tritium retention estimates

Neglecting transient wall pumping effects by implantation and saturation of a shallow near surface region with tritium, which should be of minor importance for ITER, the tritium inventory in the vacuum vessel of ITER will be mainly determined by co-deposition of tritium with carbon, and possibly beryllium, eroded from the wall on the cold surfaces of the divertor. This process has been simulated for ITER divertor conditions, using the plasma edge parameters at the strike zones, impurity release from the target (mainly chemical erosion), and near wall transport and redeposition of eroded molecules based on molecular break-up data of methane [40]. These studies predict tritium co-deposition in the range 1–20 g per pulse ($=0.3\text{--}6\times$ burnup per pulse), depending mainly on the divertor operation regime. For the nuclear testing phase of the BPP (1500 pulses per year), the corresponding per annum accumulation would be (without removal) 1.5–30 kg. Depending on the accumulation rate, 1 kg of retention would develop after 50–1000 pulses (31–600 hours of continuous operation at 1.6 pulses per hour). If an administrative limit of 1 kg were to apply (this is an arbitrary but not unreasonable assumption and consistent with safety requirements), then tritium removal action would be required after this number of pulses.

As can be seen, the variation in the range of predictions for tritium retention is large [21, 22]. With respect to the causes of the variation, attached divertor scenarios are favourable (low retention) due to a high local redeposition probability (lead to low retention), whereas detached scenarios are unfavourable, with high retention, since the low temperature conditions are more transparent for escape of the hydrocarbons formed at the target and are found in the modelling to result in thick co-deposits on side areas with large amounts of stored tritium. The estimated amount of tritium retention in these detached scenarios is consistent, to some extent, with the retention observed in present short pulse tokamaks. It is therefore conceivable that per annum retention rates in ITER will be of the order of 10 kg. At this rate of accumulation, periodic removal during a sustained operation (fluence accumulation) campaign will probably be required, if only because of the need to recover and recycle in-vessel tritium.

There is presently major uncertainty about tritium accumulation and removal in an ITER class tokamak. However, present data show the potential magnitude of in-vessel retention and a criti-

cal need to develop and test *in situ* cleaning techniques for the efficient control and removal of the co-deposited tritium in ITER. It appears likely that the need for periodic removal of tritium and subsequent wall reconditioning will be a significant factor in the ‘ready for DT plasma operation’ availability of ITER and reactor tokamaks.

1.4. Implications for ITER: summary

Five general tasks for wall conditioning in ITER have been identified.

- (1) Preparation (commissioning) for initial operation.
- (2) Recommissioning following major openings, vents, in-vessel component replacements and significant leaks.
- (3) Daily or weekly conditioning during operation.
- (4) During and between shot conditioning.
- (5) Tritium inventory removal

Wall conditioning in ITER, as in present day fusion devices, is projected to involve a variety of different methods. As has been noted above, ITER has several unique features including superconducting magnets, which produce continuous magnetic field for months at a time, long pulse operation, and a variety of wall materials. Initially, there will be no magnetic field, so most of the conditioning techniques used on present day devices can be applied to ITER during the commissioning phase and during major recommissionings. The applicable techniques include: pre-cleaning and pre-baking of individual components, *in situ* baking, glow discharge cleaning, and maybe also thin film deposition depending on operational experiences. These methods have been shown to be effective under a variety of conditions and are expected in ITER to be adequate to initially condition the machine for initial plasma operation. The plasma facing components will be baked at higher temperatures (>600 K) before installation. After assembly and prior to the first plasma, all of the plasma facing components will be baked *in situ* at 240°C followed by glow discharge cleaning ($B = 0$) and/or ECR cleaning ($B > 0$).

Once the ITER toroidal field magnets are energized, however, different techniques are required. Conditioning might be necessary after strong leaks, disruptions or other events in order to provide reproducible initial conditions and ensure a current ramp

up with low impurity influxes and no deleterious MHD, such as locked modes. The presence of full or nearly full toroidal field will preclude glow discharge cleaning. While the details of possible pulsed discharge cleaning scenarios remain to be studied, the need to implement such scenarios at nearly full toroidal field (and hence with relatively high plasma current) will likely also limit the applicability of standard pulse or Taylor discharge cleaning methods. Therefore, the principal means envisioned for between plasma operation and between pulse cleaning in ITER is ECR and/or ICRH conditioning. The ability of both of these methods to effect conditioning in present tokamaks has been demonstrated. However, the efficacy of both techniques in impurity removal and recycling control and the exact requirements for application to ITER and reactor tokamaks need more research.

Baking in ITER can be performed with or without a magnetic field and will undoubtedly be an important conditioning technique, but during routine operations it would probably have to be scheduled for periods when no experiments are planned, possibly on weekends. Diborane ‘flushing’ [1], i.e. chemically passivating oxygen by injecting B_2D_6 gas, is not affected by magnetic fields and may be a useful technique between ITER pulses.

In addition to the conditioning procedures discussed above, operation during discharges will be maintained in ITER with divertor pumping, which will provide continuous exhaust of impurities from the walls that are ionized in the SOL and swept into the divertor. Such pumping will also control the hydrogenic particle inventory in the plasma facing surfaces, particularly the graphite material. When ITER operates with its design duty cycle of a 1000 s burn every 2200 s, the total plasma on/off ratio (with start-up shutdown periods included) will reach 0.6, so *in situ* ‘plasma operation conditioning’ is expected to become (must become) the dominant conditioning (maintenance) process. In this sense, ITER operation will differ dramatically from present day experiments, where opportunities for before or between plasma operation conditioning dominate the utilization of plasma operation time.

The use of other wall conditioning techniques for ITER are more speculative. The effect of thin film deposition (such as beryllium, boron or lithium) is, at most, transient. Such transient deposition could be important for discharge initiation or real time erosion control, but the latter will require additional investigations. The use of beryllium as an ITER first wall

material may provide oxygen control, since some of this wall material will be sputtered and could act as a natural getter.

Finally, as noted above, appropriately modified wall conditioning techniques (e.g. He/O discharges or the introduction of high pressure oxygen or steam) are also being discussed for lowering the T inventory in ITER after periods of sustained plasma operation. It is, of course, quite clear that measures to remove tritium inventory will compromise the surface conditions needed for plasma operation, so a long term cycle of initial preparatory conditioning, plasma operation possibly supplemented with between operation conditioning, tritium removal, optional replacement of divertor and other plasma facing surfaces and reconditioning is envisioned.

2. Plasma control

Plasma control can be defined as the implementation of the scientific and technological understanding required to create, sustain and terminate a tokamak discharge and to optimize the parameters of the plasma. Given this all embracing scope, plasma control depends upon all of the physics basis elements presented in this paper, together with other technical considerations such as vacuum technologies and the control of pulsed electrical power systems. In a more limited sense, plasma control is frequently defined to comprise the operational implementation of the necessary control procedures and algorithms. In this sense, this section places emphasis on the physics aspects underlying the plasma control and operation methods implemented in present tokamaks and foreseen for ITER, covering (i) magnetics control; (ii) kinetics and divertor control; (iii) scenario control; and (iv) ‘advanced performance’ control involving active current profile modification. These issues are introduced by a brief history of plasma control and the present concept for ITER. The ITER plasma control design is conceptual, with a focus on identifying the physics basis and on developing a plasma control and machine protection scheme.

2.0.0.7. Plasma control: an historical overview

There has been tremendous progress in the area of plasma control since the first tokamak experiments, evolving from pre-programmed electrical pulsed power control on the first tokamaks to real time feedback control of most key plasma parameters as a routine feature of present tokamaks. A great

deal of operational experience has been accumulated in the control of important parameters such as the plasma current, plasma position, plasma shape, plasma density and plasma temperature. Control of the radial profiles of the plasma parameters is becoming well established and has led to performance enhancement, providing control of MHD instabilities and control of the power and particle exhaust characteristics of the divertor plasma. Furthermore, it is now possible to optimize the plasma operational regime to obtain operational modes with enhanced energy confinement relative to L-mode. This ‘operational regime’ control allows high beta to be obtained more regularly and reproducibly, limited in time by hardware considerations rather than by ‘uncontrolled’ or unexpected evolution of the plasma itself.

There has also been progress in plasma diagnostics and in the coupling of these diagnostics to the plasma control systems, as well as in the validation of the physics and control modelling for real time plasma control. These models will be important for optimizing the ITER plasma control system [41], since ITER will not only require a set of suitable plasma diagnostics and control actuators but also a good model of the effect of these actuators on the plasma. For complex ‘multi-variable’ control, an adequate model of the plasma response is an essential component.

Tokamak operation and control is a highly coupled problem in which the full plasma state depends on factors that are not necessarily controllable. Improved physics understanding and modelling together with actuators that can more directly affect key plasma properties will result in better control and optimization of plasma performance.

2.0.0.8. ITER control requirements

ITER control is based upon physics results and operational experience from many tokamaks, most directly those with shaped plasma cross-sections and divertors including Alcator C-MOD [42], ASDEX-Upgrade [43], DIII-D [44], JET [45] and JT-60U [46]. Many of the plasma control requirements and solutions proposed for ITER [41] are similar to those already implemented on these devices and other operating tokamaks. However, the need for ITER to operate with a burning plasma under stationary conditions for more than 1000 s, while handling more than 1 GJ of plasma thermal and magnetic energy, poses new physics and control constraints. Furthermore, ITER will be subject to a high level of regulatory scrutiny, especially with regard to reliable

control of the fusion burn conditions and provision of a guaranteed method for rapidly terminating the fusion burn under off normal conditions.

As in present day tokamaks, the ITER plasma must be initiated, controlled and stabilized magnetically (R , Z , a , κ , δ) and brought to near stationary kinetic conditions (n , T , Z_{eff} , n_{He}/n_e , n_{imp}/n_e and P_{fus}) for a period sufficient to achieve the experimental objective. The large size and high temperature of the ITER plasma means that the time to reach near stationary plasma conditions, including a fully diffused current profile, can be as long as 1000 s. Plasma control methods for ITER may, therefore, have to be ‘steady state’, possibly eliminating some transient control techniques such as current or shape ramping, gas puffing, pellet injection and wall conditioning. Such transient methods can only be important in ITER for physics experiments or in the approach to a stationary plasma condition. ITER control should ultimately be compatible with stationary plasma operation.

The high level of fusion power and the relatively short thermal time constants (~ 3 – 10 s) of the actively cooled heat removal surfaces make both reliable fusion power control and a backup fast fusion power and plasma current shutdown mechanism essential for machine component protection. Present concepts for fast shutdown in ITER focus on the injection of a massive impurity pellet ($\sim 10^{22}$ atoms) leading to a rapid fusion power and current quench on a time-scale of ≤ 300 ms (see Section 4.5 of Chapter 3).

The burning ITER plasma must be terminated in a controlled manner, in which first the fusion burn and then the plasma current are shut down without disruption. The requirements for a controlled shutdown and for disruption and VDE (vertical displacement event) avoidance throughout the plasma operation scenario are particularly important owing to the large plasma thermal and magnetic energy in burning plasma operation. Burn termination is one of the few issues for which we have no experimental physics basis and for which we will have to await ITER operation.

The requirements for the ITER plasma control system therefore comprise four major elements [41] as follows.

Plasma operation scenario sequencing: poloidal field (PF) coil system pre-magnetization, plasma initiation using EC assisted breakdown, current ramp up and shape expansion, divertor formation, auxil-

iary heating, ignition and burn, fusion power shutdown, current ramp down and termination when the maximum PF flux is reached, followed by PF reset and magnet cool down; each scenario phase may require distinct control methods, with a specific set of quantities to be controlled and a specific set of control algorithms; the present ITER control strategy is a mix of sequencing logic, pre-programmed and closed loop feedback control of the various hardware systems.

Plasma magnetics control: plasma current control, plasma shape control, plus control of selected plasma to first wall clearance gaps including the nominal divertor magnetic configuration; also non-axisymmetric error field compensation; various engineering limits of ITER are to be respected.

Plasma kinetics and divertor control: control of the H-mode transition, density, fusion power, impurity content and radiated power fraction; pressure and density limits are to be avoided; plasma profile control using auxiliary heating and current drive, divertor control using pumping, divertor gas and impurity injection; possibly stabilization of certain MHD instabilities, neoclassical modes and sawteeth, to avoid disruptions.

Fast plasma shutdown: shutdown of fusion power and current by means of impurity injection; if disruption does occur, mitigative actions should reduce thermal and electromechanical loads on the in-vessel systems; plasma control system actions will overlap hardware protection functions that have traditionally been addressed by separate interlock and/or fault detection systems.

2.1. Magnetic control and plasma disturbances

Magnetic control comprises adjusting the tokamak plasma equilibrium using the currents in the PF coils. Plasma current, position and shape measurements are derived primarily from magnetic diagnostics, such as Rogowski coils, magnetic probes, flux loops and diamagnetic loops. The ability of such a 'magnetic feedback control' to reliably and precisely adjust the tokamak discharge equilibrium is already a key factor in routine and reliable tokamak performance. Manipulating the plasma shape and current has been instrumental in experiments to determine the effect of key plasma parameters such as the $q(r)$ and $j(r)$ profiles on plasma transport and MHD stability (see Section 2.7 of Chapter 3). Mag-

netic plasma control has been refined to such a degree that it is usually regarded as a domain of engineering operation rather than plasma physics.

Magnetic control has the essential functions of stabilizing the axisymmetric vertical instability inherent in elongated cross-sections, and in keeping the plasma shape, position and current nearly fixed even in the presence of internal plasma disturbances such as sawteeth, minor disruptions, ELMs and other transients. Quantifying the expected effects of such disturbances and designing an appropriate control system to limit them is an important aspect of plasma magnetics control.

The use of superconducting coils in ITER introduces the further consideration of limiting the 'ac losses' that arise from plasma control actions. The continuous dithering of the controlled plasma parameters around a stable operating point owing to variations in the plasma parameters themselves and to repetitive plasma disturbances lead to heating of the superconducting coils and to the cryogenic structures in the magnets. The heating resulting from these effects is grouped together under the heading of 'ac losses'. Estimates of these losses for typical ITER operation indicate that they might become important during long discharges, and in certain scenarios could become a significant factor in the long term thermal balance of the coil systems. Accordingly, the ITER controller will have to limit cumulative AC losses owing to repetitive disturbances and noise.

2.1.1. Physics basis for plasma control

The underlying basis of magnetic plasma control in a tokamak is the Grad-Shafranov ideal MHD plasma equilibrium equation (see Section 2.1 of Chapter 3), which describes two dimensional magnetostatic plasma equilibria in the absence of significant mass flow, plasma rotation or pressure anisotropy. With the exception of the TSC code, all work on ITER magnetics control is based on a massless plasma assumption. The Grad-Shafranov equation determines the plasma equilibrium flux surface configuration for defined PF coil currents, plasma pressure profile, current profile and toroidal field. It is possible to determine the normalized plasma pressure, β_p , and the internal inductance l_i when the plasma is elongated [47]. Conversely, specification of these two parameters is normally sufficient to determine the static flux surface configuration accurately.

Figure 3 illustrates an ITER plasma equilibrium for the 1.5 GW fusion power burn phase. The nom-

inal internal inductance and poloidal beta assumed are $l_i = 0.9$ and $\beta_p = 0.9$ for an ELMy H-mode discharge. The equilibrium is provided by 10 PF coils located outside the toroidal field coils (see Fig. 4).

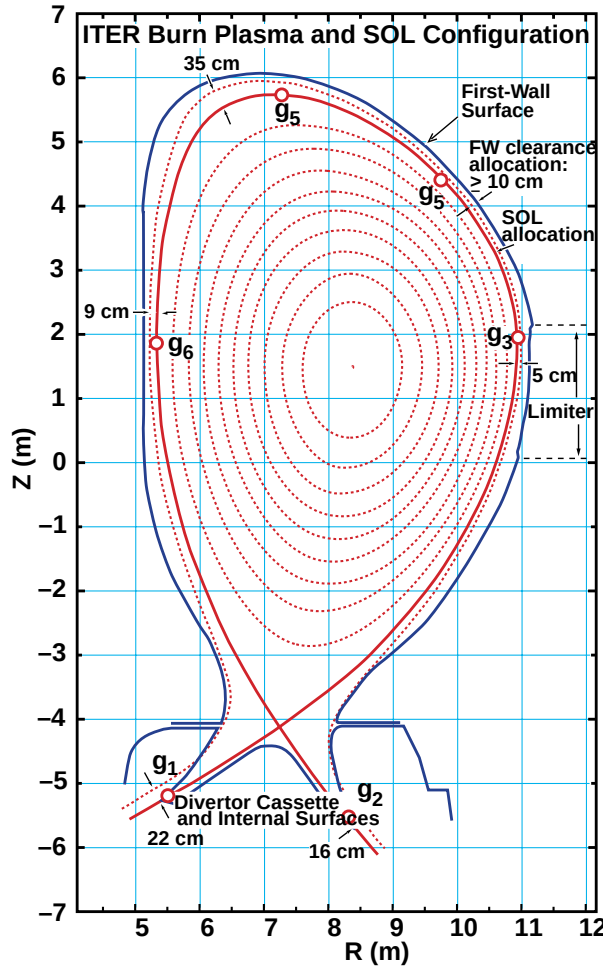


Figure 3. ITER plasma equilibrium for a 1.5 GW burn.

Figure 3 illustrates several important features for control:

- (i) the nominal burn phase ITER plasma equilibrium configuration remains close to the first wall (FW) and divertor surfaces, for reasons of efficient use of the vessel and magnet volumes;
- (ii) there is only a modest (~ 10 cm) allowance for clearance between the plasma SOL and the FW surfaces; and
- (iii) the plasma shape is specified by six ‘gaps’, labelled g_1 – g_6 in Fig. 3.

Gaps g_1 and g_2 define the location of the inboard and outboard divertor strike points, the intersections of the separatrix with the target surface: gaps

g_3 – g_6 control the shape of the plasma within the FW plasma facing surfaces. These six gap control points are used for both static equilibrium specification and as the references for dynamic control. The requirement for magnetic equilibrium control is to reduce the variations of the six gaps to within about ± 10 cm, within the SOL to FW clearance allowance.

Static equilibria form the basis for plasma scenario development (Section 2.3) and for dynamic modelling codes that self-consistently calculate the interaction between the plasma, the PF coil system and any toroidally conducting structures (see Section 4.7 of Chapter 3). There are two approaches to dynamic modelling:

- (i) calculate the evolution of the plasma equilibrium in a fully self-consistent manner based on solution of the ideal MHD pressure balance equation (done in the tokamak simulation code, see Section 4.7 of Chapter 3 and [48]); or
- (ii) prescribe constraints for equilibrium codes that employ *ad hoc* profiles to maintain correspondence between the reference and perturbed equilibrium states. When suitable constraints are imposed, the resulting simpler models can serve as tools for magnetic control design.

2.1.2. Dynamic equilibrium control and plasma disturbances

The control system must maintain the specified plasma current, shape and position parameters in spite of (i) slow evolution of the plasma temperature and resistivity, current profile and pressure as the plasma scenario proceeds, and (ii) faster transient disturbances due to changes to the current and pressure profiles produced by MHD activity, transient external disturbances such as impurity release, by the initiation of auxiliary heating, pellet injection, or flaking of first wall material. In plasma control design, three types of repetitive disturbances seen in tokamak plasmas can be considered: sawteeth, minor disruptions and ELMs. The underlying physics basis and characteristics of these disturbances are addressed in Sections 2 and 4 of Chapter 3. They affect l_i and β_p , modifying the equilibrium, which requires magnetic control action.

The effects of minor disruptions and ELMs on plasma shape, position and current are seen in all tokamaks. For minor disruptions and large ‘compound’ ELMs, there is an effect on the plasma inductance, position and current, but the discharge typi-

cally survives without any immediate control reaction. In JET, control becomes a problem with exceptionally large disturbances such as large ELMs or large sawteeth, which can even lead to a loss of equilibrium and plasma termination with a vertical displacement event (VDE) (see Section 4.3 of Chapter 3). The control following a plasma disturbance is related to the effect of the shape disturbance relative to the gap clearance tolerance. If the effects are small and if the disturbance heals rapidly, no immediate response is needed for radial position or shape control and a response may even exacerbate the effects of the disturbance because of finite feedback response time. If the shape perturbation approaches or exceeds the gap tolerance, then an effective control response following the disturbance becomes mandatory.

Because of the sensitivity of the control to the magnitude of the disturbances, quantifying them is an important issue for ITER. The expected disturbances specify the PF power supplies and PF controller. Table 1 summarizes the disturbance types and parameters that have been derived from experimental data [41], leading to assumed changes in the two equilibrium parameters. The last column lists the evolution of the disturbance, a drift, a step, a recovering sawtooth or a simple ramp, as well as the expected maximum frequency of the disturbances and the maximum number of events expected per pulse. These are required to determine the effects of the feedback control actions on the AC losses.

2.1.3. Magnetic control in present tokamaks and ITER

A PF control system similar to the ITER requirements has already been successfully implemented on all presently operating tokamaks with shaped plasma cross-sections and independently controlled PF coils: JET [49], JT-60U [50], DIII-D [44], ASDEX-Upgrade [43], Alcator C-MOD [51] and TCV [52]. While none of these experiments has a PF coil configuration with the exact coil number, position and relative coil to plasma separation of ITER, all of these experiments are able to control 'ITER like' single null diverted plasmas with separatrix elongations about 1.8. Figure 4 shows a comparison of the plasma cross-sections and PF coil geometry of ITER and ASDEX-Upgrade.

The ITER elongation ($\kappa_{95} = 1.6$, $\kappa_x \cong 1.8$) is comparable with the elongation used for routine plasma operation in ASDEX-Upgrade, C-MOD, COMPASS-D, JET and JT-60U but significantly less

than the maximum elongation obtained in DIII-D and TCV ($\kappa_x = 2.5$). The vertical control stability margin, defined by the ratio between the vertical stabilizing and destabilizing forces, $ms = Fs/Fd - 1$, is 1.7 for ITER, much larger than in DIII-D, ASDEX-Upgrade and TCV, which can operate with stability margins of a few per cent. The vertical instability growth rate in ITER (1.4 s^{-1}) is significantly lower than in current devices (over 2500 s^{-1} for COMPASS-D and TCV). The plasma triangularity in ITER ($\delta = 0.23$) is less than other tokamaks ($\delta \geq 0.8$ is routinely achieved for DIII-D and TCV). The precise control of the X-point or target strike points in a closed divertor configuration has been achieved in many tokamaks (JT-60, JET, ASDEX-Upgrade and C-MOD). Both ASDEX-Upgrade and C-MOD achieve this precise divertor control with PF coils that are relatively far removed from the X-point and the strike points. Relative control precision of the various shape and position parameters is already comparable with ITER requirements. The complexity of the ITER PF coil system (10 independently powered coils) is similar to that of existing tokamaks (18 for TCV, 16 for DIII-D and nine for JET).

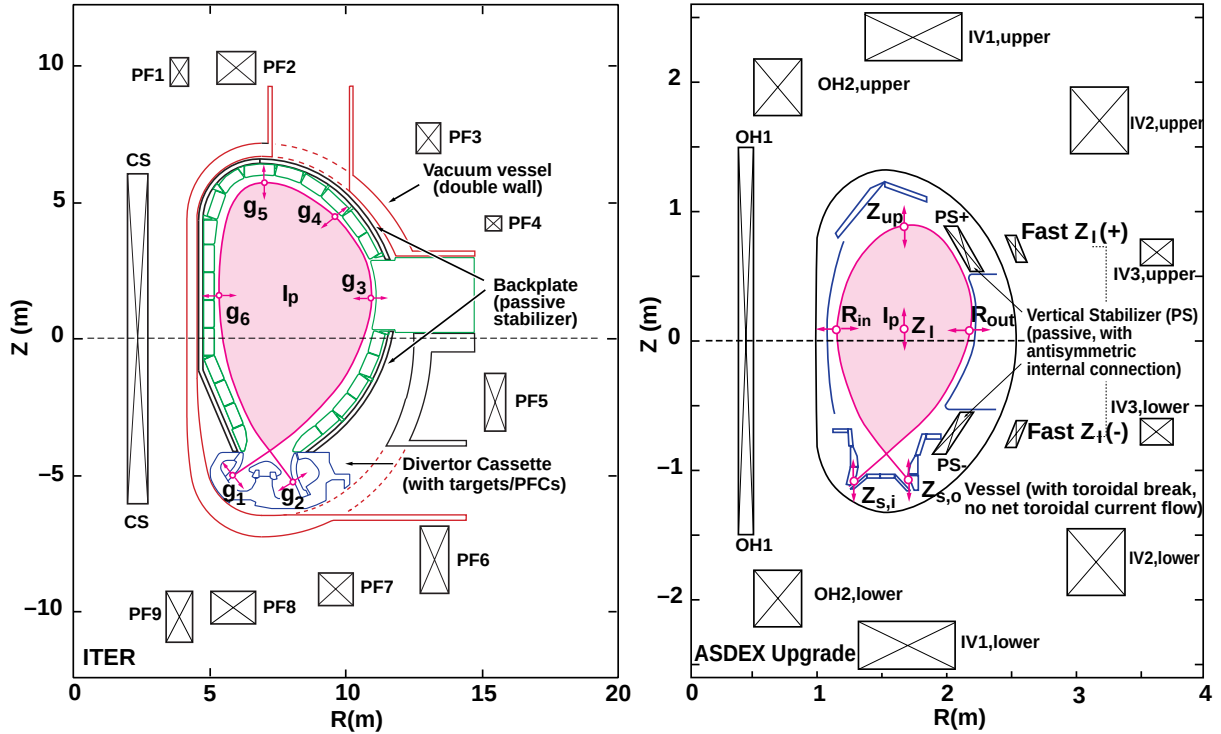
ITER magnetics control will differ from present tokamaks in two important areas: in the power required for plasma control and in the similarity of the shape and vertical position control time-scales. The local grid imposes limits on the power available for plasma control, although it could be supplemented with pulsed power from rotating machines or magnetic storage on site. The present limit on grid power for ITER control is 250 MW. Limits are also specified on the maximum rate at which the line power demand is allowed to change. The ITER control power allocation is similar to the peak control power in present experiments but the rate of change specification is more restrictive than in existing tokamaks. Restriction of the peak control power demand and rate of change of demand are, therefore, significant control design constraints.

The similarity of the time-scales for vertical position instability and more general shape control in ITER makes design of an efficient control algorithm less straightforward than in present experiments, where the 'fast' vertical control and the 'slow' equilibrium control are separated in frequency.

PID (proportional/integral/derivative) plasma magnetic controllers are adequate for present tokamaks. These controllers counteract errors in the waveforms to be tracked (controlled to follow a predetermined 'reference waveform') by applying volt-

Table 1. Plasma disturbance parameters selected for ITER control system response evaluation

Disturbance	$\Delta l_i(3)$ (decrease)	$\Delta \beta_p$	Waveform/frequency/budget
Vertical drift	0	0	0.1 m ‘control off’ drift/<1 per pulse
Minor disruption (A)	0	≤ 0.2 (decrease)	Step/0.2 Hz/<10 per pulse
Minor disruption (B)	≤ 0.1	≤ 0.2 (decrease)	Step/0.2 Hz/<10 per pulse
Sawtooth	≤ 0.05	≤ 0.05 (decrease)	Sawtooth/0.1–0.01 Hz/continuous
ELM	≤ 0.05	≤ 0.05 (decrease)	Sawtooth/2–0.2 Hz/continuous
L-mode to H-mode	~ 0.1	~ 0.2 (increase)	Ramp (5 s)/single/1 per pulse

**Figure 4.** ITER and ASDEX-Upgrade plasma and PF coil cross-sections. Plasma shape control fiducial points are indicated by the small circular symbols with bidirectional arrows. The orientation of the arrows indicates how the respective fiducial point is controlled.

ages to the PF coils that are derived from the waveform errors, their time derivatives and time integrals. For tokamaks without separately identified coil sets for particular shape parameters, physical models are used to decouple the current, shape and position, which are then controlled by individual PID feedback in JET, DIII-D, JT-60, ASDEX-Upgrade, C-MOD and TCV.

So far, a fairly primitive plasma response model is adequate to identify the decoupling controllers. The model of the dominant vertical positional instability can be quite elementary, and it is even possible

to obtain an adequate shape controller with a ‘plasmaless’ model. The dynamics of such controllers are tuned empirically, to adjust control error, dynamic response and transient stability. This empirical tuning is facilitated by the fact that shape control takes place more slowly than vertical stabilization. Ignoring the effects of passive structure currents is, therefore, possible for shape control. Fig. 5 shows the control of the position and shape parameters during the application of additional heating on ASDEX-Upgrade, illustrating the degree of control already achieved.

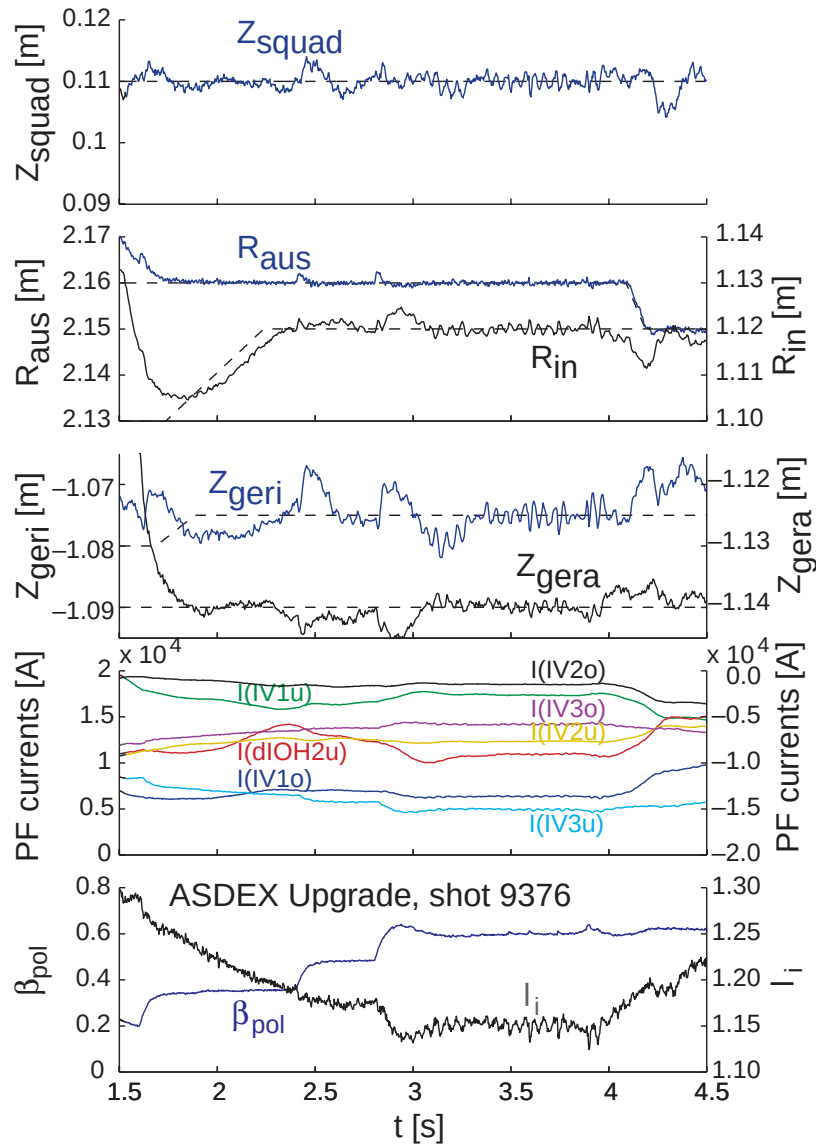


Figure 5. Plasma shape control in ASDEX-Upgrade during NBI heating with an ‘ITER like’ plasma boundary shape control algorithm (fiducial points as illustrated in Fig. 4). Dashed lines show the control programs (reference waveforms). Height of the current centroid (Z_{squad}), inner and outer major radius (R_{in} and R_{aus}) and divertor target strike point heights (Z_{geri} and Z_{gera}) are successfully controlled with transient variations $\leq \pm 5$ mm ($dR/R \leq 3 \times 10^{-3}$). Control ‘noise’ during the quasi-stationary β_p ‘flat-top’ is ~ 1 mm rms ($dR/R \sim 0.6 \times 10^{-4}$).

Magnetics control based upon this type of multi-variable PID is certainly reasonable for ITER, but may not be optimal with regard to some of the engineering constraints. Controllers of this type can produce transient peak power demands well in excess of ITER limits during simulations. These peak transient demands arise from competing interactions among the individual controllers. Modern control

optimization techniques such as linear quadratic Gaussian (LQG) optimization [53] or H-infinity (H_∞) optimization [54] techniques can design feedback controllers to provide a specified response of the multiple input/multiple output PF control system. These design methods can include constraints on additional performance requirements such as the magnet AC losses or required total power.

This optimization requires greater accuracy of the model of the system that is to be controlled. Considerable research effort has gone into deriving and comparing plasma models that are either non-linear, containing the full information of the plasma equilibrium and including different physics assumptions, or locally linearized versions of the non-linear models.

The ‘CREATE-L’ model is one linearized model system that combines a full two dimensional model of the ITER conducting structures and PF coils with a linearized two dimensional plasma equilibrium model [55]. It has been used to test candidate ITER controllers optimized in a variety of ways. This model has been subjected to systematic benchmarking tests against TCV experiments and has provided encouraging agreement for the closed loop and open loop responses, implying that the underlying physics approximations are adequate [56]. Figure 6 shows an example of the response of several plasma parameters of an SND TCV plasma when rejecting disturbances provided by square wave voltage stimulation of particular PF coils, compared with the CREATE-L model responses. It is an essential feature of these results that the CREATE-L model did not have to be adjusted.

The H_∞ procedure was used to design a controller for TCV, to test whether the CREATE-L model is adequate and whether the presence of diagnostic noise in the measurements is a problem. This controller worked exactly as designed when tested on TCV, illustrating that both points are satisfactory for a modern controller design [57].

The TSC code [48] has also shown good agreement with experimental measurements during slowly evolving quasi-stationary plasmas [58], disruptive and vertically unstable plasmas [59–62]. The TSC plasma model has become a standard against which other models can be compared. Work is underway to design a feedback controller using modern control theory and subsequently test it in a non-linear simulation of ITER, to confirm that the physics basis of the linearized model is adequate for controller design.

2.1.4. Intelligent or adaptive magnetics control

Tokamak control practice has focused on providing a sequence of predefined control procedures and algorithms for the various phases of a tokamak operation sequence. ‘Intelligent’ or adaptive magnetic control procedures have already been tested which are able to identify plasma states and state transitions, and take appropriate control actions in real

time. Model adaptive control could respect additional practical operational considerations, such as power demand, coil current and force limits.

In ITER, the plasma current, position and shape control will be embedded in a higher layer of more intelligent control. This supervisory layer will act to adjust the reference inputs to the PF controller in such a way as to keep the PF currents away from their maximum values and to adjust the current, shape and position control algorithms to optimize the control.

2.1.5. ITER controller design simulations and predicted response

The effects of controller design methodology, plasma model accuracy, PF control strategies, PF coil current and voltage limits and the effects of the passive structure configuration have been examined in an extensive set of design studies for ITER. Specific performance criteria used to define a figure of merit include variances in the divertor strike point gaps (g_1, g_2) and in the plasma to first wall gaps (g_3-g_6), in the presence of the disturbances discussed in Section 2.1.2. The undisturbed gaps g_3-g_6 allow for uncontrolled movement following a disturbance. An increase in g_3-g_6 does no immediate harm, whereas a decrease below the minimum clearance results in unacceptable plasma-wall interaction. A similar consideration applies to the divertor strike point gaps g_1 and g_2 .

A large disturbance typical of a ‘worst case’ minor disruption (see Section 4.1 of Chapter 3) is characterized during the burn phase of the discharge ($\beta_p = 0.9, l_i = 0.9$) by an instantaneous l_i drop of 0.1 together with a drop in β_p of 0.2, followed by a 5 s linear recovery of β_p . The corresponding plasma energy loss is about 0.2 GJ. Transport simulations show that thermal recovery of the pre-disruption plasma energy is only marginally possible at this energy loss level, so there is a basis for taking this minor disruption as a worst case of a recoverable disturbance. Figure 7 shows a successful simulation of a candidate H_∞ controller design controlling this disturbance, simulated using TSC. The separatrix to first wall gap responses demonstrate that the immediate reaction of the separatrix is to shrink away from the wall. The maximum change in the control points is +17 cm. The power required to bring the plasma back to the nominal separatrix location is ~ 80 MW, well within the ITER line power demand and rate of change of power limits. There is no separatrix contact with the wall.

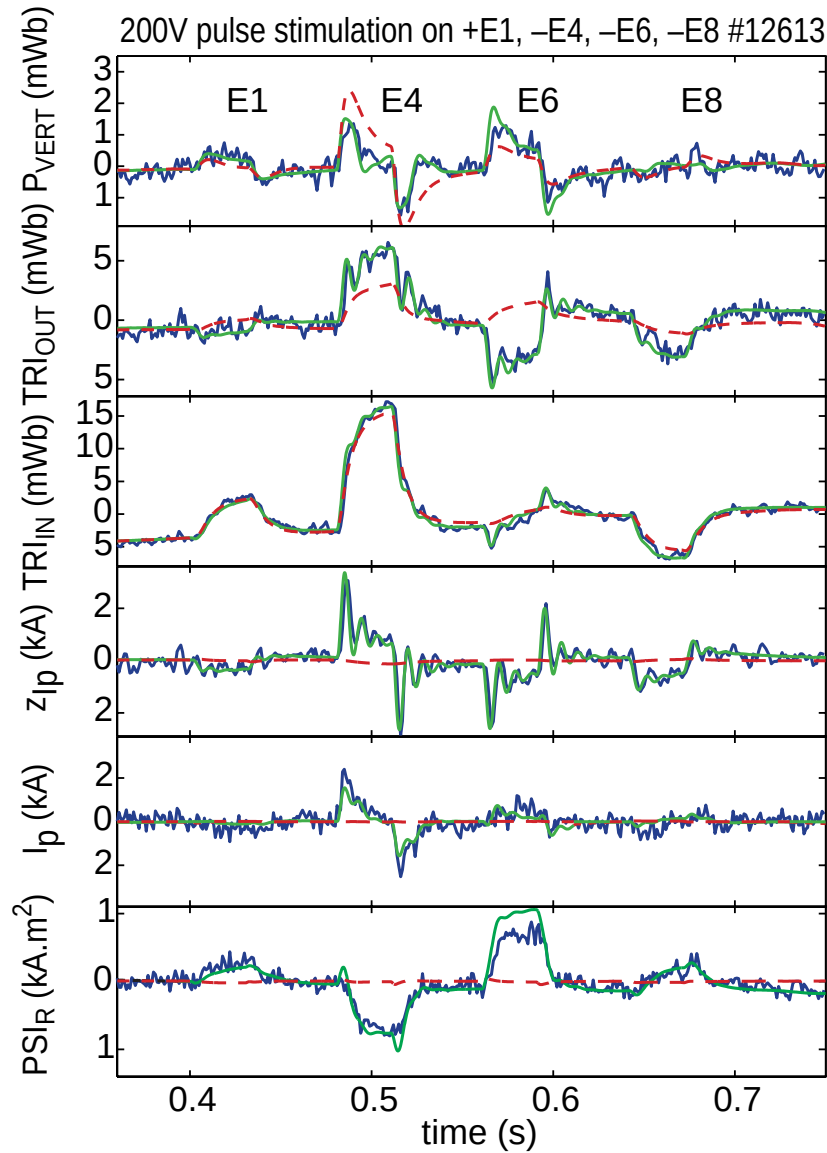


Figure 6. The closed loop evolution of TCV shape parameters during square-pulse voltage stimulation of inboard coils (dark solid and noisy line). This response shows good agreement with the nominal CREATE-L linearized model (solid grey noise free line). A model excluding the plasma response (dashed line) shows inadequate agreement, illustrating the importance of the plasma content of the PF system model.

Gap g_3 does not decrease by more than 5 cm, important for an rf antenna at this location. All nominal PF control requirements are, therefore, met in this simulation.

2.1.6. Summary and implications for ITER

The magnetic diagnostics successfully used in present tokamaks will be adequate for the magnetic control of ITER plasmas with fusion burn duration

of 1000 s or greater. Neither the longer plasma discharge duration, nor the higher plasma magnetic and kinetic energy of ITER plasmas will modify the physics basis for plasma magnetic control. However, the need to extend magnetic control of ITER plasmas to steady state would require that non-magnetic diagnostic data or true continuous magnetic measurements be added to the control system.

The physics basis for magnetic control and magnetic data interpretation in tokamaks is well under-

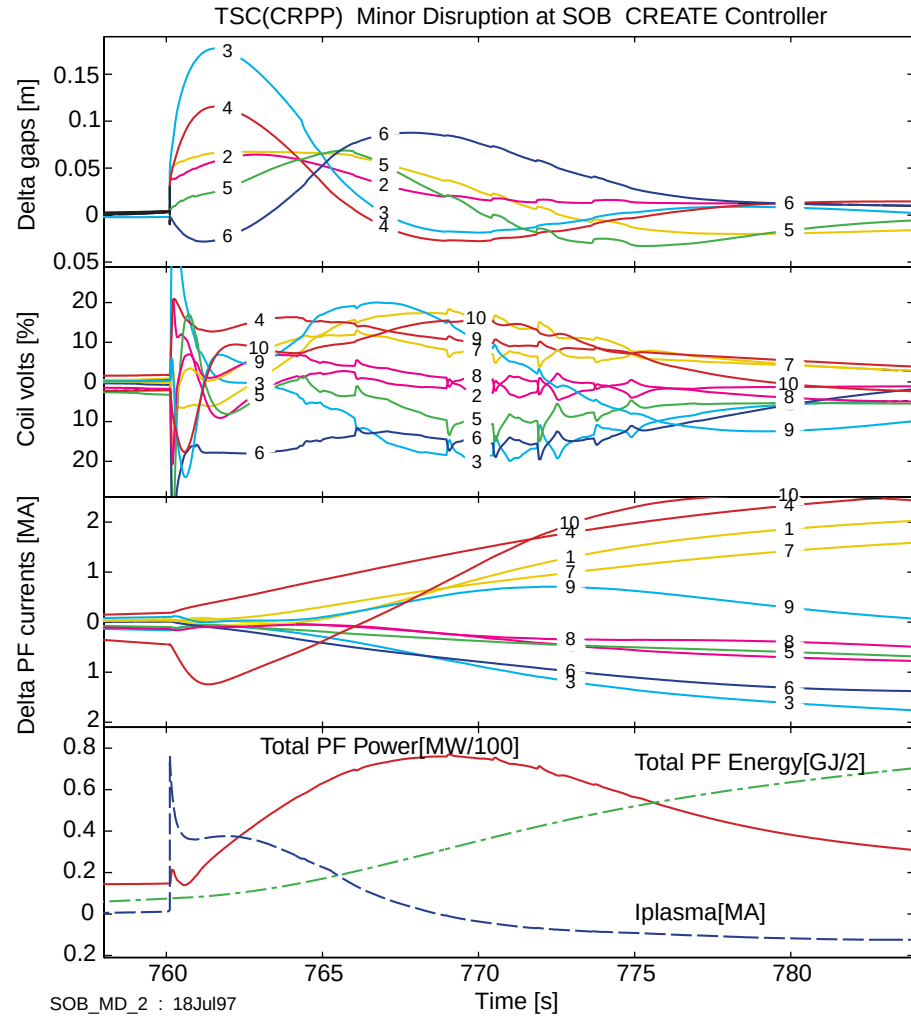


Figure 7. The evolution of the separatrix gaps (numbered 1–6; positive denotes separatrix motion away from the first wall), the PF coil voltages as a percentage of the nominal design, the PF coil current changes, the plasma current variation, the total PF control power and energy.

stood and there are no fundamental uncertainties in the extrapolation of this physics basis to ITER. A number of tokamaks have demonstrated successful plasma magnetic control with control strategies very similar to those planned for ITER. However, there are practical design considerations – power efficiency, minimization of AC losses and design of nearly optimal control responses – that argue for the application of ‘modern’ controller optimization methods. The availability of an accurate plasma response model will be a key factor for such optimization methods. Studies using present plasma equilibrium models for ITER PF control design have demonstrated their suitability for ITER design. Comparison of locally linearized models with experimental data and control system responses shows that such models

can describe the plasma and plasma control system response adequately. Tokamak operation using a high order modern controller has already been demonstrated. All of the design basis knowledge – physics, plasma response and control design methodology – needed to produce a reliable and efficient ITER plasma magnetic controller design are, therefore, available to minimize cost while meeting the control requirements.

2.2. Kinetics control and divertor control

Kinetics control comprises the establishment and sustainment of the kinetic attributes of the core and divertor regions of the tokamak discharge. At the most elementary level, the key kinetic attributes of

the core are density, temperature, impurity content, current density, and, in the case of a DT plasma, fusion power. At the same elementary level, the key attributes of the divertor region plasma are temperature, density, impurity content and ionization state, and the corresponding levels of radiated power and power conducted and convected to the material plasma facing surfaces. The physics basis considerations that apply to the core and edge/divertor plasma regions are described in Chapter 2 and Chapter 4, respectively. The subject of this Section is how these attributes are manipulated and controlled by external actuators and control algorithms during the course of a tokamak discharge. Such kinetics control now plays an increasingly important role in modern tokamaks, especially when they are operated so as to obtain maximum fusion energy gain $Q = P_{fus}/P_{in}$ [63]. In ITER, control of the plasma kinetic attributes to obtain sustained DT fusion burn at predetermined fusion power is a central and, arguably, a first priority plasma operation objective.

2.2.1. Introduction and background

Tokamak plasma control encompasses both magnetics control (described in Section 2.1) and kinetics control [41]. In most present cases of tokamak control, there is a clear distinction between magnetics control and kinetics control functions, both in terms of the actuators used (the PF coil system for magnetics control versus fuelling, heating and pumping systems for kinetics control), and in the control algorithms as well as the diagnostic sensors employed. But coordinated magnetic and kinetic control is proving to be essential in controlling current ramps to create reverse shear q profiles and their attendant internal transport barriers.

The kinetic properties of the plasma – especially the electrical conductivity, pressure and current density profile – of course ultimately affect magnetic control. Traditionally, however, these parameters are not directly manipulated for magnetic configuration control purposes. Instead, the PF control system is designed to control the plasma position, shape and poloidal flux content in a manner that endeavours to maintain acceptable magnetic control despite variation in the plasma kinetic parameters. In contrast, some degree of direct or ‘active’ kinetics control (e.g. keeping plasma density in a range that avoids disruption) is required for successful tokamak operation. In this sense, kinetics control is ultimately the *active* or dominant aspect of plasma scenario con-

trol; PF/magnetics control is a *reactive* or subordinate aspect.

Although control is usually viewed as being a mathematically linear process, optimization of fusion plasma performance requires non-linear control of kinetic properties to access plasma states with improved transport, such as the classic H-mode edge state [64] or internal transport barriers (ITBs) [65]. The H-mode edge transition requires a threshold power through the separatrix, whereas ITBs require the coordinated use of the nominal magnetics control hardware and the auxiliary heating system to ramp the plasma current, thereby controlling the current density profile. In both cases, control permits access to a transport bifurcation. ITB control serves as an example of integration of magnetic and kinetic control that will likely increase in a reactor scale device.

Recently, tokamak facilities have installed diverse and separately controllable H/D/T and impurity fuelling systems, controllable divertor pumping and neutral particle exhaust systems, between pulse and in pulse wall conditioning, as well as heating and current drive systems with controllable radial and temporal localization capabilities and with sufficient power to locally modify the inherent plasma current and temperature profiles. The availability of these increasingly capable and agile kinetics control actuators, when combined with the corresponding availability of key plasma kinetic diagnostics, has led to an increasing ability to effect more direct control of the kinetic attributes of the core and divertor plasma. This control has led to improved plasma performance as well as sustainment of near optimal performance [66]. The same control ability can also provide a basis to avoid MARFES [67] and to limit the amplitude of deleterious MHD activity to acceptable levels [68]. In this sense, kinetics control has begun to be proactive rather than just reactive. Specific examples of this type of progression in sophistication in present tokamak control practice are presented below in Section 2.2.3.

2.2.2. Kinetics control requirements for ITER

Put in its simplest terms, kinetics control must stably regulate three powers: fusion power output, power flowing across the separatrix, and power flowing to the divertor plasma facing components. There is also a key, presently empirical, constraint that the density not exceed 1.5 times the Greenwald density of $n_{20} = I_{MA}/\pi a^2$. The empirical factor of 1.5 is provisional and will hopefully be replaced by a physics

based constraint. There are also technological constraints regarding auxiliary heating power, allowable PF voltages and currents, etc.

The basic concept for fusion power output control is well known [69]. Simulation studies based on an ELMy H-mode energy confinement time show that, when a certain temperature is exceeded, fusion power will stably balance transport and radiation losses. Moreover, the fusion power so generated will increase strongly with density, thereby providing control of fusion power output. In these studies, each ion species had a diffusivity $\Delta = \chi_e = 0.5\chi_i$ and zero pinch velocity as recommended in Section 9 of Chapter 2 and as tabulated in [70]. Section 3.2 of Chapter 9 derives an analytic power–density relation for ignited ELMy H-mode discharges. Appendix A of Chapter 1 provides additional details. Control of the central plasma density by the DT fuelling rate is the most direct ‘actuator’ for fusion power control, and simple zero dimensional particle balance and fusion burnup models and also similar 1/1.5 dimensional transport simulation models that take profile and particle transport effects into account readily demonstrate that, with the diffusivities assumed above, fuelling rate control is always sufficient to effect long term fusion power control in ignited or driven burn plasmas where the operating point has positive to neutral thermal stability [69].

Figure 8 presents a POPCON (plasma operation contour) diagram for ITER under ‘standard’ ITER ELMy H-mode confinement modelling assumptions. This figure summarizes thermally stable high temperature sustained ignition operation points with 1.5 GW fusion power which exist at a volume average density $\langle n_e \rangle$ of approximately $1 \times 10^{20} \text{ m}^{-3}$ and a volume average temperature $\langle T \rangle$ of approximately 12 keV. This operational area is also consistent with maintaining plasma pressure below the nominal ITER design basis beta limit ($\beta_N = 2.5$) and being able to sustain H-mode (i.e. $P_{sep} > P_{L-H}$ in Fig. 8, where P_{sep} is the power crossing the separatrix and P_{L-H} is the threshold separatrix crossing power for an L-mode to H-mode transition, discussed in Chapter 2). As the figure shows, there is a modest sized domain involving $\sim 10\%$ variations in $\{n, T, P_{fus}\}$ space where sustained ignition is possible at $1 \leq P_{fus} \leq 1.5 \text{ GW}$ without encountering either the beta limit or loss of H-mode limit. Nonetheless, there are some experimental observations of weak central density response [71] to fuelling to be resolved before this control algorithm can be considered as fully established.

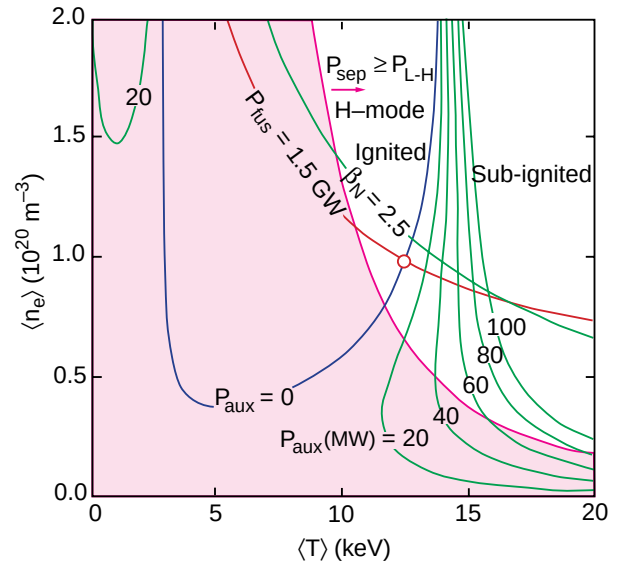


Figure 8. Representative plasma operation contour diagram for the ITER (21 MA) FDR design. Standard ELMy H-mode energy confinement applies and equilibrium thermal He levels are calculated self-consistently everywhere in the domain. However, the ignition ($P_{aux} = 0$) and finite- Q ($P_{aux} > 0$) contours are meaningful only in the H-mode domain ($P_{sep} \geq P_{L-H}$), which is the unshaded region in the figure.

With finite auxiliary power, thermally stable sub-ignited driven burn in this high temperature regime is also possible. Plasma temperatures for driven burn lie in the 13–17 keV range. In this regime, control of auxiliary heating power can also be used to supplement fuelling/density control. However, fuelling/density control is still the primary burn control actuator.

Maintenance of the power flow at the plasma edge above the level necessary for sustainment of H-mode is also required (see Chapter 2). The requirement for some minimum power at or near the separatrix to sustain H-mode (estimated to be $\sim 100 \text{ MW}$ for the ITER/FDR design) combined with a limit on maximum power conducted to the divertor targets (the ITER/FDR specification is $\leq 50 \text{ MW}$, although the targets can handle up to 100 MW on a routine sustained basis) in turn implies that significant ($\sim 50 \text{ MW}$) radiative power dispersal between the plasma edge/separatrix and the divertor targets is required.

In most present tokamaks and plasma operation regimes, explicit control of the plasma edge and divertor characteristics (target power and degree of plasma to target attachment) is optional, owing to the ability of inertially cooled divertor targets to

adequately handle the moderate power loadings and pulse durations typically encountered. For a reactor scale device, where divertor target power loadings (proportional to P/R , see Chapter 4) are inherently higher and where continuous (actively cooled) heat removal is required, it will be necessary to separately control the plasma edge and divertor plasma characteristics so as to keep divertor power and transient energy (ELM) loadings within engineering limits [72]. At the plasma edge, the power flow across the separatrix, edge density and ELM type and frequency are key attributes where control is needed or will be desirable. In the divertor, the target heat flux must be limited to acceptable values and the neutral gas pressure and exhaust will have to be adjusted to maintain constant fuel, helium and impurity densities in both the plasma core and the edge/divertor region. Impurity seeding both in the main chamber and in the divertor is envisioned to be the main tool to control divertor power fluxes. Modelling of the plasma edge and divertor region radiation and power conduction/convection characteristics indicate that the required edge to target radiative power dispersal can, in fact, be obtained (see Chapter 4), and that the resulting contamination/dilution of the core plasma by the added impurities is small enough that ignition or driven burn with adequate fusion power can still be obtained (see Chapter 2).

Representative specifications for ITER kinetics control are summarized in Table 2

The use of three independent ‘actuators’ – DT fuelling rate, impurity injection rate and P_{aux} – to independently control three key attributes – fusion power, edge conducted power, and divertor target power – leads naturally to a ‘modular’ kinetics control scheme like that shown in Fig. 9, wherein three PID (proportional/integral/derivative) feedback control loops with additional logic control (‘modules’) serve to effect control of the respective actuators [73, 74]. The DT module controls either plasma density or fusion power. Density control overrides power control if the requested density exceeds an allowable maximum (limit) density. The Ar module limits the divertor target power to less than a specified allowable power. The P_{aux} module controls the amount of auxiliary heating, in order of decreasing priority, (i) to maintain the plasma in H-mode, (ii) to maintain the plasma density below a maximum value, and (iii) to assist in the control of the fusion power.

Whether it will be possible to meet all these requirements simultaneously and reliably in ITER

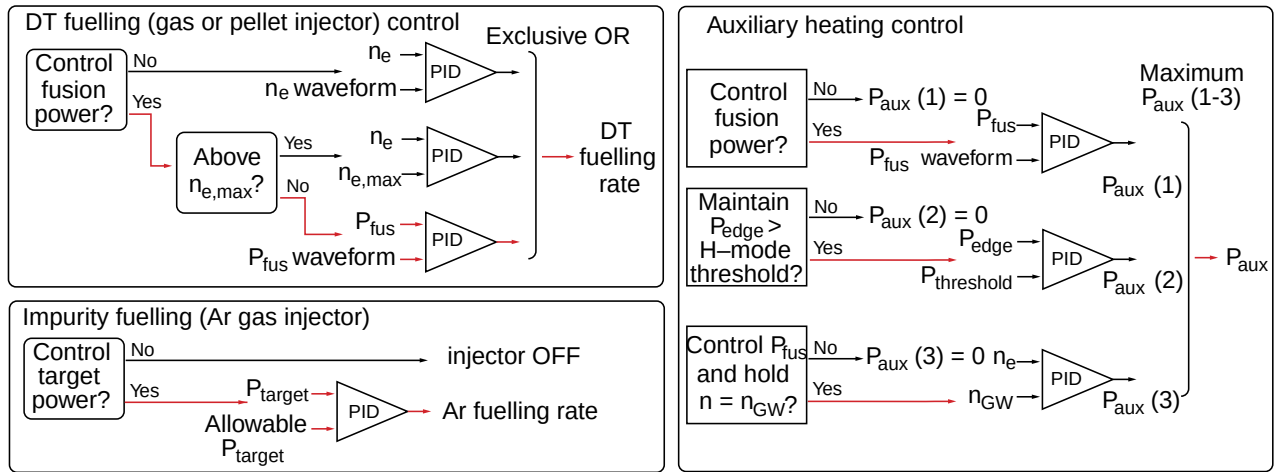
raises a number of physics and control implementation issues. The physics considerations of plasma energy and particle confinement, H-mode maintenance, DT and impurity injection, transport and exhaust, auxiliary heating and radiative dispersal of energy in the plasma boundary and divertor are reasonably well defined by this Article, and so it is possible to evaluate the efficacy of the control concept embodied in Fig. 9 with plasma simulations [74]. These simulations (an example is presented in Section 2.2.4) show that the control system shown in Fig. 9 is capable of providing effective burn and divertor power control for ELMy H-mode plasmas with operational conditions set by the ‘reference ITER physics basis’ summarized in Chapters 2–6 of this Article. These simulations also show that burn and divertor control is reasonably robust with regard to tolerance of credible plasma disturbances, modest variations in modelling assumptions, and variations in the feedback loop gain parameters. Furthermore, as will be documented in Section 2.2.3 below, all of the basic control techniques embodied in Fig. 9 (actuator utilization, feedback implementation, and demonstration of ITER equivalent closed loop control) have taken, or are taking, place in present tokamaks. So there is a reasonable basis for expecting that the type of control approach embodied in Fig. 9 will be successful in ITER.

Despite this positive outlook about ITER kinetics control, there is still a clear need for better and more precise data on the fundamental considerations that underlie Fig. 9, and there is also a need for continued exploration of kinetic control methods and optimization methodology in present tokamaks. Within the kinetic control ‘matrix’, there are both strong and weak elements involving the coupling of the core, edge and divertor kinetics to the respective actuators. While considerable success in sorting out these couplings and in designing control systems by heuristic or empirical methods (see Section 2.2.3 below) has been achieved to date in present experiments, for future reactor and ITER designs in which certain coupling factors are stronger or more critical, the application of modern control design methodologies (H-infinity or LQG optimization) could well be beneficial. These types of control design methods require a relatively accurate plasma response model plus a comprehensive model of plasma control related features of the plant/facility systems. Next, we turn to the status and growing sophistication of kinetics control in present tokamaks.

Table 2. ITER plasma kinetics control specifications¹

Plasma kinetic attribute	Nominal (for ignited or high- Q burn)	Control accuracy (%) (variance during normal operation)	Maximum or minimum
Density (10^{20} m^{-3})	0.7–1.0	$\pm 10\%$	≤ 1.3 ($\cong 1.5n_{GW}$)
Fusion power (GW)	1.0–1.5	$\pm 10\%$	≤ 1.8 (for 10 s)
Edge/separatrix power (MW)	≥ 120 ($= 1.2P_{L-H}$)	$\pm 20\%$ (?)	≥ 120 (?)
Divertor power (to target) (MW)	≤ 50	$\pm 20\%$ (?)	≤ 100 (steady state) ≤ 200 (for ~ 3 s)
Auxiliary power (MW)	0 (ignited); 0–100 (high- Q driven burn); e.g. 100 MW at $Q = 15$	0–100 MW as required (dynamic P_{fus} control in ignited plasmas or sup- plemental P_{fus} control in driven burn plasmas)	≤ 100

¹Representative parameters for $B = 5.7$ T, $I = 21$ MA, ELMy H-mode energy confinement, $\sim 9\%$ thermal He, 0.2% Ar, ‘most likely’ L–H power threshold ($\alpha = 0$, see Section 4.3 of Chapter 2).

**Figure 9.** Schematic concept for the ITER plasma kinetics control system (fusion and divertor target power control).

2.2.3. Present experience and accomplishments

All present major tokamak experiments employ some degree of kinetics control – typically control of gas fuelling input to effect density control – and a number of experiments have implemented more elaborate methods for kinetic control that play a key role in achieving specialized or optimal plasma performance. As Table 3 shows, these kinetic control features tend to be specific to each tokamak and reflect, in part, the plasma diagnostics and plasma control actuators available and the associated objectives of the experimental program. The common theme of this type of kinetics control is improvement of experimental reproducibil-

ity and sustained plasma performance. Quantities that will be relevant to ITER and have already been controlled in experiments are listed in Table 3. The intent of the Table is to provide representative examples: citations of specific experiments and references on control methods are not intended to be exhaustive.

The last section of Table 3 summarizes recent, significant experimental advances in simultaneous core and divertor control in ‘ITER like’ plasmas and in the implementation of autonomous control to replace human shot-to-shot direction of the plasma operation scenario. The concept here is to combine a plasma operation regime (or plasma state) recog-

Table 3. Plasma kinetics control experience in present tokamaks

Control category or controlled plasma attribute	Control method and/or procedure: sensor/parameter $\rightarrow$ actuator	Example/reference
Core kinetics		
Electron density (n_e or $\bar{n}_e (= \int n_e dl/l)$)	Interferometry or bremsstrahlung $\rightarrow$ fuel gas (H, D or T) injection valve	All major experiments
Electron density (n_e or $\bar{n}_e (= \int n_e dl/l)$)	Bremsstrahlung $\rightarrow$ bang-bang controller $\rightarrow$ pellet centrifuge	ASDEX-Upgrade [75]
DD fusion power (via DD neutron yield)	DD rate (neutron counter) $\rightarrow$ number of NBI sources	JT-60U [76]
W_{th} or $\beta_N (= aB\beta/I)$	W_{mhd} (diamag.) NBI duty cycle β_N (diamag.) $\rightarrow$ number NBI sources	DIII-D [77] TFTR [78]
Core radiation fraction	$P_{rad}(\text{core})/P_{heat} \rightarrow$ impurity gas injection	ASDEX-Upgrade [79]
Core radiation fraction	$P_{rad}(\text{core})/P_{heat} \rightarrow$ impurity gas injection	TEXTOR [80]
Central mass and electron density	Frequency of Alfvén eigenmode $\rightarrow$ gas injection of H and D	TCA [81]
Current profile $j(r)$ or moments	MSE or polarimetry $\rightarrow$ ECH (planned)	DIII-D [82]
Current profile $j(r)$ or moments	$l_i \rightarrow$ phasing of LH launcher modules (changes $n_{ }$ and current drive profile)	Tore-Supra [83]
Axial safety factor (q_0) or $j(r)$	MSE $\rightarrow P_{aux}$ (heat, current drive) (planned)	DIII-D [82]
neoclassical modes ($q = 2$)	SXR, ECE or Mirnov island detection $\rightarrow$ ECCD	ASDEX-Upgrade [68]
Edge kinetics and divertor		
n_e (edge)	n_0 (midplane or div.) $\rightarrow$ gas injection (div.)	DIII-D [82]
Neutral pressure (n_0)	n_0 (div.) $\rightarrow$ gas injection (div.)	ASDEX-Upgrade [79], JT-60U [84]
Neutral pressure (n_0)	Langmuir probe div. $\rightarrow$ gas injection (div)	JET [85]
Edge radiation	Bolometer (core) $\rightarrow$ gas injection (midplane)	ASDEX-Upgrade [79]
In-divertor radiation	Bolometer (div.) $\rightarrow$ gas injection (div.)	JT-60U [86]
In-divertor radiation	Bolometer (core) $\rightarrow$ gas injection (div.)	DIII-D [82]
Wall and limiter temperature	IR camera set $\rightarrow$ plasma control system, also protection interlock	Tore-Supra (planned) [87]
MARFE level	MARFE detection (localized bremsstr.) $\rightarrow$ NBI duty cycle	ASDEX-Upgrade
Fast plasma shutdown		
W_{th} and/or I_p shutdown	Trigger $\rightarrow$ massive He injection	JT-60U, DIII-D
W_{th} and I_p shutdown without VDE	Trigger $\rightarrow$ massive He injection	JT-60U [88]
W_{th} and/or I_p shutdown (typically with VDE in elongated plasmas, but with reduced halo current)	Trigger $\rightarrow$ impurity pellet (Ne, Ar, D ₂ + Kr, D ₂ + Au or Ag)	JT-60U [89,90], ASDEX-Upgrade, DIII-D, TFTR, Alcator C-Mod [91]

Table 3. (Cont.)

Control category or controlled plasma attribute	Control method and/or procedure: sensor/parameter $\rightarrow$ actuator	Example/reference
Simultaneous core and divertor		
Simultaneous control	core radiation fraction + divertor neutral pressure	ASDEX-Upgrade [79]
Simultaneous control	$\langle n_e \rangle +$ DD fusion power + $P_{rad,div}$	JT-60U [92]
Simultaneous control	$n_{edge} +$ in-divertor radiation	DIII-D [82]
Regime detection and scenario sequencing		
Plasma regime detection	P_{aux} , P_{rad} , P_{OH} , bolometer, β , l_i , $\rightarrow$ regime (plasma state)	ASDEX-Upgrade [93]
Plasma control algorithms	Plasma regime identification	JT-60U [94], ASDEX-Upgrade [95]
Plasma control algorithms	Tokamak + plasma status, scenario phase $\rightarrow$ scenario phase	ASDEX-Upgrade [96, 97]
Plasma control algorithms	Scenario phase, event $\rightarrow$ scenario phase	DIII-D [98]
Control method selection	Plasma regime $\rightarrow$ scenario phase and control mode	ASDEX-Upgrade [99]

dition capability with a control algorithm selection capability so as to make the algorithm selection state dependent. This gives the control system an autonomous capability to react to state transitions in the operation scenario sequence and to take corrective action if a state transition fails to take place as desired

ASDEX-Upgrade offers an example of successfully implementing autonomous control of ITER relevant high radiation H-mode (HRH) discharges [93, 96, 99]. In these discharges, precisely controlled edge/divertor radiation (effected by Ne injection) is used to produce an H-mode discharge with a detached divertor plasma. To achieve this HRH regime, simultaneous control of the neutral gas flux in the divertor by hydrogen puffing and of the radiative fraction P_{rad}/P_h in the main chamber by Ne seeding is used. Precise dynamic control of the Ne flux is necessary: excessive Ne or edge radiation results in a high radiation L-mode (HRL) rather than HRH discharge. Figure 10 shows the operation of the ASDEX-Upgrade control system, in which the various plasma states that lead up to initial achievement are identified by the control system and in which a reversion from the HRH mode to HRL mode is autonomously detected and corrected by control of the Ne injection rate. A plasma regime recognition algorithm and a connected scenario sequenc-

ing procedure is used, and the pre-divertor radiated power fraction is calculated from the bolometer signals depending on the discharge state. Measurements of the plasma internal inductance also enter into the regime recognition algorithm.

The kinetics control system can also be made responsible for maintaining the plasma operation in a safe ‘operational limits’ region, particularly in view of disruptions at high β_N , close to $q_{95} = 2$, or near the density limit. To accomplish this type of disruption avoidance, the control system needs a model of the dangerous regions and a procedure for how to ‘back off’ from them without causing a disruption or thermal collapse. An example of this type of multi-parameter operational domain DIII-D is during high- β , high density, high elongation and low- q operation. The zone with maximum performance is in an acute region of operational space and backing away from any single limit is likely to exacerbate another limit. In DIII-D, automation is already implemented for a single parameter, namely the approach to the high- β limit, and similar auxiliary heating control methods for beta limit disruption avoidance are now being used in all major tokamaks.

In a DIII-D disruption avoidance study [100], a parametrization of the β limit was obtained from a neural network description of the disruption bound-

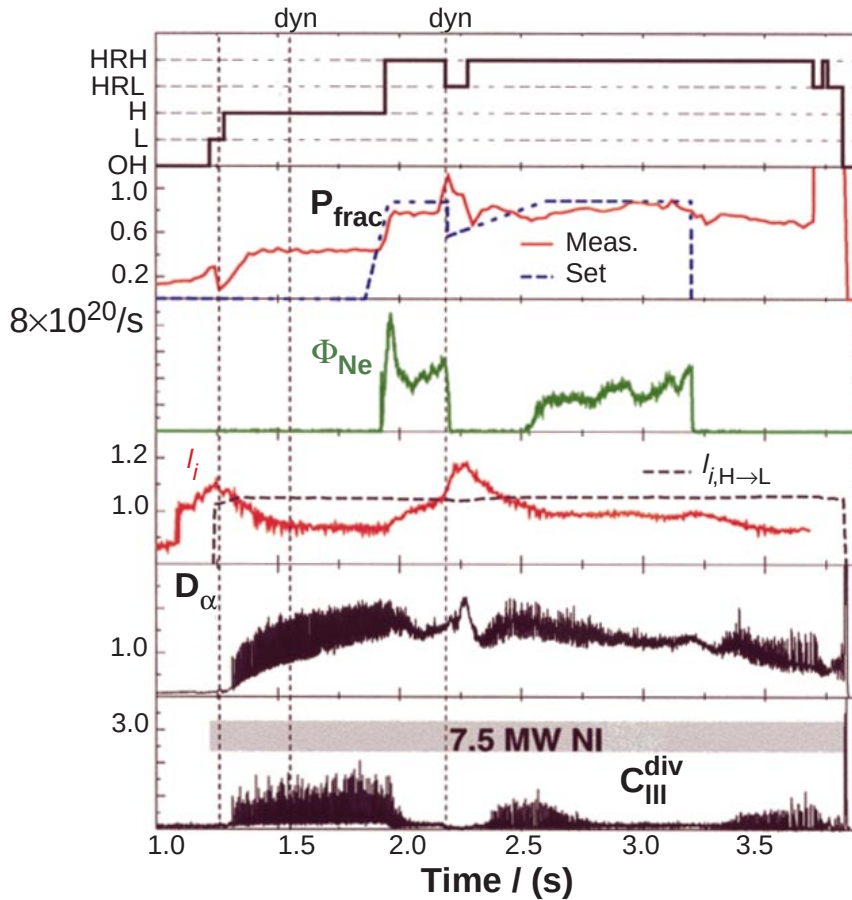


Figure 10. Supervisor autonomy and plasma regime driven kinetics control during a ASDEX-Upgrade discharge (Ref. [97]).

ary based on experimental data (see Section 4.6 of Chapter 3 for details). This neural net estimate of the beta limit was subsequently used as the basis for disruption avoidance and non-disruptive operation close to the limit was obtained. The study showed that neural network description of the β limit was appreciably more accurate than the modified Troyon limit $\beta_N = 4I_i$ (see Sections 2.1 and 4.6 of Chapter 3) in predicting onset of disruption. This finding demonstrates the advantage of using the neural net to provide *a posteriori* control or a disruption proximity estimator. A key issue for future work is how to scale the estimator to a reactor scale device, for which a set of training disruptions for the neural net will not be available.

The content of Table 3 and the brief descriptions of the regime recognition based control and disruption avoidance control presented above show that kinetic control is under active investigation. Simultaneous control of several variables and plasma state dependent switching between control algorithms

have been demonstrated: both of these demonstrations are useful for ITER. The modelling and simulation aspects of extrapolating present control experience to ITER are discussed below.

2.2.4. ITER kinetics control simulations

The feasibility of ITER kinetics control has been explored with the PRETOR 1.5 dimensional transport code with a coupled divertor model and self-consistent modelling of noble gas impurity (Ar) injection divertor power control [74]. Since the coupled plasma/divertor model is dynamic and not inherently stable, it is necessary to incorporate elementary PID controllers for DT fuelling and Ar fuelling into the overall simulation so as to be able to obtain a stable modelling result. These controllers, which are implemented in the configuration illustrated in Fig. 9 are, in effect, a representation (based on idealized plasma data and actuator response) of ITER control.

Figure 11 illustrates how the ITER kinetics control system illustrated in Fig. 9 can maintain control of fusion and divertor power during a severe plasma power balance transient in which 100 MW of auxiliary heating power was abruptly added to a plasma in steady, ignited burn. The simulation demonstrates that fusion and divertor target power control during power ramp up and ramp down are also maintained. The high frequency power transients are likely modelling artifacts, and in any case are within the transient capabilities of the divertor targets and first wall components. Refinements of this type of simulation, in which control of the Ar injection rate with a divertor temperature infrared emission monitoring system is explicitly simulated (including the non-linear emission versus temperature characteristic and instrumental noise), show that the overall divertor power control system is stable and has adequate transient response and noise level characteristics [101].

Control of the kinetic properties of ITER plasmas is of course contingent on having a corresponding set of plasma diagnostic data with adequate temporal and spatial resolution. The planned ITER diagnostics set for DT operation presented in Table 2 of Chapter 7 and in [102] is arguably equal to, or more complete than, the diagnostic sets available on the present generation of well instrumented medium and large size tokamaks. Such comprehensive diagnostics data will be necessary for implementation of proactive ‘before incident’ protection of the ITER plasma facing components that are at risk during plasma operation. In this sense, the distinction made in Table 2 of Chapter 7 between the utilization of plasma diagnostic data for plasma control, machine protection, and the acquisition of scientific data will, like the partitioning of the plasma control and machine protection functions among the ITER control system units, become increasingly blurred as the ITER plasma control system becomes fully defined.

2.2.5. Summary and implications for ITER

Plasma kinetics control in the present generation of auxiliary heated tokamaks has reached a state of sophistication in which all of the basic kinetics control requirements projected to be needed for ITER – DT fuelling rate control, impurity injection divertor target power and attachment control, auxiliary power control and kinetics control based disruption avoidance – have been successfully demonstrated. Burn control of high- Q discharges by aux-

iliary power is straightforward. Extrapolation from density control in present experiments to burn control in a fully ignited, DT burning tokamak rests on the physics mechanisms governing core density confinement, which needs an increased emphasis in confinement research. More generally, the methods used in present tokamaks for kinetics control of present H-mode plasmas are applicable and adequate for similarly robust control of ‘high temperature/low density’ ignited and driven burn ITER DT plasmas. Simulations of ITER kinetics control demonstrate that simultaneous control of fusion power and limit of divertor power to acceptable levels will be possible. The incorporation of regime identification and control reaction capabilities – already demonstrated in several present tokamaks – into the ITER plasma kinetics control system is anticipated to make ITER plasma control more reliable and robust. A comprehensive suite of plasma diagnostics will be available to support such regime and event guided kinetics control. Finally, the use of modern control system design methodologies combined with the planned availability of a detailed ITER plasma and facility operation simulator can be expected to further enhance the present expectation that plasma kinetics control in ITER will meet design objectives and fulfil the goals of the ITER/EDA Agreement.

2.3. Plasma operation scenario and related considerations

Here, specialized aspects of the plasma/PF system operation scenario and plasma control are addressed. These aspects occur during the plasma initiation, current ramp up and current shutdown phases. Plasma operation and magnetic and kinetic control during current and/or heating/burn flat-top phase are treated separately in Sections 2.1 and 2.2. Options for fast plasma energy/current shutdown – potentially needed at any time during the scenario – are also summarized herein, with further reference to Section 4.5 of Chapter 3 for details. The Section concludes with a presentation of scenario simulations for ITER and with a discussion of still open physics and plasma operation scenario integration issues.

2.3.1. Plasma initiation

ITER will, like all presently operating tokamaks, depend on ohmic (or inductive) plasma initiation effected using voltages applied to the PF coil system to produce a Townsend avalanche breakdown in a low pressure (10^{-5} Torr $\cong 10^{-3}$ Pa) filling gas,

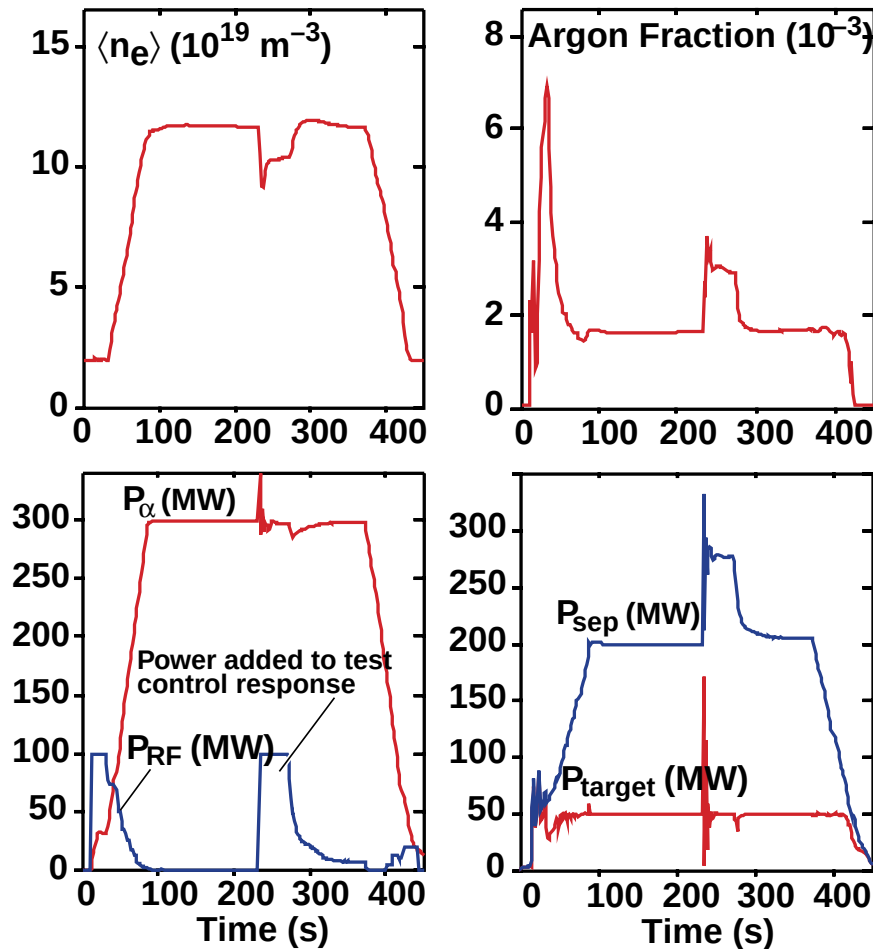


Figure 11. Simulation of burn and divertor power control during a ‘reference case’ ITER plasma operation scenario pulse (fusion burn duration intentionally limited to 400 s) that includes a 100 MW auxiliary power input transient. The kinetics control system shown in Fig. 9 successfully controls plasma and divertor power during both burn initiation and termination and also during burn power flat-top, despite the addition of 100 MW of auxiliary power.

either deuterium (or hydrogen or tritium) or helium. In ITER, the parameters for ohmic initiation are constrained by two reactor tokamak design related considerations: a somewhat limited one turn voltage and in-vessel electric field capability, about 15 V per turn and $0.3 \text{ V} \cdot \text{m}^{-1}$, imposed by terminal voltage limitations on the multi-turn superconducting PF coil system; and by the substantial in-vessel poloidal fields (error fields) that arise from induced and resistive toroidal currents that develop in the toroidally conducting ($R \sim 5 \mu\Omega$) nuclear shield module support backplate and torus vacuum vessel [103]. High error fields compromise the effectiveness of Townsend avalanche breakdown, and so must be reduced to low enough levels ($\sim 10^{-3} \text{ T}$) such that Townsend breakdown of the filling gas can be obtained. Analysis

of the Townsend avalanche and error field reduction capabilities of ITER shows that ohmic plasma initiation should be possible [104], albeit with relatively small margins and the need for careful error field and filling pressure control. Since these calculations demonstrate that avalanche margins are small and also that ohmic heating alone may fail to provide sufficient power for full ionization of low- Z impurities, present ITER plans are to provide 3 MW of EC assist during the breakdown and burnthrough phases of plasma initiation [103]. If necessary, this assist power can be increased to up to 10 MW.

Elementary analysis of the neutral gas breakdown characteristics (Townsend avalanche) shows that reliable plasma initiation in hydrogen or deuterium will be possible in ITER at this electric field.

In fact, Townsend breakdown at lower E fields and much lower one turn voltages has been demonstrated in smaller tokamaks (e.g. $0.25 \text{ V} \cdot \text{m}^{-1}$ and 3 V per turn in DIII-D [105]). However, the resulting initial buildup of ionization under these low E field conditions is relatively slow (30 ms), and careful adjustment of the filling gas pressure and field null geometry are required. The addition of modest amounts of electron cyclotron power with a resonance positioned near the centre of breakdown location (multi-pole field null) results in faster ($<3 \text{ ms}$) and more reliable breakdown with greatly increased tolerance to field null quality [105]. This latter advantage is significant, since it allows the breakdown field configuration to be adjusted to favour optimal radial equilibrium of the initial plasma (which requires a weak vertical field) rather than the high order multi-pole null that is optimal for initial breakdown. Indeed, the transformation from a breakdown plasma with negligible current to a very low current tokamak configuration has not been the topic of detailed scientific study. Nonetheless, this transition is successfully carried out by all tokamaks. The procedures set forth in [106] are representative and apply to ITER.

Reduction of the breakdown loop voltage using ECH pre-ionization has been investigated in many tokamaks (FT-1, WT-1, ISX-B, Tokapole II, CLEO, WT-2, JFT-2 and DIII-D, see [105] and references therein). Reliable breakdown at a minimum electric field of $0.15 \text{ V} \cdot \text{m}^{-1}$ (1.6 V per turn) was achieved in DIII-D with 700 kW of 60 GHz EC assist power. Alternatively, plasma current start-up without a primary voltage was demonstrated in PLT [107] by injecting lower hybrid (LH) waves. In these experiments, pre-ionization by RF heating played an essential role in reducing the loop voltage, with hydrogen gas used as the prefill gas. In JT-60U, stable plasma current start-up with $0.08 \text{ V} \cdot \text{m}^{-1}$ was obtained with helium prefill gas and LHRF heating ($1.7\text{--}2.4 \text{ GHz}$, 1.5 MW , see [108]) with no pre-ionization. Helium gas was also used in TEXTOR-94 for low voltage start-up assisted by ICRF ($350\text{--}500 \text{ kW}$, 32 MHz , second harmonic frequency regime, see [109]). As shown below, the use of helium rather than hydrogen or deuterium facilitates initiation at low voltage.

There are three principal considerations for achieving successful plasma start-up: (i) Townsend avalanche characteristics; (ii) magnetic connection length; and (iii) impurity ionization radiation ‘burn-through’. Considerations (i) and (ii) determine whether initial breakdown occurs; consideration (iii) determines whether the initial ionization produced

plasma current can be sustained and increased by application of a finite plasma loop voltage. All of these considerations are well known and we find that there are no new physics issues related to achieving plasma initiation in an ITER class tokamak. However, careful design of the start-up strategy based upon the three considerations above is required. A fourth consideration – that of producing runaway electrons during low- E field/low density start-up – has also been identified [105]. However, experimental results to date show little indication that low- E field/low density start-up produces significant runaways [108].

In what follows, the four physics design considerations for plasma initiation are further developed and the resulting implications for ITER are evaluated.

(1) Electric field and prefill gas pressure

According to Townsend avalanche theory, the ionization growth rate α (ionizations per unit length) is $\alpha = Ap \exp(-Bp/E)$, where p is the neutral gas pressure (traditionally measured in Torr; $1 \text{ Torr} = 1/760 \text{ atm} \cong 131 \text{ Pa}$), E is the electric field, and A and B are coefficients that depend on the gas species: $A = 510 \text{ m}^{-1} \cdot \text{Torr}^{-1}$, $B = 1.25 \times 10^4 \text{ V} \cdot \text{m}^{-1} \cdot \text{Torr}^{-1}$ for hydrogen or deuterium (and, presumably, for tritium); $A = 300 \text{ m}^{-1} \cdot \text{Torr}^{-1}$, $B = 3.4 \times 10^4 \text{ V} \cdot \text{m}^{-1} \cdot \text{Torr}^{-1}$ for helium. Setting $\alpha^{-1} = L$, where L is an effective connection length (average length of a magnetic field line between intersections with material surfaces; what constitutes ‘effective’ is addressed below), gives a minimum electric field $E_{\min}(p, L)$ for breakdown

$$E_{\min} (\text{V} \cdot \text{m}^{-1}) = \frac{1.25 \times 10^4 p (\text{Torr})}{\ln[510p (\text{Torr}) L (\text{m})]} \quad (1a)$$

for hydrogen, deuterium or tritium, and

$$E_{\min} (\text{V} \cdot \text{m}^{-1}) = \frac{3.4 \times 10^4 p (\text{Torr})}{\ln[300p (\text{Torr}) L (\text{m})]} \quad (1b)$$

for helium. Figure 12 shows plots of Eqs (1a) and (1b) versus p for effective connection lengths $200 \leq L (\text{m}) \leq 2000$. This range is representative of connection lengths achieved in present medium to large size tokamaks and projected for ITER (see below). The minimum value of $E_{\min}$, i.e. the lowest field at which breakdown is possible at a given L , typically occurs for $5 \times 10^{-6} \leq p (\text{Torr}) \leq 5 \times 10^{-5}$. For deuterium, and $200 \leq L (\text{m}) \leq 2000$, the corresponding range of minimum E field is $0.04 \leq E_{\min} (\text{V} \cdot \text{m}^{-1}) \leq 0.4$. For reliable breakdown, $E \geq \sim 2E_{\min}(p, L)$ is desirable.

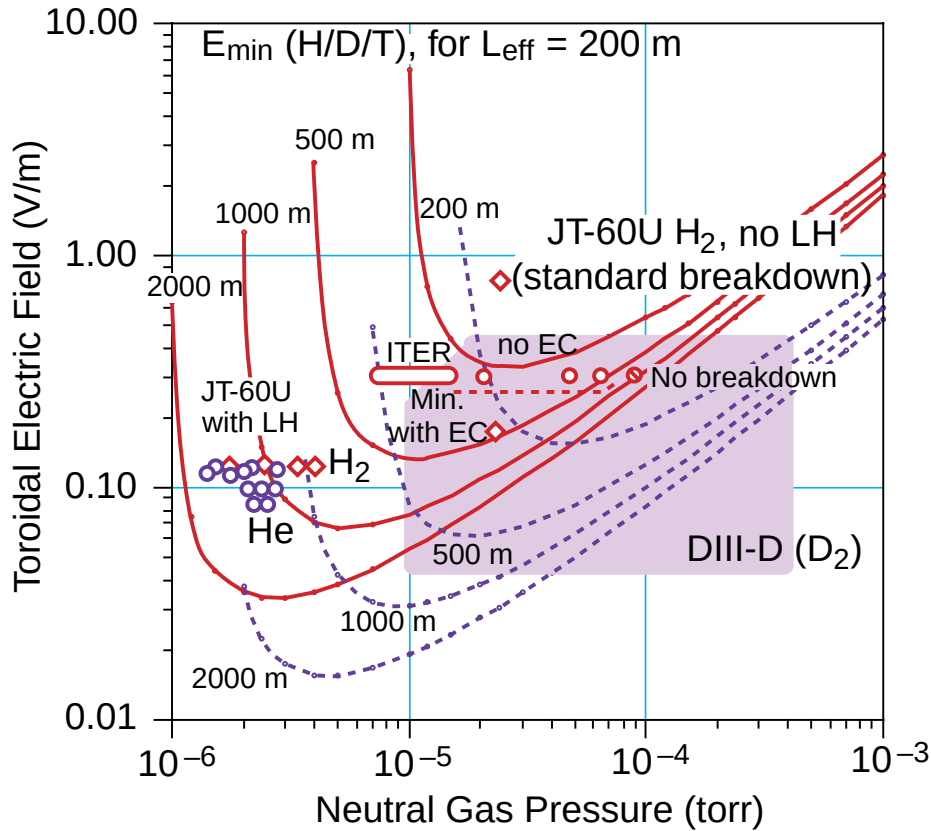


Figure 12. Minimum electric fields for Townsend avalanche breakdown in hydrogen, deuterium or tritium and in helium (dashed lines), for various connection lengths L_{eff} . Data for unassisted (ohmic) and rf assisted plasma initiation in JT-60U (Ref. [108]) and DIII-D (Ref. [105]) are superposed. Proposed start-up parameters for ITER are also shown. The shaded domain, encompassing the DIII-D data, is to facilitate identification of the data: it has no scientific significance. The dashed line labelled 'Min.' shows the lowest E field at which unassisted breakdown in DIII-D is possible.

Figure 12 demonstrates that at minimum E_{min} (i.e. for $p \sim 10^{-5}$ Torr), helium should theoretically reduce the breakdown electric field by a factor of 2.5 relative to the field for hydrogen. Figure 12 also shows selected experimental data for ohmic and rf assisted breakdown experiments in DIII-D and JT-60U. In the DIII-D experiments with D_2 , the minimum E field for breakdown was $0.25 \text{ V} \cdot \text{m}^{-1}$, and the effective connection length inferred from the high pressure limit for breakdown was 500 m ($\cong 50 \times 2\pi R$). With the addition of EC assist (700 kW, 60 GHz), breakdown at $0.15 \text{ V} \cdot \text{m}^{-1}$ was possible. In JT-60U with LH assist (up to 1.5 MW at 1.7–2.4 GHz), the minimum E field for plasma initiation was $0.12 \text{ V} \cdot \text{m}^{-1}$ for H_2 and $0.08 \text{ V} \cdot \text{m}^{-1}$ for He, both achieved at prefll pressures of $2\text{--}3 \times 10^{-6}$ Torr. Normal unassisted breakdown in JT-60U takes place at higher pressures (2×10^{-5} Torr) that are more typ-

ical of standard tokamak practice, with E fields of about $0.8 \text{ V} \cdot \text{m}^{-1}$. The proposed pressure range for ITER start-up is $0.7\text{--}2.0 \times 10^{-5}$ Torr. As demonstrated by Fig. 12, this range should yield reliable Townsend breakdown in D_2 for $L \geq \sim 300$ m. Overall, the ITER start-up parameters are rather similar to those demonstrated in DIII-D.

Helium prefll offers two additional advantages for low E field plasma initiation:

- (i) better controllability of the prefll gas pressure compared to hydrogen (especially at low pressure), because wall absorption of helium is much lower; and
- (ii) charge exchange energy losses are lower for helium, which improves the power balance after the initial ionization phase of the breakdown (see below).

(2) Connection length and allowable stray field

The effective length L that enters into the Townsend avalanche growth ($n_e \propto \exp(\alpha L)$) is set by the ratio of poloidal (perpendicular) and toroidal fields. In tokamaks with toroidally conductive vacuum vessels, poloidal ‘error’ fields at start-up arise primarily from the eddy currents that application of the one turn voltage produces. There may also be error fields produced by a non-ideal ohmic heating coil (e.g. a finite length central solenoid). Application of a simple ‘free drift’ model shows that the length L for an electron starting at the centre of a toroidal region with minor radius a_{eff} to drift to the wall is

$$L = a_{eff} B_T / B_\perp \quad (2)$$

where B_T is the toroidal magnetic field, and a_{eff} is the distance to the wall along the direction of $B_\perp$. For avalanche growth, $\alpha L \geq 1$ is required. This is for a simple ‘zero dimensional’ drift model: comparisons of magnetic connection lengths with the connection lengths inferred from observations of the avalanche growth time and recent re-investigations of data from DIII-D and JET [108] show that

$$L_{eff} \cong 0.25 a_{eff} B_T / B_\perp \quad (3)$$

gives a better estimate of the effective connection length for avalanche growth. The factor of 0.25 can be understood from the facts that not all electrons start from the initiation region centre and that $B_\perp$ obtained in a typical multi-pole null increases more rapidly than r/a_{eff} . This means that the effective radius of the start-up region is somewhat smaller than the actual radius of the low field ‘breakdown’ region.

For ITER start-up with a 0.5 m radius null region positioned near the outboard limiter, the estimated $B_\perp$ is 0.002 T and the effective connection length from Eq. (3) is ≥ 300 m. The E_{min} data for D₂ plotted in Fig. 12 show that reliable ohmic initiation should be possible for pressures of about 10^{-5} Torr, albeit with little additional margin beyond the experimental ‘safety’ factor of 4 ($1/0.25$) that is incorporated into Eq. (3).

Since error fields arising from induced vessel currents tend to increase linearly with applied voltage, and since higher error fields (lower L) lead to a higher E_{min} (Fig. 12), an $EB_T/B_\perp$ requirement also arises. The empirical scaling for the allowable error field for reliable ohmic breakdown in JET [110] is

$$EB_T/B_\perp \geq 1000 \text{ V} \cdot \text{m}^{-1}. \quad (4)$$

This requirement is marginally satisfied in ITER [104]. Start-up assist relaxes the requirement significantly: the DIII-D EC assist plasmas obtain reliable start-up at $EB_T/B_\perp \sim 150 \text{ V} \cdot \text{m}^{-1}$. In JT-60U, $EB_T/B_\perp \sim 300 \text{ V} \cdot \text{m}^{-1}$ is obtained for plasma initiation with helium and LHRF heating. In circumstances where there is an appreciable contribution to $B_\perp$ from current flowing in the vessel, the steady poloidal field within the vessel must be such as to cancel fields from the vessel current when they saturate and the applied loop voltage becomes fully available for breakdown. Breakdown will occur a vessel L/R time after voltage initiation ($L/R \approx 2$ s for ITER/FDR).

(3) Impurity burnthrough

It is well known that the survival of the plasma produced following initial breakdown and hydrogen ionization depends critically upon achieving subsequent ionization of low- Z impurities such as carbon or oxygen. Until these impurities (which can easily constitute 10% of the initial plasma ion density) become nearly fully ionized, they radiate strongly and, hence, prevent the plasma from heating to temperatures >100 eV, where it is possible to sustain (and raise) the plasma current with reasonable applied voltages. The resulting (successful) transition to a nearly fully ionized impurity state is termed ‘impurity burnthrough’: failure to achieve full burnthrough results in being unable to maintain the initial plasma current, and the discharge fails.

Reduction of the initial influx of wall released low- Z impurities, especially oxygen, by wall conditioning (see Section 1) is a primary consideration for start-up optimization. Given that low- Z contamination is reduced to acceptable levels, the secondary requirement is to have enough plasma power input such that a favourable global power balance ($P_{in} > P_{ion} + P_{rad}$) is obtained following initial Townsend ionization. For ohmic start-up, this requires sufficient loop voltage, and in present tokamaks it is usually the ‘ohmic burnthrough’ requirement that determines the applied start-up voltage rather than the Townsend avalanche initiation voltage requirement. In a simple, zero dimensional picture, the required burnthrough electric field has no size or magnetic field scaling, but is linearly proportional to the neutral density, suggesting that ohmic burnthrough in ITER will be similar to present tokamaks and will be easiest at low fill pressures. In ITER and reactor tokamaks, the option to apply enough voltage

to attain the electric field required for ohmic burnthrough is not available. (The standard $0.8 \text{ V} \cdot \text{m}^{-1}$ used in JT-60U corresponds to a loop voltage of 40 V for ITER.) In addition, even when marginal ohmic burnthrough is possible, the prolonged plasma current dwell period that follows Townsend avalanche initiation – during which the plasma temperature remains below 100 eV – results in resistive flux consumption (resistive V-s loss) that can be avoided if higher input power can be provided.

Simulations of ITER breakdown and impurity burnthrough with a zero dimensional model [104] show that with a post-avalanche electron density of $n_e \approx 3 \times 10^{18} \text{ m}^{-3}$ and no impurities, discharge survival without EC assist is marginal. With $n_e \approx 1.5 \times 10^{18} \text{ m}^{-3}$, a beryllium impurity level of about 2% is tolerable without EC start-up assist. In contrast, with 2 MW of externally added power (e.g. EC), $n_e \approx 5 \times 10^{18} \text{ m}^{-3}$ with up to 5% Be is tolerable. For 5% C, 5 MW of absorbed power provides reliable start-up. These calculations lead to an ITER requirement for 3 MW EC power for plasma start-up assist. A 3 s pulse duration and two frequencies, $\sim 90 \text{ GHz}$ and $\sim 110 \text{ GHz}$, are specified to ensure that the EC power absorption radius falls within the cross-section of start-up plasmas positioned on the outboard limiter for on-axis toroidal fields of 4.0–5.7 T [1]. In this regard, the DIII-D and JT-60U experiments show that for burnthrough assist, it is (not surprisingly) essential to have the rf absorption region fall within the start-up plasma boundary. In contrast, having an rf absorption resonance or absorption region anywhere within the torus volume is usually sufficient to facilitate Townsend breakdown assist.

(4) Suppression of runaway electron generation

Runaway electron generation is possible during low voltage plasma initiation because low prefill gas pressure is required. High electron temperature at low prefill pressure is claimed to generate runaway electrons because the cross-section for collisions between electrons and hydrogen becomes small at $T_e > 4\text{--}5 \text{ eV}$ and high energy electrons can gain energy from the applied electric field between collisions. The threshold condition of runaway electron generation is proposed as [105]

$$E/p > (2\text{--}2.5) \times 10^4 \text{ V} \cdot \text{m}^{-1} \cdot \text{Torr}^{-1}. \quad (5)$$

However, runaway electron generation was not observed for an E/p of $\sim 8 \times 10^4 \text{ V} \cdot \text{m}^{-1} \text{ Torr}^{-1}$ in JT-60U, even in the presence of superthermal elec-

trons (in the energy range 80–200 keV) during LH heating. The possible causes are:

- (i) lower resistive electric field than the Dreicer field at a high electron temperature of $\sim 100 \text{ eV}$;
- (ii) not enough time for the superthermal electrons to be accelerated to runaway energies; and
- (iii) magnetic fluctuations during plasma initiation (see Section 4.4 of Chapter 3).

In summary, the physics design considerations embodied in categories (1)–(4) above have been evaluated for ITER and ohmic/inductive plasma initiation by Townsend avalanche in 10^{-5} Torr D_2 gas with an applied electric field of $0.3 \text{ V} \cdot \text{m}^{-1}$ and have been found to be feasible. Acceptably low error fields, $\sim 0.002 \text{ T}$, for such start-up can be achieved in a 0.5 m radius multi-pole field null positioned near the outboard limiter. The planned addition of 3 MW of EC assist power that is resonant in the start-up plasma will facilitate both reliable breakdown over a wider range of error field and pressure conditions and also reliable impurity ionization following initial Townsend + EC assisted breakdown. Substitution of He for D_2 as the initial filling gas is also projected to provide a further margin for reliable plasma initiation.

2.3.2. Plasma current ramp up

Initial breakdown and impurity burnthrough in present medium and large size tokamaks typically results in an initial plasma current of 20–100 kA. The corresponding initial current in ITER at 1 s after initiation will be about 0.5 MA [104]. In all cases, a further increase of the current is required to obtain the ‘flat-top’ current (21 MA in ITER/FDR) needed for full plasma performance. This increase is typically accomplished by a more or less linear (versus time) ramping up of the plasma current. In addition, in elongated cross-section tokamaks with divertors, and in ITER, the current ramp up phase is accompanied by a plasma elongation increase and/or cross-section expansion that includes a transition from an initial limiter defined low current start-up plasma to a final full current/full cross-section divertor defined plasma. Fig. 13 illustrates the current ramp up and plasma cross-section expansion sequence proposed for ITER.

There is usually some increase of plasma density – typically effected by gas injection, see Section 2.2

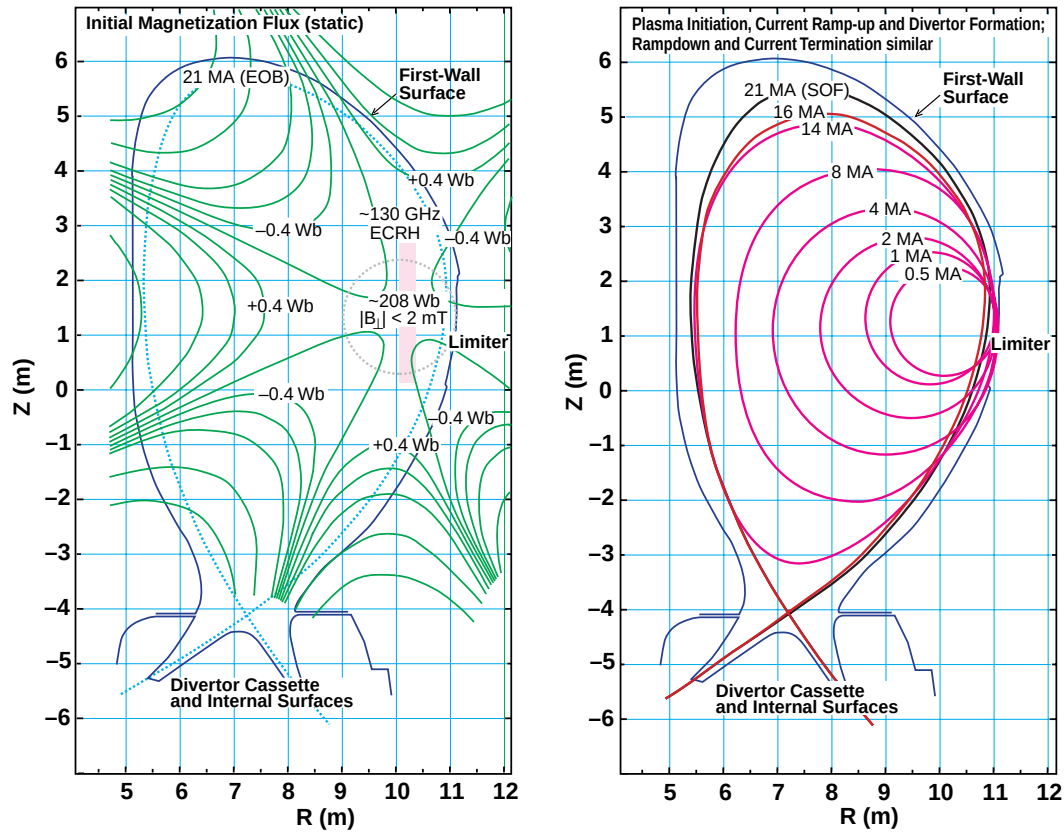


Figure 13. Plasma initiation and current ramp up configurations in ITER (Ref. [103]).

of this Chapter – during the current rise. While this portion of the plasma scenario is typically described as the ‘current ramp up’ phase, in present divertor tokamaks it actually involves a rather carefully coordinated sequence of plasma current increase, magnetic configuration development and plasma densification that ensures that the various MHD and kinetic/power balance operational limits that govern plasma stability, confinement and power and particle exhaust are successfully skirted. During this process, it is also usually necessary to implement a ramp up strategy that avoids unnecessary plasma resistive loss (resistive V-s consumption), and thus arrive at the final flat-top current state with a PF flux variation reserve for inductively sustaining the plasma current through the balance of the scenario.

The principal physics and plasma operation considerations that enter into the design of the current ramp up phase of the scenario are (1) MHD stability, (2) plasma densification, and (3) resistive flux (resistive V-s) consumption. In addition, it is of course necessary for the PF coil system to provide

- (1) the one turn voltage needed to raise the current;

- (2) ‘slow’ quasi-static equilibrium control to achieve the desired plasma configuration evolution; and
- (3) ‘fast’ dynamic equilibrium control to stabilize the plasma vertical position and compensate for the effects of plasma disturbances (minor disruptions, ELMs, etc.) that occur during the current ramp up.

These plasma magnetic control matters are addressed using the same PF system design procedures and control methods described in Section 2.1.

The more generic physics issues that apply to the design of the current ramp up phase in present tokamaks and ITER are addressed below.

2.3.2.1. MHD stability and operation limits during current ramp up

The immediate requirement is to achieve current ramp up without encountering a major disruption. As presented in Chapter 3, there are two causes of major disruption:

- (i) growing internal MHD instability, and

- (ii) density limit/plasma edge radiation power balance initiated current profile peaking.

During plasma current ramp up, the resistive tearing mode instability, which produces transient magnetic reconnection localized at the $q = m/n$ rational surface, is the principal potential category (i) trigger for major disruption. This type of instability develops if the plasma current density profile becomes too broad, and is thus associated with low values of the dimensionless plasma internal inductance l_i . In contrast, too high values of the plasma density, which lead to excessive plasma edge cooling, current profile shrinkage and, hence, high values of l_i , result in category (ii) ‘density limit’ disruption.

The experimental manifestation of these two constraints on plasma operation is embodied in the well known l_i - q operation space diagram, as illustrated for JET in Fig. 14. As the Figure shows, successful ramp up requires negotiating a trajectory in the l_i - q domain that avoids the lower ‘sawtooth like’ tearing mode internal instability boundary and the upper density limit edge power balance boundary. The essential requirement is to maintain the internal inductance in the approximate range 0.8–1.2, with a final refinement to 0.9–1.0 near $q = 3$. The principal experimental means of controlling the inductance are current ramp rate (higher dI/dt reduces inductance; lower dI/dt raises inductance) and gas injection, which both directly control the density and also cool the edge, thus raising the inductance. These ‘control means’ lead to the well known strategy for successful ramp up that combines a gradually decreasing rate of current rise with carefully controlled gas injection that suppresses edge current without unduly raising the plasma density. Greater precision in controlling the start-up trajectory is needed near rational q values, especially $q = 3$, where the range of allowable l_i is relatively small.

Double tearing mode instabilities occur during current ramp up at low l_i when the ramp rate is excessive. When q_a passes a rational value, strong growth of a tearing mode with $m/n = q_a$ is observed. The mode growth rate is sufficiently fast that it can lock toroidally, often with a toroidal phase determined by error fields (see Section 2.5 of Chapter 3), and be preserved during the rest of the discharge, where it can ultimately produce a disruptive termination. Increase in l_i is essential to avoid this type of problem. As has been noted above, inductance control by gas puffing has been demonstrated in many tokamaks, and current profile control by LHCD, and

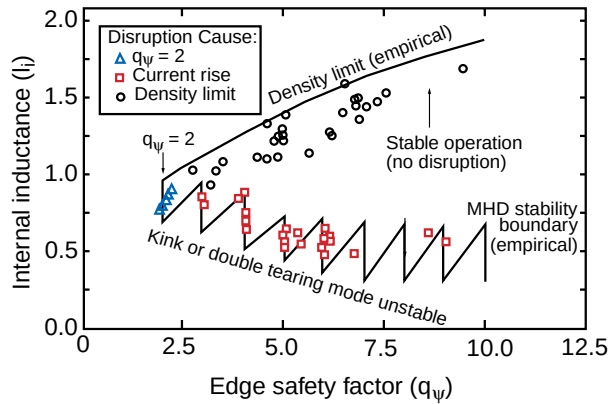


Figure 14. Empirical stability diagram for limiter plasmas in JET, with cause of MHD activity or disruption indicated. Data and empirical limit boundaries reproduced from Ref. [111].

by the use of an ergodic divertor, has been demonstrated in Tore-Supra [112]. The use of NBI – which provides toroidal rotation drive – to avoid mode locking has been demonstrated in JT-60 [113] and other tokamaks.

The tearing mode stability boundaries plotted in Fig. 14 are semi-empirical and hence provide an indicative rather than definitive characterization of the actual stability boundary, which must ultimately be determined during experimental operation.

The rate of current rise that can be obtained in a large minor radius plasma with appreciable edge temperature is limited by the tendency of current to concentrate at the plasma edge and lower l_i excessively. However, this difficulty can be minimized by using a (nearly) constant q ramp up scenario, wherein the minor plasma radius is gradually increased, adding a new current carrying layer around the original high temperature plasma core without decreasing l_i . This type of expanding cross-section scenario is illustrated in Fig. 13. Current ramp up rates of up to $2 \text{ MA} \cdot \text{s}^{-1}$ have been obtained in JT-60U with this method [114]. This rate can be contrasted with the rate of $\sim 0.5 \text{ MA} \cdot \text{s}^{-1}$ that is more typically used in ‘full radius’ JT-60U or JET plasma current ramp up.

Figure 15 illustrates a simulation of the ITER plasma current rise phase implemented with the expanding radius/increasing elongation configuration expansion sequence illustrated in Fig. 13. The initial dI/dt is $0.2 \text{ MA} \cdot \text{s}^{-1}$. Current ramp rate decreases as the ramp up continues, ultimately falling to $0.05 \text{ MA} \cdot \text{s}^{-1}$ prior to attainment of current flat-top. An instability free trajectory through l_i - q space

is achieved at start of current flat-top (SOF) and maintained during the DT burn phase (SOB to EOB). Following end of burn, the internal inductance rises but remains within acceptable limits (to avoid density limit disruption) during the plasma current ramp down (see Section 2.3.3 below).

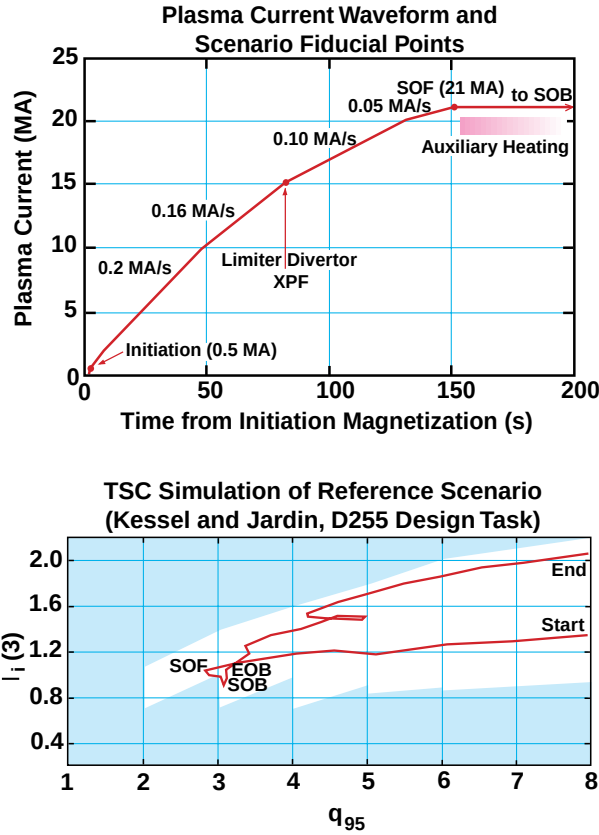


Figure 15. Waveform for ITER plasma current ramp up to start of flat-top and simulation with a tokamak simulation code plasma scenario model of the resulting trajectory in l_i - q space. The simulation also shows the l_i - q variation during the ignition, burn and current ramp down phase. TSC simulation by C. Kessel and S. Jardin for the ITER Joint Central Team (see Ref. [103]).

In a given tokamak there is an optimal current ramp up time and/or current rise waveform that results in reaching a fully penetrated quasi-stationary equilibrium current profile at the end of the ramp up with near-minimum resistive V-s consumption (see below). Slower current ramps increase resistive losses and, hence, are undesirable; faster ramps can achieve current ‘flat-top’, albeit with a broadened non-equilibrium profile, with decreased inductive and resistive flux consumption. However, the ‘flux savings’ produced by this strategy are ultimately expended when the current profile finally

equilibrates, and the use of auxiliary heating during current ramp up serves only to delay rather than reduce ultimate resistive V-s consumption. In contrast, non-inductive current drive during ramp up can reduce net V-s consumption and also relax the requirement for fast plasma current ramp up. In JT-60U, plasma current start-up from plasma breakdown to 1 MA plasma current with a rate of $\sim 0.2 \text{ MA}\cdot\text{s}^{-1}$ has been demonstrated using a low electric field of $\sim 0.1 \text{ V}\cdot\text{m}^{-1}$ [108].

2.3.2.2. Resistive flux consumption

The plasma resistive flux consumption ($\Delta\Psi_{res}$) during the current ramp up phase is an important scenario design basis consideration. Here, the underlying physics basis is well understood: there is an irreducible minimum resistive (Poynting formalism) flux consumption [115] that is described by empirical ‘Ejima’ formula

$$\Delta\Psi_{res} = C_{Ejima} \mu_0 R_0 I_p \quad (6)$$

where C_{Ejima} is a coefficient that depends weakly on the precise details of the evolution of the plasma profile quantities, such as the impurities and the electron temperature, and R_0 is the plasma major radius. The theoretical minimum value of C_{Ejima} is about 0.4 for a $q = 3$ plasma and for design purposes, $C_{Ejima} = 0.45$ constitutes a practical estimate of the minimum flux consumption. Figure 16 shows the data that support this basis: flux consumption for $q = 3$ plasmas typically falls in the range $0.4 \leq C_{Ejima} \leq 0.5$.

Greater than minimum flux consumption can occur in situations where the current ramp rate is too slow and excessive resistive losses accumulate. Greater than minimum resistive loss ($C_{Ejima} > 0.45$) can also occur in large tokamaks (ITER) in start-up situations with a non-optimal minor radius expansion. Finally, it is worth noting that adding auxiliary heating during current ramp up does *not* reduce the total resistive loss that is accrued when a fully penetrated current profile (i.e. $l_i \cong 0.9$ for a $q = 3$ plasma) is achieved. Heating does delay the accrual of full resistive loss and, of course, increases the time to achieve full current profile equilibration appropriate for ELMy H-mode discharges.

2.3.3. Plasma current ramp down

Two types of disruptions are observed during the plasma current ramp down. One is the disruption

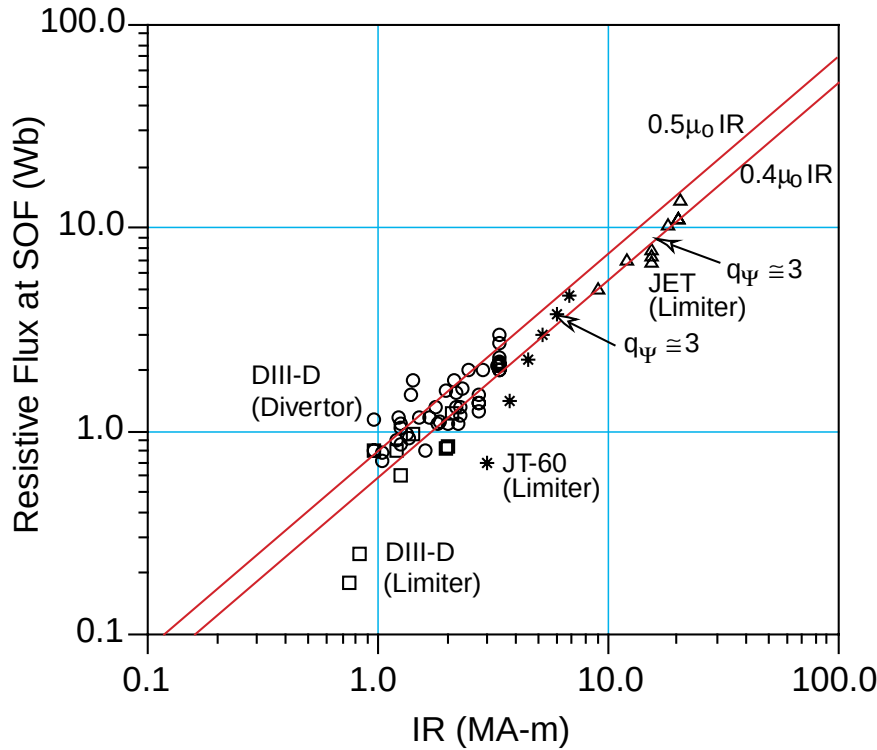


Figure 16. Resistive flux consumption in various limiter and divertor tokamaks at start of current flat-top.

that is seen to occur close to, or even above, the Greenwald density

$$n_{GW} = I_p / \pi a^2 [10^{20}, \text{MA}, \text{m}]. \quad (7)$$

The other disruption occurs at very low density and is observed at high l_i and occurs at densities below $0.2n_{GW}$. Thus, there is a wide density window of 0.2 – $1.0n_{GW}$ to get a stable plasma ramp down for hydrogen plasmas. These two types of disruption are caused by the same tearing instability due to the peaking of the current profile. The following three methods have been demonstrated for obtaining stable ramp down.

- (1) The plasma minor radius is decreased to suppress the increase in l_i during the plasma current ramp down. This is the method that is used conventionally on non-circular tokamaks, together with a small ramp down rate.
- (2) MHD instability is excited by external helical fields to reduce the increase in l_i during ramp down, which has avoided high- l_i disruption [114].

- (3) The density window can be expanded by further heating and/or an intense helium puffing. The density limit of the helium plasma during the plasma current ramp down is about two times higher than that of the hydrogenic plasma, and the maximum ramp down rate of $-6 \text{ MA} \cdot \text{s}^{-1}$ was obtained from 2 MA to 0 MA [114,116]. The reduction of the charge exchange loss is considered to be a possible cause of the higher density limit for helium plasmas. This method relies on fast positional control of the plasma at the neutral point.

2.3.4. Fast plasma shutdown: requirements and options

Normal plasma shutdown in ITER from a current of 21 MA requires approximately 200 s to complete: this shutdown time is set by both the current ramp down rate limitations necessary to avoid a high- l_i disruption and by the time required to exhaust the plasma particle inventory in a well controlled manner. Current shutdown in shorter times is projected to lead to disruption and the voltage capabilities of the ITER PF system limit more rapid

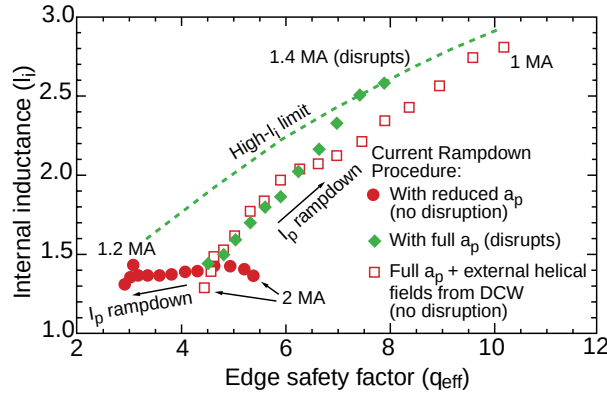


Figure 17. Comparison of plasma current shutdown trajectories in l_i - q space (cf. Figs 3 and 4) in JT-60U with ‘standard’ full minor radius current ramp down procedure (disrupts owing to high l_i), with minor radius contraction (disruption avoided), and with standard ramp down with external helical fields added (l_i decreased, disruption avoided).

current shutdown to about 50 s. Fast shutdown for purposes of machine protection or other plasma operation need will require some form of impurity injection to enhance plasma radiation cooling and current ramp down. Massive deuterium or helium gas injection, liquid hydrogen jet injection or solid low- Z pellet injection (D_2 , Li, Be, etc.) are proposed as candidates for fast plasma energy and current shutdown in ITER (see Section 4.5 of Chapter 3). All of these means appear to be capable of rapidly terminating a full performance ITER plasma in a few seconds or less. However, to what extent such a shutdown can be radiative rather than disruptive, to what extent vertical instabilities and halo currents during shutdown can be avoided, and to what extent significant runaway electron current conversion during shutdown can be avoided are all matters of ongoing study for reactor scale tokamaks and ITER. These matters, especially the concerns about runaway conversion, are more fully discussed in Section 4.5 of Chapter 3.

2.3.5. ITER scenario concept and simulations

Plasma operation in ITER will be conducted within the framework of an inductively driven and controlled plasma operation scenario. The scenario concept is identical to that employed in the present generation of shaped cross-section divertor tokamaks. Figures 18 and 19 illustrate the overall scenario concept and show a summary of the plasma current/shape/configuration evolution that the scenario will incorporate.

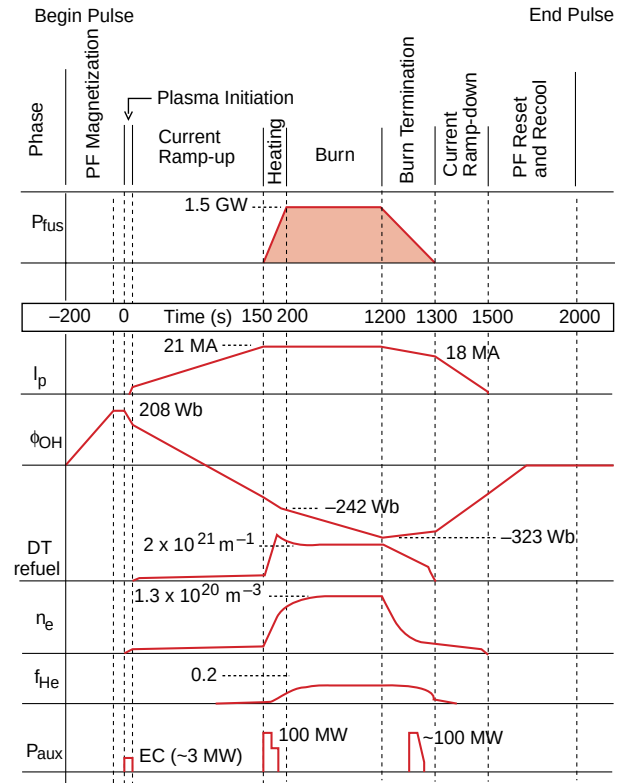


Figure 18. Plasma operation scenario waveforms for 1.5 GW inductively sustained ignited burn in ITER.

Features of the scenario include

- (i) a 530 Wb PF system flux swing;
- (ii) inductive plasma initiation (Townsend avalanche breakdown with EC assist) in a high order multipole field null positioned near the outboard port mounted limiter;
- (iii) minor radius and elongation expansion of the start-up plasma on the limiter prior to divertor formation at $I_p \sim 15$ MA; and
- (iv) maintenance of a well controlled single null divertor configuration during the heating/burn/burn termination phases.

Burn termination occurs in a diverted configuration to allow the plasma particle inventory to be exhausted by the divertor pumping. Current termination is effected following burn termination with a minor radius and elongation contraction on the limiter. The configuration evolution sequence during current ramp down is the reverse of the ramp up evolution. Final current termination occurs in a circular plasma on the outboard limiter.

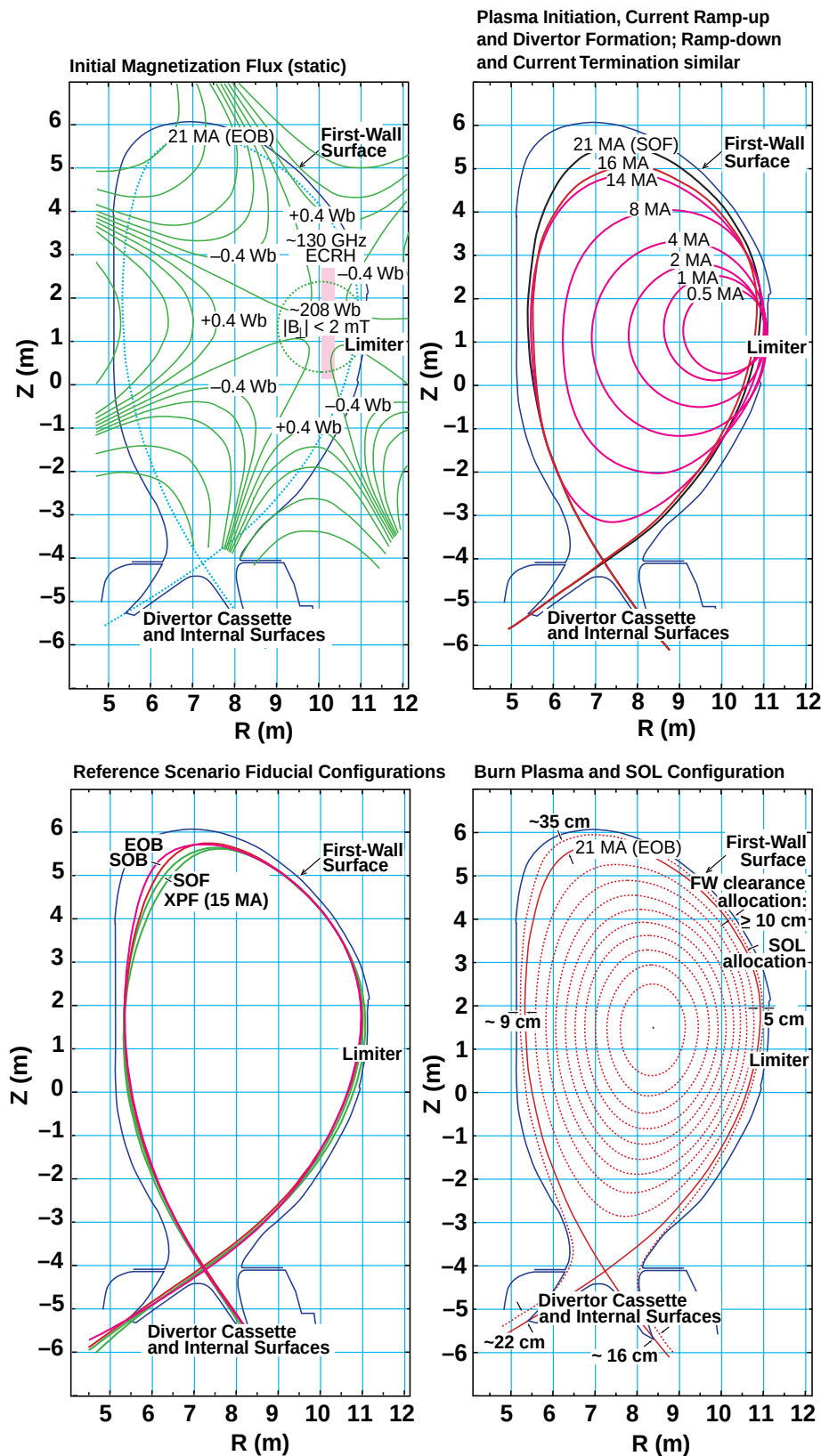


Figure 19. Plasma configurations and configuration evolution for the 21 MA sustained ignition scenario shown in Fig. 18.

Simulations of the plasma start-up and shutdown dynamics show that MHD stability (trajectory in the q - l_i domain) and the edge plasma power balance needed to avoid a density limit disruption are satisfied with acceptable margins (Fig. 15). The resistive flux (V-s) consumption during the start-up and current ramp up phase falls within the physics design basis of $0.45\mu_0 R_0 I_p$ ($\cong 100$ Wb), and ≥ 80 Wb of PF system flux swing will be available for sustaining the 21 MA plasma current during fusion burn. For the estimated burn phase plasma resistive voltage, this flux swing will provide a 1300 s burn.

The scenario design basis is predicated upon a ‘reference case’ plasma with $I_p = 21$ MA, $\beta_p = 0.9$ and $l_i(3) = 0.9$. The sizing of the PF coils and power supplies allows equilibrium control and, in most cases, ≥ 1000 s inductively sustained burn to be obtained for 21 MA plasmas with $0.7 \leq \beta_p \leq 1.2$ and $0.7 \leq l_i \leq 1.1$. Operation at 24 MA ($q_{95} \cong 2.6$) with $\beta_p \cong 0.8$ and $l_i \cong 0.8$ (1.5 GW fusion power) is also feasible. Inductively sustained burn at 24 MA is about 500 s. The scenario will also support operation with ohmic and auxiliary heated DD plasmas during initial plasma commissioning, and extended pulse inductively sustained driven burn operation with reduced plasma current (e.g. 6000 s burn at ~ 1 GW with $I_p = 17$ MA and 100 MW H/CD power). True steady state operation may also be possible with a reversed shear plasma at $I_p \sim 12$ MA (see Sections 2.4 and 2.7 of Chapter 3). The trade-off that exists among plasma current, scenario basis (ignited, driven burn, non-inductive steady state) and burn duration is summarized in Fig. 20. As the figure demonstrates, the scenario capabilities span a range of plasma currents from 12 to 24 MA and burn durations from 500 s to steady state. There is also a general tendency for the longer duration driven and steady state scenarios to require ‘enhanced’ confinement ($H_H > 1$) and beta limit ($\beta_N > 2.2$) performance relative to the performance required for nominal 21 MA ignition in ELMy H-mode.

2.4. Advanced tokamaks: control of the current and pressure profiles

This subsection deals with the control issues for current density, pressure profile, and toroidal rotation profile needed to attain either (or both) an internal transport barrier or steady state operation with $\beta \geq 0.03$ coupled with a high bootstrap current fraction. Both operational regimes represent an improvement with respect to the ELMy H-mode

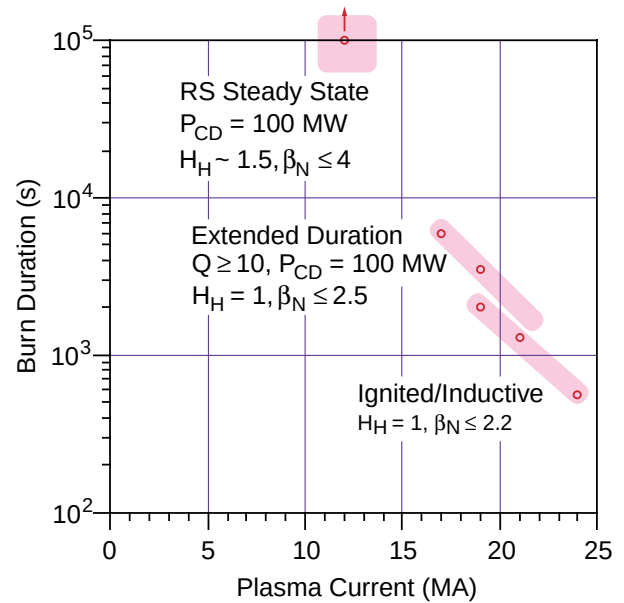


Figure 20. ITER plasma operation scenario capabilities.

assumed for nominal ITER operation and are frequently described in terms of the achievement of ‘advanced performance’ or ‘advanced tokamak’ operation [117]. Present examples of this type of plasma operation are the transient regimes with optimized magnetic shear as obtained in DIII-D [118], TFTR [119], JET [120] and JT-60U [121], as well as the steady state lower hybrid current drive (LHCD) sustained hot electron LHEP mode in Tore-Supra [122] and similar steady state LHCD sustained optimized shear plasmas in JT-60U [123].

The theoretical conceptual basis that initially motivated advanced tokamak research is the observation that a relatively high q_{95} discharge with a controlled current density profile that leads to a reverse shear q profile can achieve a high bootstrap fraction, congruence between the bootstrap and required plasma current density, and ideal MHD beta limit values attractive for a steady state reactor [124]. Operationally, combined magnetic and kinetic control – plasma current ramps during auxiliary heating – was employed to create transient reverse shear q profiles for experimental investigations, such as access to an MHD second stable core configuration [125]. A key experimental finding was that reverse shear q profiles lead to the spontaneous development of internal transport barriers within which ion thermal transport was reduced to neoclassical levels. The confinement and MHD stability enhancement aspects of such regimes are addressed, respectively, in Section 3.4 of Chapter 2 and Section 2.7 of Chapter 3.

Transport barriers, if controlled and sustained in a reactor scale plasma with $\beta \geq 0.03$, will reduce the facility size and cost needed to attain ignition. The β value assures reasonable fusion power generation for a given facility. Control of reverse shear plasmas constitutes an ‘emerging subject’, both in terms of physics understanding and in terms of experimental implementation, particularly implementation of control in the formal sense, wherein shear profile and/or barrier characteristics are actively monitored and controlled by non-inductive means [126]. Generally, reverse shear plasmas require a nearby conducting wall to attain MHD stability with $\beta \geq 0.03$. Experiments indicate that control, either driven plasma rotation [127] or magnetic feedback [128] (or both), is needed to realize steady operation in this ‘wall stabilized’ regime.

2.4.1. Candidate parameters for ‘advanced performance’ operation in ITER

As indicated above, one can distinguish two varieties of ‘advanced performance’ operation in reactor scale plasmas. The first variety is the creation and sustainment of internal transport barriers in reverse (or otherwise optimized) shear profiles. These barriers promise to provide enhanced confinement, thereby facilitating attainment of ignition or high- Q driven burn in reactor scale plasmas. Presently, transient inductive means (current ramping) constitutes the actuator, which controls the shear profile. Because ‘transient’ is defined with respect to the current density relaxation time, reverse shear q profiles and internal transport barriers can be sustained for many energy confinement times, and, ideally, for times that are comparable to the thermal He equilibration time (typically about 100 s in ITER). The size and temperature of an ignited or near ignited plasma in ITER are such that the current density relaxation time will be about 500 s, so even ‘transient’ advanced performance established by current ramping will make it possible to achieve quasi-stationary ignition or driven burn operation and allow study of energy and particle transport effects on a quasi-stationary basis, including meaningful investigation of He buildup and removal.

The second variety of ‘advanced performance’ operation lies in the achievement of true steady state, high- β plasma operation, wherein the plasma current is sustained wholly by non-inductive means. High bootstrap fraction implies low plasma currents, and such plasmas must operate in the ‘wall stabilized’

regime [127] to achieve $\beta \geq 0.03$. Here, as is summarized in Table 4, self-consistent reverse shear plasma operation scenario analyses that include consideration of likely plasma temperature and density profiles and achievable current drive efficiencies show the *possibility* of obtaining non-inductively sustained steady state operation in a reactor scale plasma with a plasma current of approximately 12 MA and a bootstrap current fraction (I_{BS}/I_p) that is typically $\geq 80\%$ [129].

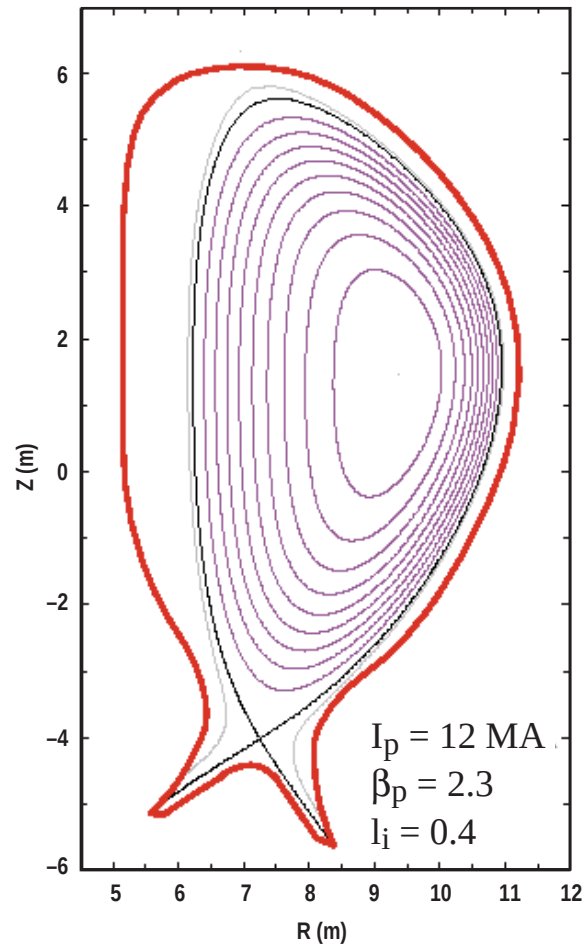


Figure 21. A 12 MA ITER reversed shear equilibrium, which is feasible from the point of view of both fast alpha ripple losses and vertical control.

The parameters given in Table 4 are predicated on a reduced minor radius reverse shear plasma configuration with increased elongation and triangularity (see, for example, Fig. 21). Such plasmas demonstrate the shape flexibility possible with reactor scale poloidal field coils for non-chamber filling discharges, which remain well matched in terms of separatrix configuration to the ITER divertor geometry. Recent theoretical studies indicate a higher beta limit with

Table 4. Advanced steady state operating modes in ITER

Parameter	Fully optimized	H-mode like density profile	Low density reversed shear
$P_{fus} P_{CD}$ (MW)	1500 100	1500 100	1000 100
Q_{CD}	15	15	10
I_p (MA) q_{95}	12 4.95	12 4.95	12 4.95
$R_0 a$ (m)	8.66 2.32	8.66 2.32	8.66 2.32
$k_{95\%} d_{95\%}$	2.00 0.44	2.00 0.44	2.00 0.44
f_{BS}	94%	79%	71%
γ_{cd} ($10^{20} \text{ A} \cdot \text{W}^{-1} \cdot \text{m}^{-2}$)	0.06	0.21	0.21
$\beta_N \beta_{toroidal}$	2.9 2.8%	3.8 3.7%	3.6 3.5%
$\langle T_e \rangle_n T_{e0}$ (keV)	10.2 18.1	12.5 25.9	15.8 32.9
$\langle n_e \rangle n_{e0}$ (10^{20} m^{-3})	1.0 1.7	1.0 1.1	0.71 0.78
$\langle n_e \rangle / n_{GW}$	1.4	1.4	1.0
τ_E (s)	2.35	2.46	2.84
$\tau_E / t_{ITER93-H}$	1.05	1.22	1.26
$\tau_E / t_{ITER89-P}$	2.0	2.26	2.38

elongation and triangularity [130] for reverse shear q profiles. Poloidal field system calculations show that such plasmas have acceptable vertical position control and can be realized by dynamic evolution of the ITER poloidal field currents [131]. Fast alpha particle toroidal ripple loss calculations, as reported in Section 3 of Chapter 5, find acceptable losses when ferromagnetic shims are used to reduce field ripple.

The results summarized in Table 4 are based upon assumed plasma density and temperature (and hence pressure) profiles. Both fully optimized profiles and ‘H-mode like’ density and temperature profile cases are examined. Figure 22 shows the respective profiles: the optimized profiles are L-mode like in the edge region; and the H-mode profiles have nearly flat density profile and an edge temperature pedestal.

The different profile assumptions lead to substantially different bootstrap current profiles and corresponding differences in the non-inductive current drive requirements (Fig. 23). The optimized profiles lead to excellent bootstrap current congruence with the total current profile, while the H-mode profiles require more off-axis current drive and also some reverse current drive at the plasma edge. Bootstrap current alignment is not as favourable as the optimal profile case, and the current drive efficiency requirement is about three times higher than for the optimal profile case.

In Table 4, a total of 100 MW of auxiliary power is assumed to be available to both heat and drive current. The required current drive figure of merit,

γ_{CD} is computed for each case using the formula

$$\gamma_{CD} \equiv \langle n_e \rangle I_{CD} R_0 / P_{CD}$$

where $\langle n_e \rangle$ is the volume averaged electron density, I_{CD} is the current that must be driven non-inductively, R_0 is the major radius of the plasma, and P_{CD} is the power available to drive the current. In the scenario with H-mode like profiles, a current drive figure of merit of $\gamma_{CD} \geq 0.21 \times 10^{20} \text{ A} \cdot \text{W}^{-1} \cdot \text{m}^{-2}$ is required. Such values of γ_{CD} have been achieved with lower hybrid current drive, and are generally consistent with theoretical projections for electron cyclotron or neutral beam current drive in ITER (see Chapter 6).

Table 4 demonstrates that some enhancement in energy confinement and MHD stability relative to the ‘standard’ ITER ELMy H-mode case assumptions of $H_H = 1$ and $\beta_N = 2.2$ is also required. For the H-mode like profiles case, the respective confinement enhancement factor is $H_H = 1.2$ and $\beta_N = 3.8$. These values suggest that the required confinement enhancement is relatively modest. However, the required β_N is about 1.5 times the wall at infinity ideal MHD limit, and, hence, resistive wall mode (external kink mode in the presence of a resistive wall) stabilization through plasma rotation or active magnetic feedback will be required (see Section 2.4 of Chapter 3). And, while the global increase in confinement is modest, recent modelling studies indicate that transport reduction within the barrier must be appreciable [101].

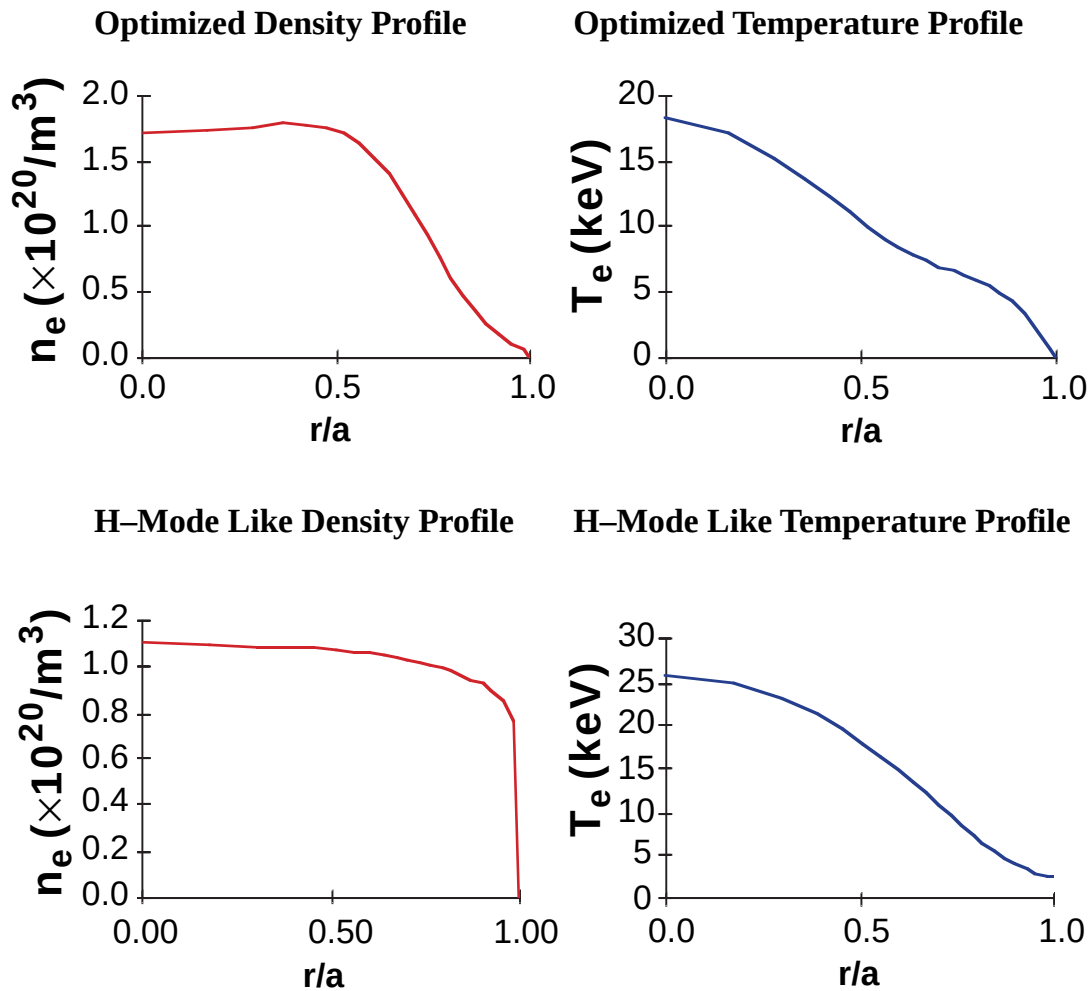


Figure 22. Density and temperature profiles assumed for the optimized profile and H-mode like profile cases in Table 4.

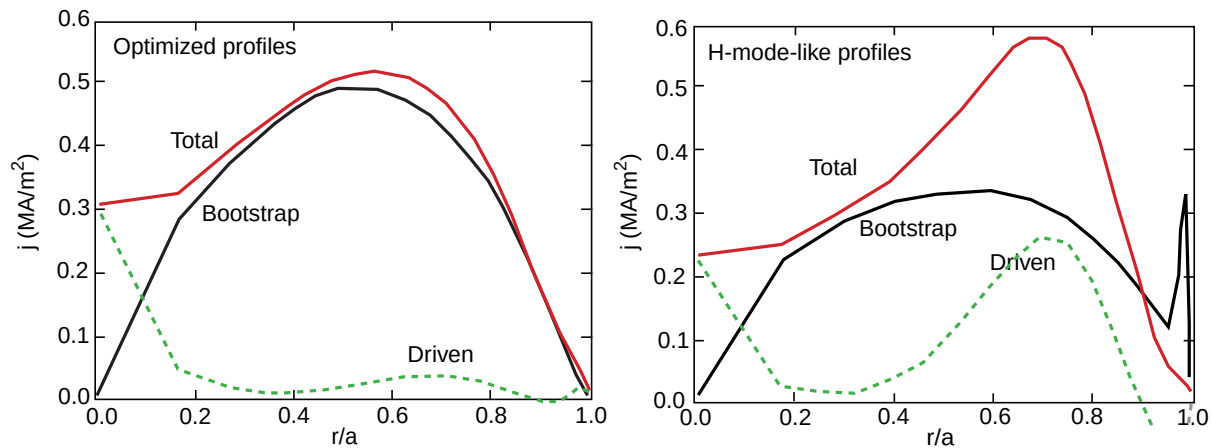


Figure 23. Current density profiles for the optimized and H-mode like density/temperature profiles in Fig. 22. Note the higher off-axis current drive requirement and the negative edge current drive needed for the H-mode like case.

Finally, both of the ITER advanced steady state scenarios discussed above assume that operation will be possible in ITER at densities above the Greenwald density limit. The final scenario in Table 4 addresses this issue by considering operation with the volume averaged density equal to Greenwald limit. Otherwise, the profiles are the same as assumed for the scenario with H-mode like profiles. It is possible to compensate for the lower density to an extent by raising the temperature. However, this pushes the operating point past the peak in $\langle\sigma v\rangle/T^2$, so that the plasma reactivity and the total fusion power are correspondingly reduced (to 1000 MW in the low density scenario). Confinement and beta enhancement factors are otherwise similar to the higher density H-mode like scenario. It is quite clear that high density ($\geq n_{GW}$) is an essential requirement for successful steady state operation in ITER at rated fusion power.

2.4.2. Advanced performance control in present experiments and projections to ITER

The parameters presented in the previous section are intended to illustrate the characteristics of candidate steady state operation modes for ITER and to illustrate the uncertainties in realizing advanced performance in reactor scale plasmas. Nonetheless, certain aspects of advanced performance operation and control can already be foreseen. These operation and control considerations are as follows.

- (i) Setting up of the required current profile and transport barrier during the low- β and initial burn phases of the discharge.
- (ii) Control of the transport barrier, burn and pressure gradients throughout the evolution of the discharge towards steady state equilibrium.
- (iii) Maintenance of the current density profile on its relaxation time-scale.

All the tokamaks cited have found satisfactory recipes for addressing aspect (i); setting up the target q profile with a limited risk of disruption. In particular, the current rise rates are relatively fast, but remain essentially free of MHD and skin currents because the plasma cross-section also increases. These techniques can be extrapolated to ITER [131].

Experimental attempts to control transport barriers at fixed current density – aspect (ii) – are

just getting underway. Potential control tools are the chosen q profile, the auxiliary heating profile, its relative deposition to electrons and ions, and the toroidal rotation profile arising from angular momentum deposition by NBI auxiliary heating systems. The excellent global confinement seen in reversed shear discharges has motivated extensive analysis of the experimental data. This analysis has focused on the paradigm of turbulence suppression via $\mathbf{E} \times \mathbf{B}$ flow shear [132]. The $\mathbf{E} \times \mathbf{B}$ flow shear (which is produced mainly by internal pressure gradients) can give rise to transport barriers. Barriers to both electron [121] and ion [133, 134] transport have been reported, and it has been inferred from transport analysis of TFTR and DIII-D discharges that the ion thermal transport is reduced to neoclassical levels in the reversed shear region. Such turbulence suppression is expected on theoretical grounds [135] and is currently the subject of much theoretical research. There has been some success in this area through the use of IBW waves to create sheared poloidal flows within the plasma, which, in turn, created barriers to the transport of particles and energy [136, 137]. Further progress in the demonstration of the formation and active control transport barriers could greatly enhance the performance of ITER advanced steady state operating modes and the attractiveness of a steady state demonstration reactor based on these operating modes.

Maintaining the optimized current profile – aspect (iii) – during the relaxation towards a steady state equilibrium with zero applied electric field and a high bootstrap current fraction will require a large amount of experimental work on present long pulse machines and also, presumably, on ITER, where magnetic relaxation times are comparable to the standard inductive pulse duration. Current drive by non-thermal power is the main control tool [126]. In Tore-Supra, initial attempts to control the current density profile have been partly successful and produced LH driven steady state discharges with a required internal inductance [138]. This involves actuating either the launched LH power or the phasing of the launcher in order to modify the wave power deposition pattern. In JT-60U, generating and maintaining the negative central magnetic shear were demonstrated by LHCD alone [123] and a negative central shear has been sustained for 7.5 s. When the steep internal transport barrier was formed, a high bootstrap current component carrying 70% of the total plasma current was transiently induced [139].

2.4.3. Simulations of ITER current drive and plasma operating point control

Simulations of ITER plasmas elucidates whether control of steady state discharges will be possible. Assuming the density and temperature profiles from the optimized operating point of Table 4, the ACCOME code has been used to develop scenarios in which a combination of 50 MW of LH and 40 MW of neutral beam power can, in combination with the bootstrap effect, support the full plasma current, providing a total plasma current profile that is stable to ballooning modes and $n = 1, 2$ and 3 ideal kink modes up to $\beta_N = 3.0$ even in the absence of a conducting wall [140].

Time dependent scenarios involving LH and fast wave power have been studied [126, 141] using a thermal diffusivity that replicates experimental data and reflects $\mathbf{E} \times \mathbf{B}$ shear stabilization [142]. In an advanced scenario of steady state operation at 13 MA plasma current, off-axis LHCD in the region $r/a = 0.5$ – 0.8 is used, together with on-axis FWCD and bootstrap current, to sustain the plasma current profile. A consistent scheme is demonstrated with a fusion power output of about 1.5 GW in steady state [126, 141].

The sensitivity of these results has been studied. In the case of strongly non-linear dependence of the heat transport on magnetic shear, feedback schemes may be necessary to control the discharge against MHD collapse and undesired evolution to other steady state equilibria due to the strong coupling between the current profile, heat transport, alpha particle heating, bootstrap current, and resistive current diffusion [141]. Thus, access to optimized MHD stable profiles and prescribed fusion yields may require simultaneous control, on a slow resistive time-scale, of the off-axis (LH) current generation, of the central heating and current drive power (e.g. FWCD), and of the plasma fuel density. It is found that real time calculations of the internal electric field from magnetic reconstruction are very advantageous for control purposes in order to avoid strong relaxation oscillations and collapses during transitions of the transport coefficients.

2.4.4. Summary and future R&D recommendations

Present experimental investigations of advanced performance have tended to concentrate on attainment of the required plasma conditions and optimization of plasma performance rather than on exploring explicit control of the shear and/or pres-

sure profiles or transport barrier properties. This present limitation on the study of control can be ascribed, to a great degree, to the fact that experimental means – e.g. powerful rf heating and current drive systems with good radial deposition control and agility – to actively control the current and pressure profiles are only now becoming available in the present generation of tokamak experiments. So, at present, understanding of underlying physics mechanisms that control internal transport barrier formation and efficacy is lacking, and empirical data on active shear profile and barrier control are only now starting to become available. Hence the physics design of advanced performance scenarios for ITER and for the control of such advanced performance operation are provisional.

However, design of a reactor scale facility can incorporate the basic hardware provisions anticipated to be necessary to support steady state operation sustained by non-inductive current drive, bootstrap current, rotation drive, and active $n = 1$ magnetic feedback.

There are also significant uncertainties as to how well reverse shear plasmas can be controlled in a steady state regime. For example, plasma rotation affects wall stabilization and whether ‘active’ stabilization of the resistive wall MHD mode will be required to obtain adequate β values. Both remain as open physics issues to be resolved by future research. It is clear, however, that at a minimum, accurate real time measurement of the $q(r)$ or $j(r)$ profiles, and also of the electron and ion $p(r)$ profiles, will be critical to successful control. Moreover, candidate current drive systems will have to have some degree of active control capability that can be used to first optimize the current profile and then to respond to changes in the bootstrap current profile that confinement effects produce. Finally, the time-scale of ultimate relaxation to true equilibrium steady state is expected to approach 10 000 s [126], so control means must be capable, essentially, of operating indefinitely.

References

- [1] Winter, J.J., J. Nucl. Mater. **176–177** (1990) 14.
- [2] McCracken, G.M., Stott, P.E., Nucl. Fusion **19** (1979) 889.
- [3] Winter, J.J., Plasma Phys. Control. Fusion **38** (1996) 1503.
- [4] Lazarus, E.A., et al., Phys. Rev. Lett. **77** (1996) 2714.

- [5] Strachan, J.D., et al., Phys. Rev. Lett. **58** (1987) 1004.
- [6] Koide, Y., JT-60U Team, Phys. Plasmas **4** (1997) 1623.
- [7] Jackson, G.L., et al., Phys. Rev. Lett. **67** (1991) 3098.
- [8] Pontau, A.E., Morse, D.J., J. Nucl. Mater. **141–143** (1986) 124.
- [9] Campbell, D.J., et al., ‘The influence of beryllium on plasma operation in JET’ (submitted to Fusion Eng. Des., 1999).
- [10] Oren, L., Taylor, R.J., Nucl. Fusion **17** (1977) 1143.
- [11] Dylla, H.F., J. Nucl. Mater. **145–146** (1987) 48.
- [12] Jackson, G.L., et al., Nucl. Fusion **30** (1990) 2305.
- [13] Winter, J.J., J. Nucl. Mater. **145–147** (1987) 131.
- [14] Winter, J.J., et al., Phys. Rev. Lett. **71** (1993) 1549.
- [15] Strachan, J.D., et al., J. Nucl. Mater. **217** (1994) 145.
- [16] Nakashima, Y., et al., J. Nucl. Mater. **162–164** (1989) 812.
- [17] Esser, H.G., et al., J. Nucl. Mater. **241–243** (1997) 861.
- [18] Keilhacker, M., JET Team, in Symposium on Fusion Engineering 1998 (Proc. 17th IEEE/NPSS Symp. San Diego, 1999), Vol. 1, IEEE, Piscataway (1999) 3.
- [19] Maingi, R., et al., Nucl. Fusion **36** (1996) 245.
- [20] Mioduszewski, P.K., et al., J. Nucl. Mater. **220–222** (1995) 91.
- [21] Federici, G., et al., J. Nucl. Mater. **266–269** (1999) 14.
- [22] Skinner, C.H., et al., Nucl. Fusion **39** (1999) 271.
- [23] Longhurst, G.R., et al., in TAMP4 User’s Manual, Rep. EGG-FSP-10315, Idaho National Engineering and Environmental Laboratory (1992).
- [24] Skinner, C.H., et al., J. Vac. Sci. Technol. A **14** (1996) 3267.
- [25] Skinner, C.H., TFTR Team, J. Nucl. Mater. **241–243** (1997) 214.
- [26] Mueller, D., et al., J. Nucl. Mater. **241–243** (1997) 897.
- [27] Winter, J.J., J. Nucl. Mater. **145–147** (1987) 131.
- [28] Sartori, R., et al., J. Nucl. Mater. **176–177** (1990) 624.
- [29] Coad, J.P., et al., J. Nucl. Mater. **162–164** (1989) 533.
- [30] Coad, J.P., et al., J. Nucl. Mater. **241–243** (1997) 408.
- [31] Campbell, D.J., JET Team, J. Nucl. Mater. **241–243** (1997) 379.
- [32] Coad, J.P., Report of the ITER Contract: Deuterium Content of Material Redeposited in Tokamaks for the period Jan. 1996 to Jan. 1997, JET Rep. JET-IR(97) 02, 3 (1997).
- [33] Andrew, P., et al., Nucl. Fusion **33** (1993) 1389.
- [34] Andrew, P., et al., Tritium retention and clean-up in JET, to be published in Fusion Engineering and Design.
- [35] Miya, N., et al., Fusion Technol. **26** (1994) 507.
- [36] Masaki, K., et al., Fusion Eng. Des. **31** (1996) 181.
- [37] Taylor, P.L., et al., J. Fusion Energy **12** (1993) 35.
- [38] Franzen, P., et al., in Controlled Fusion and Plasma Physics (Proc. 24th Eur. Conf. Berchtesgaden 1997), Vol. 21A, Part IV, European Physical Society, Geneva (1997) 1429.
- [39] Franzen, P., et al., Nucl. Fusion **37** (1997) 1375.
- [40] Brooks, J.N., et al., J. Nucl. Mater. **241–243** (1997) 294.
- [41] Wesley, J., et al., Fusion Technol. **32** (1997) 495.
- [42] Porkolab, M., et al., in Plasma Physics and Controlled Fusion Research 1994 (Proc. 15th Int. Conf. Seville, 1995), Vol. 1, IAEA, Vienna (1995) 123.
- [43] Köppendorffer, W., et al., in Plasma Physics and Controlled Nuclear Fusion Research 1994 (Proc. 15th Int. Conf. Seville, 1995), Vol. 1, IAEA, Vienna (1995) 241.
- [44] Stambaugh, R.D., DIII-D Team, in Plasma Physics and Controlled Fusion Research 1994 (Proc. 15th Int. Conf. Seville, 1995), Vol. 1, IAEA, Vienna (1995) 83.
- [45] JET Team (presented by D. Stork), in Plasma Physics and Controlled Fusion Research 1994 (Proc. 15th Int. Conf. Seville, 1995), Vol. 1, IAEA, Vienna (1995) 51.
- [46] JT-60U Team (presented by M. Kikuchi), in Plasma Physics and Controlled Fusion Research 1994 (Proc. 15th Int. Conf. Seville, 1995), Vol. 1, IAEA, Vienna (1995) 31.
- [47] Lao, L.L., et al., Nucl. Fusion **25** (1985) 1421.
- [48] Jardin, S.C., et al., J. Comput. Physics **66** (1986) 481.
- [49] Puppini, S., et al., in Fusion Technology 1996 (Proc. 19th Symp. Lisbon, 1997), Vol. 1, Elsevier, Amsterdam (1997) 949.
- [50] Toyoaki, K., et al., Fusion Technol. **32** (1997) 404.
- [51] Horne, S., et al., in Fusion Engineering 1993 (Proc. 15th IEEE Symp. Hyannis, 1994), Vol. 1, IEEE, Piscataway (1994) 242.
- [52] Lister, J.B., et al., Fusion Technol. **32** (1997) 321.
- [53] Doyle, J.C., Stein, G., IEEE Trans. Autom. Control **AC26** (1981) 4.
- [54] Maciejowski, J.M., Multivariable Feedback Design, Addison-Wesley, Reading, MA (1989).
- [55] Albanese, F., Villone, R., Nucl. Fusion **38** (1998) 723.
- [56] Vyas, P., et al., Nucl. Fusion **38** (1998) 1043.
- [57] Ariola, M., et al., Fusion Technol. **36** (1999) 126.
- [58] Jardin, S.C., et al., Nucl. Fusion **33** (1993) 371.
- [59] Jardin, S.C., et al., Nucl. Fusion **27** (1987) 569.

- [60] Merrill, B.J., et al., *Fusion Eng. Des.* **15** (1991) 163.
- [61] Sayer, R.O., et al., *Nucl. Fusion* **33** (1993) 969.
- [62] Ambrosino, G., et al., in *Fusion Technology 1996* (Proc. 19th Symp. Lisbon, 1997), Vol. 1, Elsevier, Amsterdam (1997) 735.
- [63] Söldner, F.X., et al., *Nucl. Fusion* **39** (1999) 407.
- [64] Wagner, F., et al., *Phys. Rev. Lett.* **49** (1982) 1408.
- [65] Kusama, Y., JT-60 Team, *Phys. Plasmas* **6** (1999) 1935.
- [66] Gruber, O., et al., in *Fusion Energy 1998* (Proc. 17th Int. Conf. Yokohama, 1999), IAEA, Vienna (1999), Paper IAEA-CN-69/OV4/3. (NF MS 6686, to be published in Sept.)
- [67] Mahdavi, M.A., General Atomics, personal communication, 1999.
- [68] Zohm, H., ASDEX-Upgrade Team, in *Fusion Energy 1998* (Proc. 17th Int. Conf. Yokohama, 1999), Vol. 1, IAEA, Vienna (1999), Paper IAEA-CN-69/Post Deadline.
- [69] Boucher, D., Wesley, J., in *Controlled Fusion and Plasma Physics* (Proc. 22nd Eur. Conf. Montpellier, 1994), Vol. 18B, Part II, European Physical Society, Geneva (1994) 524.
- [70] ITER Joint Central Team, *Physics Design Description Document/Plasma Performance*, ITER Engineering Design Activity, Garching (1998).
- [71] Baylor, L.R., et al., in *Controlled Fusion and Plasma Physics* (Proc. 22nd Eur. Conf. Bournemouth, 1995), Vol. 19C, Part I, European Physical Society, Geneva (1994) 113.
- [72] Tamai, H., et al., *J. Plasma Fusion Res.* **74** (1996) 1336.
- [73] Wesley, J., et al., in *Fusion Energy 1998* (Proc. 17th Int. Conf. Yokohama, 1999), IAEA, Vienna (1999) Paper IAEA-F1-CN-69/ITER P1/3. (NF MS 6610, to be published.)
- [74] Boucher, D., et al., in *Controlled Fusion and Plasma Physics* (Proc. 24th Eur. Conf. Berchtesgaden, 1997), Vol. 21A, Part III, European Physical Society, Geneva (1997) 953.
- [75] Lang, P., et al., *Nucl. Fusion* **36** (1996) 1531.
- [76] Neyatani, Y., et al., *Fusion Eng. Des.* **36** (1997) 429.
- [77] Taylor, T., et al., in *Plasma Physics and Controlled Nuclear Fusion Research 1992* (Proc. 14th Int. Conf. Würzburg, 1993), Vol. 1, IAEA, Vienna (1993) 167.
- [78] Lawson, J.E., et al., in *Fusion Technology 1994* (18th Symp. Karlsruhe, 1995), Vol. 1, Elsevier, Amsterdam (1995) 739.
- [79] Kallenbach, A., et al., *Nucl. Fusion* **35** (1995) 1231.
- [80] Samm, U., et al., *Plasma Phys. Control. Fusion* **35** (1993) Supplement (12)B, B167.
- [81] Dudok De Wit, T., et al., *Nucl. Fusion* **30** (1990) 1493.
- [82] Walker, M., et al., in *Fusion Engineering 1995* (16th IEEE/NPSS Symp. Piscataway, 1996), Vol. 2, IEEE/NPSS, Illinois (1996) 885.
- [83] Van Houtte, D., Equipe Tore-Supra, in *Fusion Engineering 1995* (16th IEEE/NPSS Symp. Piscataway, 1996), Vol. 1, IEEE/NPSS, Illinois (1996) 730.
- [84] Tamai, H., et al., *J. Nucl. Mater.* **266–269** (1999) 1219.
- [85] Gadeberg, M., et al., in *Fusion Technology 1994* (18th Symp. Karlsruhe, 1995), Vol. 1, Elsevier, Amsterdam (1995) 695.
- [86] Tamai, H., et al., *Fusion Eng. Des.* **39–40** (1998) 163.
- [87] Lipa, M., et al., in *Fusion Engineering 1997* (17th IEEE/NPSS Symp. Piscataway, 1998), Vol. 1, IEEE/NPSS, San Diego (1998) 353.
- [88] Yoshino, R., et al., *Nucl. Fusion* **37** (1997) 1161.
- [89] Yoshino, R., et al., *Plasma Phys. Control. Fusion* **39** (1997) 313.
- [90] Yoshino, R., et al., in *Plasma Physics and Controlled Nuclear Fusion Research 1994* (Proc. 15th Int. Conf. Seville, 1995), Vol. 1, IAEA, Vienna (1994) 685.
- [91] Granetz, R., et al., *Nucl. Fusion* **36** (1996) 545.
- [92] Fukuda, T., JT-60U Team, in *Fusion Technology 1998* (Proc. 20th Symp. Marseille, 1999). *Fusion Eng. Des.* In press.
- [93] Franzen, P., et al., in *Controlled Fusion and Plasma Physics* (Proc. 24th Eur. Conf. Kiev, 1996), Vol. 20C, Part I, European Physical Society, Geneva (1996) A021.
- [94] Kimura, T., et al., *Fusion Technol.* **32** (1997) 404.
- [95] Mertens, V., et al., *Fusion Technol.* **32** (1997) 459.
- [96] Raupp, G., et al., in *Fusion Technology 1992* (17th Symp. Roma, 1993), Vol. 2, North Holland, Amsterdam (1993) 1072.
- [97] Raupp, G., et al., *Fusion Technol.* **32** (1997) 444.
- [98] Ferron, J.B., et al., in *Fusion Engineering 1995* (16th IEEE/NPSS Symp. Piscataway, 1996), Vol. 2, IEEE, Piscataway (1996) 870.
- [99] Zehetbauer, T., et al., in *Fusion Technology 1996* (19th Symp. Lisboa, 1997), Vol. 1, Elsevier, Amsterdam (1997) 925.
- [100] Wróblewski, D., et al., *Nucl. Fusion* **37** (1997) 725.
- [101] Boucher, D., et al., in *Fusion Energy 1998* (Proc. 17th Int. Conf. Yokohama, 1999), IAEA, Vienna (1999) Paper IAEA-CN-69/ITERP1/09.
- [102] Costley, A., et al., *Overview of the ITER Diagnostic System*, in *Diagnostics for Experimental Thermonuclear Fusion Reactors*, Plenum Press, New York (1998) 41.
- [103] ITER JCT, *Technical Basis for the ITER Final Design Report, Cost Review and Safety Analysis*

- (FDR), in ITER/EDA Documentation Series. To be published, IAEA, Vienna (1999).
- [104] Lloyd, B., et al., Plasma Phys. Control. Fusion **38** (1996) 1627.
 - [105] Lloyd, B., et al., Nucl. Fusion **31** (1991) 2031.
 - [106] Lazarus, E.A., et al., Nucl. Fusion **38** (1998) 1083.
 - [107] Jobes, F., et al., Phys. Rev. Lett. **52** (1984) 1005.
 - [108] Yoshino, R., Seki, M., Plasma Phys. Control. Fusion **39** (1997) 205.
 - [109] Koch, R., in Fusion Energy 1996 (Proc. 16th Int. Conf. Montreal, 1999), Vol. 1, IAEA, Vienna (1996) 633.
 - [110] Tanga, A., in Tokamak Start-up (ed. U. Knoepfel), Plenum Press, New York (1986) 159.
 - [111] Wesson, J., et al., Nucl. Fusion **29** (1989) 641.
 - [112] Edery, D., et al., in Plasma Physics and Controlled Nuclear Fusion Research 1992 (Proc. 14th Int. Conf. Würzburg, 1993), Vol. 1, IAEA, Vienna (1992) 279.
 - [113] Ninomiya, H., et al., Nucl. Fusion **28** (1988) 1275.
 - [114] Yoshino, R., et al., in Plasma Physics and Controlled Nuclear Fusion Research 1992 (Proc. 14th Int. Conf. Würzburg, 1993), Vol. 3, IAEA, Vienna (1992) 405.
 - [115] Ejima, S., et al., Nucl. Fusion **22** (1982) 1313.
 - [116] Yoshino, R., et al., Nucl. Fusion **33** (1993) 1599.
 - [117] Taylor, T.S., Plasma Phys. Control. Fusion **39** (1997) B47.
 - [118] Ferron, J.R., et al., Phys. Fluids B **5** (1993) 2532.
 - [119] Levinton, F.M., et al., Phys. Rev. Lett. **75** (1995) 4417.
 - [120] JET Team (presented by C. Gormezano), in Fusion Energy 1996 (Proc. 16th Int. Conf. Montreal, 1997), Vol. 1, IAEA, Vienna (1996) 487.
 - [121] Fujita, T., et al., Phys. Rev. Lett. **78** (1997) 2377.
 - [122] Hoang, G.T., et al., Nucl. Fusion **34** (1994) 75.
 - [123] Ide, S., et al., in Fusion Energy 1996 (Proc. 16th Int. Conf. Montreal, 1997), Vol. 3, IAEA, Vienna (1996) 253.
 - [124] Kessel, C., et al., Phys. Rev. Lett. **72** (1994) 1212.
 - [125] DIII-D Team (presented by R. Stambaugh), in Plasma Physics and Controlled Nuclear Fusion Research 1994 (Proc. 15th Int. Conf. Seville, 1995), Vol. 1, IAEA, Vienna (1994) 83.
 - [126] Moreau, D., Voitsekhovitch, I., Nucl. Fusion **39** (1999) 685.
 - [127] Garofalo, A.M., et al., Phys. Rev. Lett. **82** (1999) 3811.
 - [128] Navratil, G.A., et al., Phys. Plasmas **5** (1998) 1855.
 - [129] Boucher, D., et al., in Fusion Energy 1996 (Proc. 16th Int. Conf. Montreal, 1997), Vol. 2, IAEA, Vienna (1996) 945.
 - [130] Turnbull, A.D., et al., Nucl. Fusion **38** (1998) 1467.
 - [131] Kessel, C., in Appendix D, Section 4.7, ITER EDA Design Description Documents (available according to the intellectual property provisions of the ITER Agreement), IAEA, Vienna (1998).
 - [132] Burrell, K.H., Phys. Plasmas **4** (1997) 1499.
 - [133] Levinton, F.M., et al., Phys. Rev. Lett. **75** (1995) 4417.
 - [134] Strait, E.J., Phys. Rev. Lett. **75** (1995) 4421.
 - [135] Waltz, R., et al., Phys. Plasmas **2** (1995) 2408.
 - [136] Craddock, G.G., et al., Phys. Plasmas **1** (1994) 1944.
 - [137] Ono, M., Phys. Fluids B **5** (1993) 241.
 - [138] Wijnands, T., et al., Nucl. Fusion **37** (1997) 777.
 - [139] Ide, S., et al., Plasma Phys. Control. Fusion **38** (1996) 1645.
 - [140] Bonoli, P., MIT, personal communication, 1998.
 - [141] Litaudon, X., et al., in Fusion Energy 1996 (Proc. 16th Int. Conf. Montreal, 1997), Vol. 1, IAEA, Vienna (1996) 669.
 - [142] Voitsekhovitch, I., et al., in Fusion Energy 1998 (Proc. 17th Int. Conf. Yokohama, 1999), IAEA, Vienna (1998) Paper IAEA-CN-69/THP2/14.
- (Manuscript received 27 November 1998
Final manuscript accepted 12 August 1999)
- E-mail address of J. Wesley:
wesley@apollo.gat.com
- Subject classification: M0, Td; M0, Ti;
P0, Td; P0, Ti